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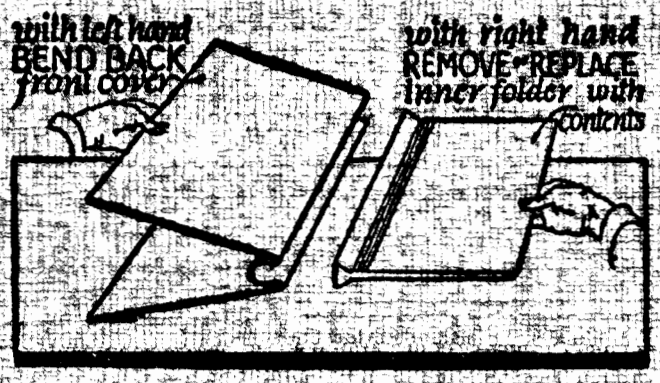

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By Henry Arthur Mosler

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PACER - A DIGITAL COMPUTER EXECUTIVE PROGRAM FOR PROCESS SIMULATION AND DESIGN

A Thesis

Submitted to the Faculty

of

Purdue University

by

Henry Arthur Mosler

In Partial Fulfillment of the

Requirements for the Degree

of

Master of Science

in

Chemical Engineering

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TABLE OF CONTENTS

	Page
LIST OF TABLES	vi
LIST OF ILLUSTRATIONS	vii
ABSTRACT	viii
A GENERAL DISCUSSION OF PACER	1
INPUT DATA FORMAT	8
Title Card	8
Run Control Card	8
Dimensions Card	10
Process Matrix Control Card	10
Process Matrix Vector	11
Equipment Parameters Control Card	11
Equipment Parameters Vector	12
Equipment Control Parameters Control Card	12
Equipment Control Parameters Vector	12
Stream Variables Control Card	13
Stream Variables Vector	13
Stream Control Variables Control Card	14
Stream Control Variables Vector	14
Stream Variables Test Vector Control Card	14
Stream Variables Test Vector	15
Stream Control Variables Test Vector Control Card	15
Stream Control Variables Test Vector	15
Preferred Stream Vector Control Card	15
Preferred Stream Vector	16
THE PACER EXECUTIVE PROGRAM	17
PACE	20
DREAD	21
DCHECK	23
DPRINT	24
RPRINT	25
SPRINT	26
CHANGE	27
MAJOR	28
SCAN	30
SSUB(N)	31

TABLE OF CONTENTS - CONTINUED

	Page
EQUIP	32
EQCALL	33
GUESS1	34
GUESS2	36
SETUP	39
TEST	41
GOOF	42
THE EQUIPMENT SUBROUTINE	43
UNAME1	45
UNAME2	46
UNAME3	47
UNAME4	48
FUTURE WORK	50
Equipment Subroutines	50
Convergence	51
Optimization	51
SUMMARY	53
BIBLIOGRAPHY	55
APPENDICES	56
A - PACER FORTRAN Statements	56
PACER	56a
DREAD	58
DCHECK	62
DPRINT	63
RPRINT	65
SPRINT	67
CHANGE	69
MAJOR	70
SCAN	73
SSUB(N)	75
EQUIP	76
EQCALL	78
GUESS1	80
GUESS2	83
SETUP	88
TEST	94
GOOF	96
DEBUG	98
UNAME1	100
UNAME2	102

TABLE OF CONTENTS - CONTINUED

	Page
UNAME3	104
UNAME4	106
DUMMY SUBROUTINES	109
B - Example Problems	111
Example 1	112
Example 2	123
Example 3	130
Example 4	136
Example 5	140
C - Nomenclature	144

LIST OF TABLES

Table		Page
1.	Input Data Summary	9
2.	Main Calling Routine	18
3.	MAJOR Calling Sequence	19
4.	Results for Example 4	139
5.	Results of Example 5, Run 3	143

LIST OF ILLUSTRATIONS

Figure		Page
1.	Program Logic for Primary Loop	3
2.	Program Logic for Recycle Calculations	4
3.	Illustrative Flow Diagram	36a
4.	Graphical Representation of the Flash Function	49a
5.	Flow Diagram for Example 1	113
6.	Flow Diagram for Example 2	124
7.	Flow Diagram for Example 3, Run 1	131
8.	Flow Diagram for Example 3, Run 2	132
9.	Flow Diagram for Example 4	137
10.	Flow Diagram for Example 5	141

ABSTRACT

Mosler, Henry Arthur, M.S.Ch.E., Purdue University, January, 1964. PACER - A DIGITAL COMPUTER EXECUTIVE PROGRAM FOR PROCESS SIMULATION AND DESIGN. Major Professor: Paul T. Shannon.

Industry today uses almost exclusively the continuous process for mass production. The use of the computer has made the calculations of complex equipment easier. Using individual equipment programs created an input-output bottleneck in that only one equipment unit could be evaluated at a time.

PACER eliminates the bottleneck by using the equipment programs as canned subroutines built into the executive program. Using PACER, an entire process design can be analyzed with one pass at the computer. This executive program does a large amount of internal bookkeeping and has the ability to do recycle calculations. PACER uses an iterative trial and error solution to solve the recycle problem. By first minimizing the number of equipment which must be calculated in an iterative solution, computing time is minimized.

Several example problems have been included to illustrate the use of PACER. Future work appears to be necessary in three areas, increasing the equipment subroutine library, studying in more detail the problem of convergence and finally, introducing into PACER a method of optimizing a given design.

A GENERAL DISCUSSION OF PACER

Industry today uses almost exclusively the continuous process for mass production. The process consists of streams entering equipment which refine, separate, react or alter the streams in some analogous manner. A realistic analysis of such a system is usually complex. The high speed computer is an invaluable tool in the study and solution of such a design problem.

If an accurate model of a given equipment can be found, then the analysis can be programmed for solution on a computer. Thus, for a given process realistically simulated, the product streams can be accurately predicted for given feed streams.

An executive system is the second step in such an analysis. When a stream is operated on by a series of equipment reacting and/or separating the different components of the stream, a step by step solution even with the use of a computer becomes unwieldy. When part of the stream is recycled in a process, it is impossible to calculate directly the final products of the process. Other methods must be used such as solving a set of simultaneous equations or using an iterative trial and error solution. PACER is designed to evaluate this type of problem given three pieces of information, the feed streams to the process, the process diagram, and finally a theoretical model for each equipment unit appearing in the process. A given design can then be analyzed with one pass at the computer. PACER uses an iterative solution for the problem of recycle.

PACER has a second advantage over the individual programming of a

possible design. If a second design is to be studied using the same equipment, but a different flow diagram or feed conditions, the individual program must be completely rewritten. With PACER it is only necessary to change the input data. PACER reduces the time necessary for a complete examination of different designs of a process. In addition introduction of human error is minimized.

When a different problem is to be analyzed using different equipment, at most,new equipment subroutines may have to be written if they are not already available in the equipment subroutine library. The executive part of the program remains fixed since it can analyze any given process operating at any given set of conditions with any given feed streams.

The next natural step would be the optimization of a design for a certain process. At the present time PACER is unable to do this.

A brief description of the PACER method of solution follows. Many of the basic ideas used in PACER were first presented by M. T. Tayyabkhan.(14) Figures 1 and 2 show the basic logic of the PACER system. The process flow diagram is numerically encoded as the Process Matrix (KPM) and treated as input data. This information is placed in the memory of PACER relating each equipment and the streams flowing to and from the equipment. The numbering of equipment units and streams is arbitrary. No two units of equipment may be numbered the same. This also applies to any two streams. The basic rule of calculation is if all input streams to an equipment are known or have been calculated previously, then the output streams from this piece of equipment can then be calculated. The function of the equipment subroutine is the above calculation. A different equipment subroutine must be available for each different type of

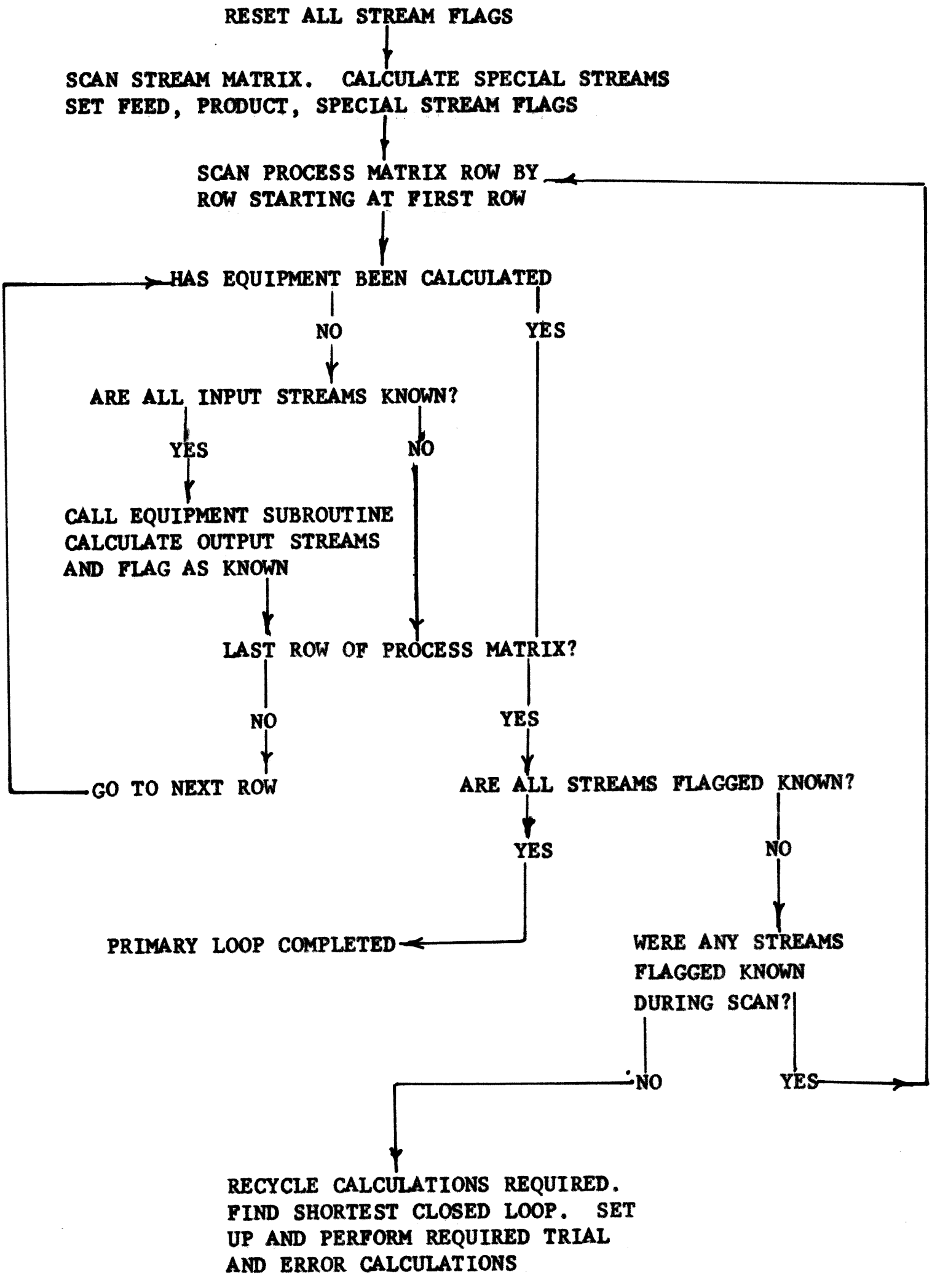

FIGURE 1

PROGRAM LOGIC FOR PRIMARY LOOP (13)

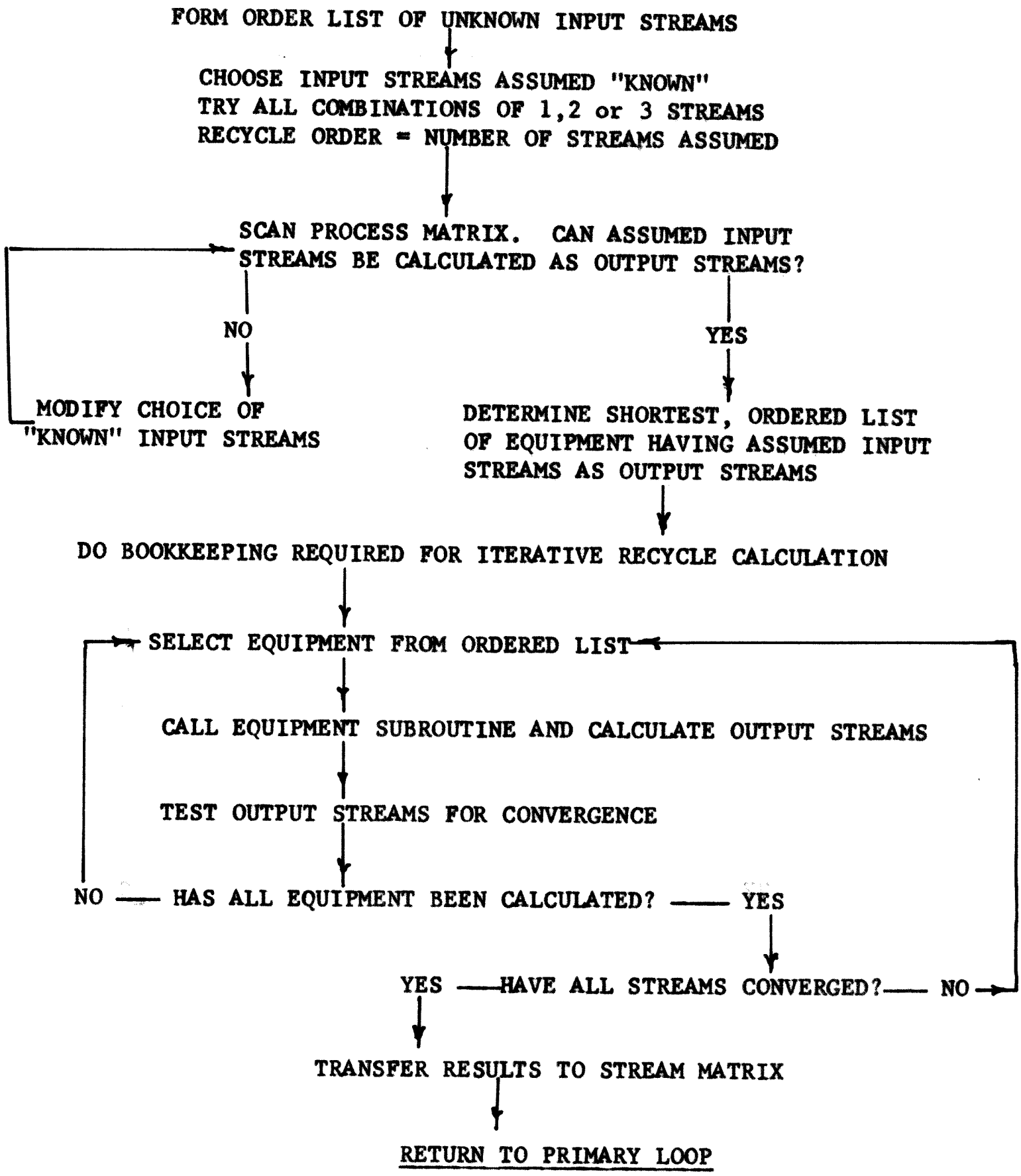

FIGURE 2

PROGRAM LOGIC FOR RECYCLE CALCULATIONS (13)

operation performed, i.e., each different equipment calculation.

Operation of the executive program is independent of the system of numbering equipment units and streams. In each row of the Process Matrix (KPM), an equipment number appears, followed by the input stream numbers (positive numbers) and then by the output stream numbers (negative numbers) associated with this piece of equipment. A one to one correspondence exists between the Process Matrix and the process flow diagram. The method of calculation does not depend on the order of equipment in the Process Matrix.

The current status of calculation of each stream and equipment is denoted by its flag (KSFLAG or KEFLAG). Initially all equipment flags (KEFLAG) and all stream flags (KSFLAG) are set to zero. All known streams are flagged greater than zero (i.e., feed streams). Because product streams by definition are not input streams to any equipment in the process, they are also flagged greater than zero. All unused equipment are flagged greater than zero. The Process Matrix (KPM) is then scanned row by row to find equipment for which the output streams can be calculated (i.e., all input streams are flagged greater than zero). When such an equipment is found, the proper equipment subroutine is called to calculate the output streams of this equipment. The output streams are flagged greater than zero since they are now known. The equipment unit is also flagged greater than zero. Scanning the Process Matrix continues.

The Process Matrix is continually rescanned until no more equipment units can be found which can be calculated directly. If the process has one or more recycle streams, the method described above will be unable to completely solve the problem. PACER recognizes this situation when one or more equipment units are flagged zero, but none of these

units has all of its input streams known. For such a problem, an iterative trial and error solution is used.

Certain prerequisites must be satisfied before the iterative solution can be used. PACER assumes an unknown input stream known and then rescans the Process Matrix to determine if any more equipment can now be calculated. If so, is the stream, which has temporarily been assumed known, an output of one of the units which can now be calculated? If both of the above conditions are not satisfied, PACER returns to the unknown stream list to assume known another stream. Combinations of unknown streams are assumed known first one, then two, and finally three at a time in an attempt to find a starting point for the trial and error calculations. Only in a system of great complexity will the problem be greater than third order recycle. PACER is not presently equipped to handle such a situation, but one is rarely encountered.

When the necessary conditions for an iterative solution are satisfied, a list (KE2) is made of all equipment which can now be calculated. PACER removes from this list all equipment unnecessary in using an iterative solution. A more detailed description is given in the section on GUESS2. The initial values of the variables for the assumed known streams are used in performing the first iterative loop. Upon completion of the loop new values for the assumed known streams are calculated. These results are used in performing the next loop in the iterative solution. The results of each loop are tested for convergence by comparing them with the results of the previous loop. When this absolute difference for each stream is less than the value specified in the input data, convergence has been obtained.

Upon the convergence of an iterative solution, the Process Matrix

(KPM) is again scanned to determine if any new equipment can now be calculated directly. In this manner the problem is solved.

When solution of the present problem has been completed, PACER reads in the next set of data to begin work on another problem. If the new problem is related to the one presently solved, a complete set of data need not be provided, but only that information about the latter which differs from the former.

In addition to the numbering restrictions mentioned previously, all PACER equipment numbers must be positive and less than 31. All PACER stream numbers must be positive and less than 61. The format for stream variable vectors (SN), stream control variable vectors (SNC), equipment parameter vectors (EN), and equipment control parameter vectors (ENC) is fixed. The most serious restriction probably is that the number of equipment must be less than 31 and the number of streams less than 61. If such is not the case and the problem cannot be broken down because of recycle streams, the dimension statements of the PACER system will have to be increased. PACER has been written so that only the dimension statements need to be changed to solve a larger problem. The programming of an entirely new problem is only as big as the job of writing the new Process Matrix providing the equipment subroutines are in the library. Even if this is not the case and certain new equipment subroutines must be written, the amount of time and energy necessary in programming the problem has been reduced to a minimum. Thus, PACER enables design problems to be more quickly examined. The PACER solution will be as accurate as is the equipment simulation model. No accuracy is lost by using the PACER system in comparison with the separate programming and solution of each individual problem.

INPUT DATA FORMAT

Sample sets of input data are shown in Appendix B, EXAMPLE PROBLEMS. The data deck must appear in the order shown in Table 1. A detailed discussion of data input follows.

Title Card (12A6)

Any alpha-numeric identification appearing on the title card in columns 1-72 will be printed at the beginning of the results.

Run Control Card (5I5)

Column Number	Variable	Comment
1-5	KRUN	the run number
6-10	KSETS	≥ 0, intermediate printing results < 0, no intermediate printing $=-1$, subroutines DCHECK, RPRINT, and CHANGE are not called by PACE.
11-15	KCLEAN	$=0$, all matrices cleared before reading in data $\neq 0$, matrices not cleared
16-20	LOOPS	maximum number of trial and error iterative loops to be performed in trying to obtain convergence.
21-25	NOGO	$=0$, if recycle calculations do not converge, final set of answers will be accepted as correct. $\neq 0$, if recycle calculations do not converge, other sets of assumed input streams will be tried in an attempt to obtain convergence. This can force termination by subroutine GOOF.

TABLE 1

Input Data Summary

	Order of Input Data	Format	Comment
1.	Title Card	12A6	Always required.
2.	Run Control Card	5I5	Always required.
3.	Dimension Card	9I5	Always required.
4.	Process Matrix Control Card	I5	Always required.
5.	Process Matrix	I5,A6,4X,I5,5X,9I5	
6.	Equipment Parameters Control Card	I5	Always required.
7.	Equipment Parameters Matrix	5F14.4	
8.	Equipment Control Parameters Control Card	I5	Always required.
9.	Equipment Control Parameters Matrix	5F14.4	
10.	Stream Variables Control Card	I5	Always required.
11.	Stream Variables Matrix	5F14.4	
12.	Stream Control Variables Control Card	I5	Always required.
13.	Stream Control Variables Matrix	5F14.4	
14.	Stream Variables Test Vector Control Card	I5	Always required.
15.	Stream Variables Test Vector	5F14.4	
16.	Stream Control Variables Test Vector Control Card	I5	Always required.
17.	Stream Control Variables Test Vector	5F14.4	
18.	Preferred Stream Vector Control Card	I5	Always required.
19.	Preferred Stream Vector	10I5	

Dimensions Card (9I5)

Column Number	Variable	Comment
1-5	N2MAX	number of rows in Process Matrix
6-10	N3MAX	number of columns in Process Matrix
11-15	NEMAX	maximum equipment number
16-20	NSMAX	maximum row number
21-25	NELMAX	length of equipment parameters row vector
26-30	NECLMX	length of equipment control parameters row vector
31-35	NSLMAX	length of stream variables row vector
36-40	NSCLMX	length of stream control variables row vector
41-45	NPSMAX	maximum number of preferred streams to be supplied

The ranges of above variables are:

$$1 \leq \text{N2MAX} \leq 30$$
$$3 \leq \text{N3MAX} \leq 10$$
$$1 \leq \text{NEMAX} \leq 30$$
$$2 \leq \text{NSMAX} \leq 60$$
$$2 \leq \text{NELMAX} \leq 10$$
$$2 \leq \text{NECLMX} \leq 10$$
$$5 \leq \text{NSLMAX} \leq 20$$
$$4 \leq \text{NSCLMX} \leq 10$$
$$0 \leq \text{NPSMAX} \leq 10$$

Process Matrix Control Card (I5)

The number of Process Matrix vectors to be read (NOKPM) is given in columns 1-5. If NOKPM =0, the next card to be read is the equipment parameters control card.

Process Matrix Vector (I5,A6,4X,I5,5X,9I5)

Column Number	Variable	Comment
1-5	KPM(NEX,1)	PACER equipment number. NEX is also the equipment number.
6-11	NAME(NEX)	Equipment subroutine name.
12-15	-	blank
16-20	NEXEQN(NEX)	External equipment number.
21-25	-	blank
26-30		
31-35		
36-40		
41-45		Associated streams numbers with equipment (NEX). N3=2,·····, 10. Input streams are positive and are given first. Output streams are negative and are given immediately after the input streams numbers.
46-50	KPM(NEX,N3)	
51-55		
56-60		
61-65		
65-70		

Equipment Parameters Control Card (I5)

The number of equipment parameters vectors to be read (NOEN) is given in columns 1-5. If NOEN=0, the next card to be read is the equipment control parameters control card.

Equipment Parameters Vector (5F14.4)

Column Number	Variable	Comment
1-4	EN(NEN,1)	PACER equipment number. NEN is also the PACER equipment number.
15-28	EN(NEN,2)	PACER external equipment flag.
29-42	EN(NEN,K3)	Other pieces of information determined by the equipment subroutine. K3=3,····,5.
43-56		
57-70		

If the EN vector is longer than 5 words, the additional words are placed on consecutive cards using the format (5F14.4). PACER determines the number of cards per vector from the length of equipment parameters vector (NELMAX) specified in the Dimension card.

Equipment Control Parameters Control Card (I5)

The number of equipment control parameters vectors to be read (NOENC) is given in columns 1-5. If NOENC=0, the next card to be read is the streams variables control card.

Equipment Control Parameters Vector (5F14.4)

Column Number	Variable	Comment
1-14	ENC(NENC,1)	PACER equipment number. NENC is also the PACER equipment number.
15-28	ENC(NENC,K3)	Other pieces of required information determined by the equipment subroutine. K3=2,····,5.
29-42		
43-56		
57-70		

If the ENC vector is longer than 5 words, the additional words are placed on consecutive cards using the format (5F14.4). PACER determines the number of cards per vector from the length of the equipment control parameters vector (NECLMX) specified in the DIMENSION card.

Stream Variables Control Card (I5)

The number of stream variables vectors to be read (NOSN) is given in columns 1-5. If NOSN=0, the next card to be read in is the stream control variables control card.

Stream Variables Vector (5F14.4)

Column Number	Variable	Comment
1-14	SN(NSN,1)	PACER stream number. NSN is also the PACER stream number.
15-28	SN(NSN,2)	PACER external stream flag. =0, if stream is inter-equipment stream. =1, if stream is feed stream. =2, if stream is product stream. =3, if stream is special feed stream.
29-42	SN(NSN,3)	Total flow rate of stream.
-	SN(NSN,K3)	NOCOMP is the number of components in the stream. The next NOCOMP entries are the fractions of each component. K3=4,·····, NOCOMP+3.
-	SN(NSN,K3)	The next NOCOMP entries are the flow rates of each component in the stream. K3=NOCOMP+4,·····, 2*NOCOMP+3.

If the SN vector is longer than 5 words the additional words are placed on other cards using the format (5F14.4). PACER determines the number of cards per vector from the length of the stream variables vector (NSLMAX) specified in the DIMENSION card.

Stream Control Variables Control Card (I5)

The number of stream control variables vectors to be read (NOSNC) is given in Columns 1-5. If NOSNC=0, the next card to be read is the stream variables test vector control card.

Stream Control Variables Vector (5F14.4)

Column Number	Variable	Comment
1-14	SNC(NSNC,1)	PACER stream number. NSNC is also the PACER stream number.
15-28	SNC(NSNC,2)	PACER external stream flag. Same system as for the SN vector.
29-42	SNC(NSNC,3)	Key component. When this is unnecessary, leave this word blank.
43-56	SNC(NSNC,4)	Number of components in the stream (COMPNO). NOCOMP=COMPNO.
57-70	SNC(NSNC,5)	-

The remainder of the SNC vector is available for any information required by equipment subroutines. If the SNC vector is longer than 5 words, the additional words are placed on other cards using the format (5F14.4). PACER determines the number of cards per vector from the length of the stream control variables vector (NECLMX) specified in the DIMENSION card.

Stream Variables Test Vector Control Card (I5)

NDELS appears in columns 1-5. If NDELS $>$ 0, the entire stream variables test vector of NSLMAX words is read. If NDELS $\leq$ 0, the next card to be read is the stream control variables control card.

Stream Variables Test Vector (5F14.4)

This vector (DELS) is used to test for convergence of the stream variables vectors in a trial and error iterative solution. In recycle calculations for convergence to occur, the absolute difference between the previous calculated value of an SN vector and the present calculated value of the same SN vector, must be less than the value specified in the DELS vector. The length of the DELS vector is the same as the length of an SN vector (NSLMAX).

Stream Control Variables Test Vector Control Card (I5)

NDELSC appears in columns 1-5. If NDELSC>0, the entire stream control variables test vector of NSCLMX words is read. If NDELSC≤ 0, the next card to be read is the preferred stream vector control card.

Stream Control Variables Test Vector (5F14.4)

This vector (DELSC) is used to test for convergence of the stream control variables vectors in a trial and error iterative solution. In recycle calculations for convergence to occur, the absolute difference, between the previous values of an SNC vector previously calculated and the present values of the same SNC vector, must be less than the values specified in the DELSC vector. The length of the DELSC vector is the same as the length of an SNC vector (NSCLMX).

Preferred Stream Vector Control Card (I5)

The number of preferred streams to be read (NOKPS) is given in columns 1-5. If NOKPS>0, NOKPS words are read from the preferred stream vector card. If NOKPS≤ 0, the program begins the calculations.

Preferred Stream Vector (10I5)

A list of preferred streams to be used in recycle calculations is given in the preferred stream vector (KPS). NOKPS is the number of preferred streams in the vector.

THE PACER EXECUTIVE PROGRAM

PACER is an executive system written in FORTRAN to be used on an IBM 7090 digital computer or its equivalent. PACER performs process design, energy, and material balance calculations. Table 2 shows the calling sequence of PACE, the main calling routine, and gives a brief description of each subroutine function. Table 3 shows the calling sequence of MAJOR, the main subroutine for performing all calculations.

TABLE 2

Main Calling Routine

Calling Sequence	Subroutine Function
PACE	Main calling program.
DREAD	Reads input data.
DCHECK	(dummy) When written, will check input data for errors.
DPRINT	Prints out the input data.
MAJOR	Performs equipment and stream calculations either directly or by a trial and error iterative solution.
RPRINT	Prints results of MAJOR.
SPRINT	Prints dimensions of problem and stream variables matrix.
CHANGE	(dummy) When written, will be able to vary the input parameters to simulate an unsteady-state problem.

TABLE 3

MAJOR Calling Sequence

Calling Sequence	Subroutine Function
MAJOR	Calling subroutine for performing all calculations.
SCAN	Flags all streams and unused equipment.
SSUB(N)	(dummy) May be used to compute values of special feed streams which are changing.
EQUIP	When an equipment can be calculated directly, EQUIP is called to do the bookkeeping before and after the equipment is calculated.
EQCALL	Calls the proper equipment subroutine for a given equipment.
GUESS1	Forms list of unknown streams when calculations cannot be done directly. GUESS1 chooses combinations of these streams 1, 2, and 3 at a time and assumes them known temporarily.
GUESS2	With the known plus the assumed known input streams, GUESS2 examines the unknown equipment to find a set of equipment on which to use a trial and error iterative solution.
SETUP	Does the bookkeeping for a trial and error iterative solution.
EQCALL	Calls the proper equipment subroutine for a given equipment.
TEST	Tests iterative solution for convergence.
GOOF	Prints all important matrices if problem is greater than third order recycle or if the iterative solution will not converge within the specified maximum number of loops and NOGO≠0.

PACE

PACE is the main calling program and calls the following subroutines:

DREAD
DCHECK
DPRINT
MAJOR
RPRINT
SPRINT
CHANGE

If KSETS = -1, the following subroutines are not called:

DCHECK
RPRINT
CHANGE

If KSETS > 0, intermediate values useful in debugging are printed.

If KBACK > 0, some of the input data is changed by subroutine CHANGE, and the problem is again solved for the new set of data as determined by CHANGE.

If KBACK ≤ 0, control is transferred back to DREAD, and another set of data is read in to be solved. When no more data remain, the job is terminated.

DREAD

This subroutine is called by PACE. DREAD reads and stores the input data. The data is read in the following order:

*title card (TITLE)
*control variables card (KRUN, KSETS, KCLEAN, LOOPS, NOGO)
*dimension variables card (N2MAX, N3MAX, NEMAX, NSMAX, NELMAX, NECLMX, NSLMAX, NSCLMX, NPSMAX)
*process matrix control card (NOKPM)
process matrix (KPMR)
*equipment parameters matrix control card (NOEN)
equipment parameters matrix (EN)
*equipment control parameters matrix control card (NOENC)
equipment control parameters matrix (ENC)
*stream variables matrix control card (NOSN)
stream variables matrix (SN)
*stream control variables matrix control card (NOSNC)
stream control variables matrix (SNC)
*stream variables test vector control card (NDELS)
stream variables test vector (DELS)
*stream control variables test vector control card (NDELSC)
stream control variables test vector (DELSC)
*preferred streams vector control card (NOKPS)
preferred streams vector (KPS)

A more detailed discussion on input is given in the section Input Data Format. All starred (*) data must appear as input.

If KCLEAN = 0, all matrices, into which the input data is read, will be cleared of previous information prior to the reading of this set of data. If KCLEAN≠0, this previous information is not cleared.

As each row of the process matrix is read, certain information is stored separately. The NECALL vector stores the knowledge of what type of process each equipment performs. The NAME vector contains the PACER equipment name for each equipment. The NEXEQN vector contains the external equipment number for each equipment. The KPM matrix is the flow diagram with each row vector of the KPM matrix representing one equipment and its associated input and output streams. The first entry in each KPM row is the PACER equipment number. This is followed by the input and

output streams.

After all input data is read, the stream-equipment matrix (KSEM) is formed. Each row vector contains information about a stream. The first entry in each KSEM row is the stream number. Next the equipment which has the stream as an output stream is given, followed by the equipment which has the stream as an input stream.

DCHECK

This subroutine is called by PACE. At present DCHECK is a dummy subroutine and does nothing. When written, it will check input data for errors.

DPRINT

This subroutine is called by PACE. DPRINT prints out the input data in the following order:

run number (KRUN)
title (TITLE)
process matrix (KPM, NAME, NEXEQN)
stream-equipment matrix (KSEM)
control variables (KSETS, KCLEAN, LOOPS, NOGO)
preferred stream vector (KPS)
dimension variables (N2MAX, N3MAX, NEMAX, NSMAX, NSLMAX, NSCLMX, NELMAX, NECLMX)
stream variables test vector (DELS)
stream control variables test vector (DELSC)
stream variables matrix (SN)
stream control variables matrix (SNC)
equipment parameters matrix (EN)
equipment control parameters matrix (ENC)

RPRINT

This subroutine is called by PACE. After the calculations have been completed for a given set of data, if KSETS ≠ -1, RPRINT is called to print the results. The results are printed in the following order:

run number (KRUN)
title (TITLE)
process matrix (KPM, NAME, NEXEQN)
preferred streams vector (KPS)
equipment parameters row vector for each equipment (EN)
equipment control parameters row vector for each equipment (ENC)
stream variables row vector for each stream associated with the above equipment (SN)
stream control variables row vector for each stream associated with the above equipment (SNC)

SPRINT

This subroutine is called by PACE. SPRINT is a shortened print routine for the results. The data is printed out in the following order:

dimension variables (N2MAX, N3MAX, NEMAX, NSMAX, NSLMAX, NSCLMX, NELMAX, NECLMX
stream variables matrix (SN)

CHANGE

This subroutine is called by PACE. At present a dummy subroutine, CHANGE will eventually be used for internal variation of problem parameters. This subroutine is called only when KSETS $\neq$ -1. If CHANGE is used to vary a parameter, the indicator, KBACK, will be set equal to a positive number. Control will then be transferred to DCHECK by PACE, and the new problem will be solved.

MAJOR

This subroutine is called by PACE and can call subroutines SCAN, EQUIP, and GUESS1. After the input data is read by DREAD, checked by DCHECK, and printed by DPRINT, PACE calls MAJOR to perform material and energy balances for the given problem. Upon completion of MAJOR, all streams have been calculated unless the problem is one of greater magnitude than third order recycle. Recycle order is determined by the minimum number of unknown streams which must be assumed known so that the KE2 list of equipments can be formed. A trial and error iterative solution may be used if the streams which have been assumed known are all output streams of the equipment contained in list KE2.

After setting all equipment flags (KEFLAG) and all stream flags (KSFLAG) equal to zero, MAJOR calls subroutine SCAN. SCAN flags feed streams to 7 and product streams to 10. If any special feed streams are contained in the problem, they are flagged 8, and then SCAN calls subroutine SSUB(N) to calculate each special feed stream. Control is then returned to MAJOR.

The process matrix (KPM) is scanned to find equipment having all input streams known, that is all input streams are flagged greater than zero. When such an equipment is found, the output streams can be calculated directly. I2 is the number of the equipment being calculated directly. MAJOR calls EQUIP to do the bookkeeping before and after the proper equipment subroutine is called to calculate directly the output streams given the input streams.

After an equipment has been calculated directly, the equipment (NE) is flagged 1, and its output streams are flagged 1. A flag of 1 signifies that the equipment or stream has been calculated.

After the process matrix (KPM) has been scanned, if an equipment is calculated directly, KES is set to 1, and the process matrix (KPM) is rescanned. This rescanning is continued until no more equipment can be found having all of its input streams known.

The equipment flag list (KEFLAG) is then scanned to determine if any equipment is still flagged zero signifying that the output streams from this equipment are not yet known. If all equipment have been calculated, the problem has been completely solved, and control is transferred back to the main calling routine, PACE, to have the results printed.

If some equipment are still flagged zero, a trial and error iterative solution is required. MAJOR then calls subroutine GUESS1 to begin the trial and error calculations. If, upon return, KRET2=2, no more equipment have been calculated successfully by GUESS1, and subroutine GOOF has printed the status of all important matrices. Control is then transferred back to PACE to have the partial results printed.

If at least one more equipment has been calculated, KRET2≠2. In this case control is transferred back to statement #7 in subroutine MAJOR. The process matrix (KPM) is again scanned to determine if any equipment, not yet calculated, has all of its input streams now known. Control is cycled between MAJOR and GUESS1 until all equipments have been calculated and the problem successfully completed, or until KRET2=2, signifying that a complete solution to the problem was not possible. In either case control is finally transferred back to the main calling routine, PACE.

SCAN

This subroutine is called by MAJOR and can call subroutine SSUB(N). The stream type is signified by the second element in the stream variables row vector. If this entry is 1, the stream is a feed stream. If 2, the stream is a product stream. If 3, the stream is a special feed stream. If 0, the stream is an inter-equipment stream. A feed stream has its flag (KSFLAG) set to 7. Product streams are flagged 10. If any special feed streams are present in the process, they are flagged 8, and SCAN calls subroutine SSUB(N) to calculate each special feed stream. SSUB(N) then prints the row vector of the special feed stream from the stream variables matrix (SN) and from the stream control variables matrix (SNC). Control is then returned to SCAN. Unused equipment are flagged 99. Control is then transferred back to MAJOR.

SSUB(N)

This subroutine is called by SCAN. The purpose of SSUB(N) is to calculate any special feed stream prior to the start of material and energy balances by MAJOR. N is the stream number of the special feed stream. At present, SSUB(N) is a dummy routine. The space is available to simulate a transient problem where one or more of the streams is changing as a function of another parameter such as time. After the special feed stream has been calculated, and its row vectors from the stream variables matrix (SN) and from the stream control variables matrix (SNC) are printed, control is returned to SCAN.

EQUIP

After an equipment has been found with all input streams known, MAJOR sets I2 equal to the number of this equipment and then calls subroutine EQUIP. EQUIP calls subroutine EQCALL.

Each input stream to this equipment (I2) has its row vector in the stream variables matrix (SN) stored as a STRMI row vector. Each input stream also has its row vector in the stream control variables matrix (SNC) stored as a STRMCI row vector. Each output stream from this equipment (I2) has its row vector in the stream variables matrix (SN) stored as a STRMO row vector. Each output stream also has its row vector in the stream control variables matrix (SNC) stored as a STRMCO row vector. The number of input streams to this equipment is NIN, and the number of output streams is NOUT. The STRMI, STRMCI, STRMO, and STRMCO matrices only contain information about streams entering or leaving this equipment (I2) which is about to be calculated.

After storing this information, EQUIP calls subroutine EQCALL to call the proper equipment subroutine for this equipment (I2). The equipment subroutine calculates the output streams for this equipment (I2) given the input streams.

Control is then returned to EQUIP where the output stream variables (STRMO) and the output stream control variables (STRMCO) are stored in their appropriate places in the stream variables matrix (SN) and stream control variables matrix (SNC). Control is then returned to MAJOR.

EQCALL

This subroutine is called by EQUIP or SETUP. NE is the number of the equipment being calculated. The NECALL vector is used to determine which equipment subroutine is to be called. After the equipment has been calculated by the proper equipment subroutine, control is returned to EQCALL and then back to the calling subroutine, EQUIP or SETUP.

If new equipment subroutine names are to be used, EQCALL and DREAD must be modified to include the new names.

GUESS1

This subroutine is called by MAJOR when it is found necessary to use trial and error in an attempt to solve the problem. By assuming certain streams known, that is temporarily flagging them greater than zero, it may then be possible to use trial and error methods to converge on a solution.

The KUS list is made of all unknown inter-equipment streams (those still flagged zero). If any preferred streams were given in the input data, the preferred streams appearing in the KUS list of unknown streams are moved to the beginning of the list. The KUS list contains NUSMAX unknown streams. All possible combinations of these NUSMAX streams are assumed known, first one, then two, and finally three at a time. These streams are called KCV1, KCV2, and KCV3 and flagged 2. GUESS1 then calls subroutine GUESS2 to examine the list of unknown equipment to search for a list of equipment which can now be calculated and which have all of the assumed known streams as output streams. If this list does not exist, KRET is set to 2, and control is then returned to GUESS1.

New combinations of unknown streams are picked until the above mentioned equipment list can be found at which time the iterative trial and error solution is performed.

There are three possibilities for GUESS1 to exit from the cycle of picking combinations of unknown streams. If a satisfactory combination is found, and the calculations converge within the maximum number of iteration loops (LOOPS) specified in the control variables card, KRET is set to zero. Control is returned to GUESS1, and then to MAJOR.

Second, if all possible combinations are tried, but the problem appears to be greater than third order, GUESS1 will call subroutine GOOF to print

the present status of all equipment and stream flags. KRET is set to 2, and control is returned to MAJOR.

A third possibility is that the problem is not greater than a third order recycle, but convergence was not obtained within the specified maximum number of iteration loops (LOOPS). If the control variable (NOGO) equals zero, the final set of iterative values will be accepted as correct although convergence has not been obtained. If NOGO≠0, control will transfer back to GUESS1 in an attempt to find a combination of streams to assume known so that the problem will converge within the specified limits. In this event subroutine GOOF may still be called although the problem was not greater than third order recycle.

GUESS2

This subroutine is called by GUESS1 to see if it is possible to find a list (KE2) of unknown equipment which can now be calculated using the temporarily assumed known streams. These assumed streams must all be output streams of the equipment in the KE2 list. GUESS2 calls subroutine SETUP when the above conditions are met.

The process matrix (KPM) is scanned, and each equipment still unknown is examined to see if all of its input streams are now either known or assumed known. When this is the case, this equipment is placed in the KE2 list, and each output stream from this equipment is flagged known. Each stream flag (KSFLAG)is set to 3. The unknown equipment are continually rescanned until no more equipment can be added to the KE2 list. NE2MAX is the number of equipment in the KE2 list.

If any assumed known stream is not an output stream of an equipment in the KE2 list, a trial and error iterative solution is not possible. In this case all equipment and streams are reflagged to their former values. Control is then transferred back to GUESS1 to pick a new combination of streams to be assumed known .

If all assumed streams are output streams of the equipment in the KE2 list, an iterative solution is possible. All equipment and streams which have been flagged 3 are reflagged zero.

The next step is to find the shortest list of equipment with which to perform the iterative solution. All branch equipment not essential to the iterative solution are eliminated.

The example in Figure 3 shows a first order recycle problem. Stream 1 is a feed stream and is known. No equipment in this diagram can be calculated directly, because no equipment has all of its input streams

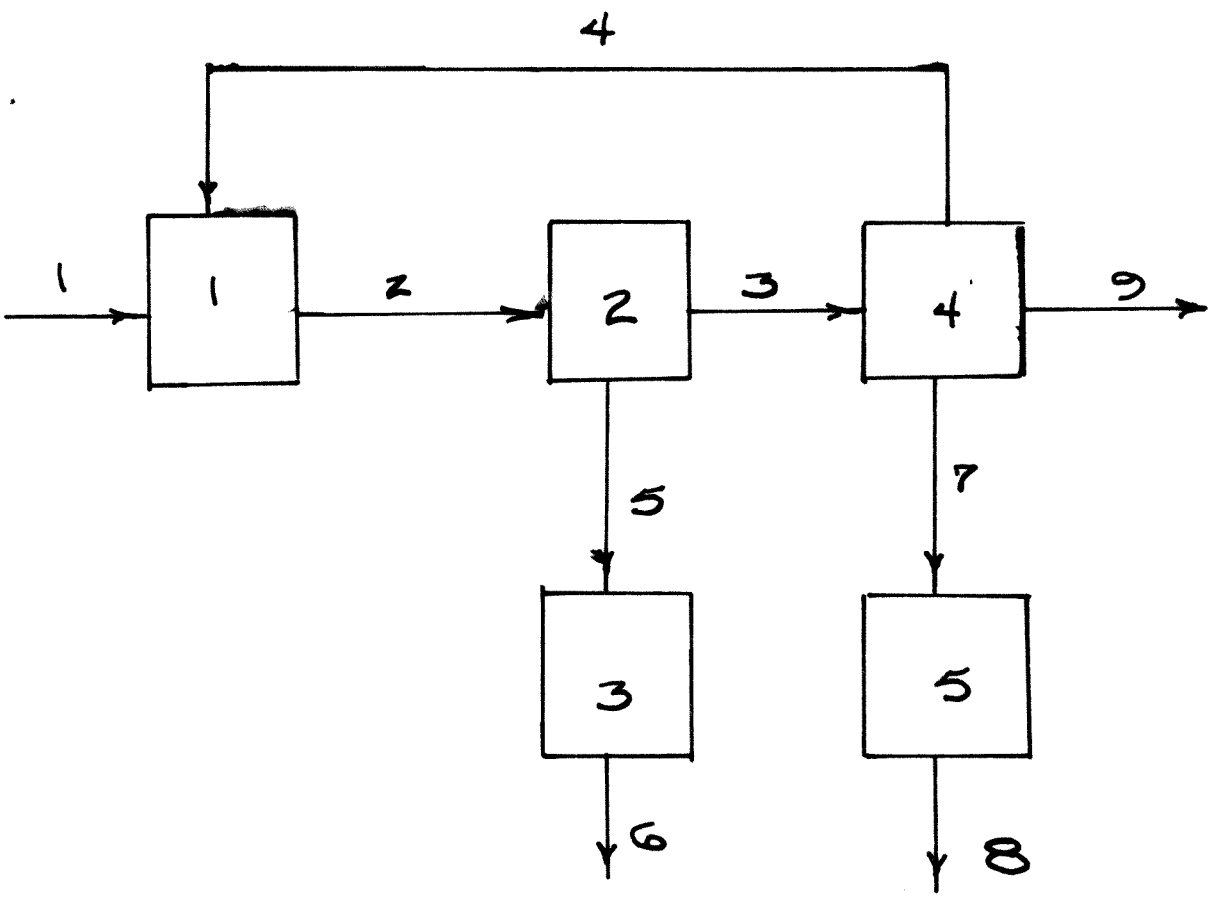

FIGURE 3

Illustrative Flow Diagram

known. The problem is first order because only one stream need be assumed known to meet the requirements for an iterative solution. If either stream 2, 3, or 4 assumed known by GUESS1, the KE2 list formed will contain all five equipment, and an iteration solution may be used. When performing the iterative solution, the shortest list will not contain equipment 3 or 5.

For purposes of illustration let stream 2 be known. The KE2 list is then: 2,3,4,5,1. The shortest list (KE3) is formed by tracing the unknown input streams back through the flow diagram. The initial equipment in the KE3 list are the equipment for which each of the assumed known streams is an output stream. Since stream 2 has been assumed known, equipment 1 is the first entry in list KE3. Tracing backward through the flow diagram, the complete KE3 list is: 1,4,2. All equipment listed in KE3 are flagged 4. All output streams from equipment in list KE3 are also flagged 4. The shortest list is then reorganized as list KE1 containing all the equipment in list KE3 but appearing in the sequence of the KE2 list. The KE1 list is: 2,4,1. GUESS2 now calls subroutine SETUP to perform the trial and error iterative solution.

When control returns to GUESS2, three possibilities exist. If KRET3=0, convergence has been obtained within the maximum number of iterative loops (LOOPS). All equipment in the KE1 list are flagged 1, and all output streams from these equipment are flagged 1. GUESS2 sets KRET=0, and control is returned to GUESS1.

If KRET3=-2, the problem did not converge, but the non-convergent final answers have been accepted as correct because NOGO=0. All of the above mentioned equipment and streams are flagged 13. SETUP has set KRET=0, and control is returned to GUESS1.

If KRET3=2, the iterative calculations did not converge and the final set of answers is not accepted because NOGO≠0. GUESS2 sets KRET=2. Control is then returned to GUESS1 to find another combination of streams to assume known such that convergence might be obtained. This is the same return to GUESS1 as if list KE2 could not be successfully formed with the given assumed known streams in the first place.

SETUP

This subroutine is called by GUESS2. SETUP does the bookkeeping for the trial and error iterative calculations and calls EQCALL and TEST. Every input and output stream from each equipment in list KE1 has its row vector in the stream variables matrix (SN) stored in the corresponding row of the STRM matrix and has its row vector in the stream control variables matrix (SNC) stored in the corresponging row of the STRMC matrix. SETUP sets LOOP=1 and LIMIT2=0.

SETUP uses the following procedure in performing the calculations. For a given equipment, each input stream has its row vector from the STRM matrix stored as a STRMI row vector and has its row vector in the STRMC matrix stored as a STRMCI row vector. Each output stream from this equipment has its row vector from the STRM matrix stored as a STRMO row vector and has its row vector from the STRMC matrix stored as a STRMCO row vector. The number of input streams to this equipment is NIN, and the number of output streams is NOUT. NE is the number of the equipment being calculated.

After storing this information, SETUP calls subroutine EQCALL to call the proper equipment subroutine for equipment NE. The equipment subroutine calculates the output streams for this equipment given the input streams. Each output stream has its stream variables vector and stream control variables vector stored as a STRMO row vector and a STRMCO row vector respectively. Control is then returned to SETUP, and SETUP in turn calls subroutine TEST to test for convergence.

If LIMIT=0 upon return of control to SETUP, convergence has been obtained for this equipment (NE). If not, 1 is added to LIMIT2. The recently calculated STRMO and STRMCO row vectors replace their older

counter-parts in the STRM and STRMC matrices.

After every equipment in list KE1 has been calculated in the manner described above, one iteration has been completed. If LIMIT2=0, convergence has been obtained for each equipment in the KE1 list. Each output stream from each equipment in list KE1 has its STRM row vector stored in the SN matrix and has its STRMC row vector stored in the SNC matrix. SETUP sets K15=1 and KRET3=0, and control is returned to GUESS2.

If LIMIT2>0, convergence has not been obtained for one or more equipment in list KE1. If the number of iterations completed (LOOP) is less than the maximum specified (LOOPS), control is transferred back to statement #18 in SETUP, and another iteration is performed. If LOOP = LOOPS the maximum number of iteration loops specified have already been performed, but since LIMIT2>0, convergence has not yet been obtained for all of the equipment listed in KE1. If NOGO=0, the final set of iterative values will be accepted as correct. SETUP sets K15=3, and the STRM and STRMC row vectors are stored as before. The equipment in list KE1 and their respective output streams are flagged 13 indicating they are now known. SETUP sets KRET3=-2, and control is returned to GUESS2.

If NOGO≠0, the equipment in list KE1 and their respective output streams are reflagged zero. SETUP sets K15=2 and KRET3=2. Control is then returned to GUESS2 to find another combination of streams to assume known such that convergence may be obtained within the specified maximum number of iteration loops (LOOPS).

TEST

After the output streams have been calculated for an equipment (NE) from list KE1, SETUP calls TEST to test for convergence. If the absolute difference, between the respective elements in the STRMO and STRMCO row vectors just calculated and the corresponding STRM and STRMC row vectors calculated one iteration back, are less than the convergence test vectors DELS and DELSC, the iterative solution has converged for this equipment. If convergence has been obtained, LIMIT=0 when control returns to SETUP. If convergence has not been obtained, LIMIT > 0.

GOOF

This subroutine is called by GUESS1 when the problem appears to be greater than a third order recycle or if the iterative solution would not converge within the specified maximum number of loops (LOOPS) and NOGO≠0.

GOOF prints:

- run number (KRUN)
- unknown stream list (KUS)
- equipment flags (KEFLAG)
- stream flags (KSFLAG)
- process matrix (KPM, NAME, NEXEQN)
- preferred stream vector (KPS)
- stream variables matrix (SN)
- stream control variables matrix (SNC)
- equipment parameters matrix (EN)
- equipment control parameters matrix (ENC)

Control is then returned to GUESS1.

THE EQUIPMENT SUBROUTINE

An equipment is by definition a unit which takes one or more input streams and performs some type of operation on them in producing one or more output streams. Prior to calling an equipment subroutine to calculate the output streams of an equipment, given the input streams, certain information is stored by the executive program. If the equipment is being calculated directly, because all input streams are really known, subroutine EQUIP does the bookkeeping. If the equipment is being calculated as part of a trial and error iterative solution, subroutine SETUP does the bookkeeping. The required information is stored in the same manner by both subroutines.

NE is the PACER number of the equipment being calculated. There are NIN input streams to this piece of equipment and NOUT output streams from this equipment. Each input stream has its SN vector stored as a STRMI vector and has its SNC vector stored as a STRMCI vector. Each output stream has its SN vector stored as a STRMO vector and has its SNC vector stored as a STRMCO vector. The first row of the STRMI matrix contains the SN vector for the first stream entering equipment (NE) as listed by the Process Matrix reading from left to right. The second row contains information about the second input stream, etc. The first row of the STRMO matrix contains the SN vector for the first stream listed as leaving the equipment. The second row contains information about the second output stream, etc. The order of the rows of the

STRMCI and STRMCO matrices are determined similarly.

After all the information is stored as described above, EQCALL calls the proper equipment subroutine to calculate the output streams from equipment (NE) given the input streams. The equipment subroutine performs the desired calculations, and must store the newly calculated output stream vectors in certain predetermined locations. Upon return of control from the equipment subroutine to the executive program, these newly calculated streams must appear in the STRMO and STRMCO matrices. The first row of the STRMO matrix must contain the SN vector for the first output stream as listed by the Process Matrix. The second row contains the SN vector for the second output stream, etc. The order of information stored in the STRMCO matrix is determined similarly.

Output information storage is the only restriction PACER places on an equipment subroutine. With this in mind, an equipment subroutine can be written simulating any desired process.

UNAME1

This subroutine simulates a mixer which takes NIN input streams and produces NOUT output streams. The input streams may have different compositions and different flow rates. All output streams will have the same composition, but the flow rates of the output streams need not necessarily be equal. The fraction of the overall output in a given stream is specified in the equipment parameters vector for the equipment (NE) being calculated. Identical components must appear in each stream variables vector (SN) in the same order even if the magnitude of one or more components in a given stream is zero. The number of components in each stream (NOCOMP=COMPNO) is the fourth word of the stream control variables vector (SNC).

The format of the input data for this equipment (NE) is as follows:

EN(NE,1)= PACER equipment number. NE also is the PACER equipment number.

EN(NE,2)= PACER external equipment flag.

EN(NE,K2)= The fraction of the overall output in each output stream, 1 to NOUT. K2=3,····, NOUT+2.

ENC(NE,1)= PACER equipment number.

ENC(NE,2)= PACER external equipment flag.

UNAME2

This subroutine simulates a distillation column where one input stream is split into two output streams, an overheads and a bottoms. The fraction of each component in the feed appearing in the overhead is given in the equipment parameters vector (EN) for this equipment (NE). The first output stream listed in the Process Matrix vector for this piece of equipment is the overheads. The second output stream listed is the bottoms.

The format of the input data for this equipment (NE) is as follows:

EN(NE,1)= PACER equipment number. NE is also the PACER equipment number.

EN(NE,2)= PACER external equipment flag.

EN(NE,3)= Ficticious heat load.

EN(NE,J)= The fraction of the amount of component 1 to NOCOMP appearing in the overhead. J=4,····, NOCOMP+3.

ENC(NE,1)= PACER equipment number.

ENC(NE,2)= PACER external equipment flag.

ENC(NE,3)= First constant in heat equation.

ENC(NE,4)= Second constant in heat equation.

ENC(NE,5)= Third constant in heat equation.

UNAME3

This subroutine simulates a mixer which takes NIN input streams and produces NOUT output streams. The input streams may have different compositions and different flow rates. All output streams will have the same composition. The total output is split equally between the NOUT output streams. Identical components must appear in each stream variables vector (SN) in the same order even if the magnitude of one or more components in a given stream is zero. The number of components in each stream (NOCOMP=COMPNO) is the fourth word of the stream control variables vector (SNC).

The format of the input data for this equipment (NE) is as follows:

EN(NE,1)= PACER equipment number. NE is also the PACER equipment number.

EN(NE,2)= PACER external equipment flag.

ENC(NE,1)= PACER equipment number.

ENC(NE,2)= PACER external equipment flag.

UNAME4

This subroutine simulates a flash distillation where one feed stream is split into two output streams in equilibrium with each other, one vapor and the second liquid. The equilibrium value of each component must be known at the conditions of the flash. Complete miscibility of all components is assumed, and activity corrections are not used.

UNAME4 first certifies that a fraction of the feed is flashed greater than zero but less than one. This occurs when the temperature of the flash is less than the dew point but greater than the bubble point. This corresponds to:

$$\sum_{i=1}^{NOCOMP} Z_i/K_i > 1 \qquad (1)$$

and

$$\sum_{i=1}^{NOCOMP} Z_i K_i > 1 \qquad (2)$$

where Z_i is the mole fraction of the i^{th} component in the feed and K_i is the equilibrium value of the i^{th} component at the conditions of the flash.

If the vapor and liquid product streams are at equilibrium,

$$V = \sum_i V_i = \sum_i \frac{K_i Z_i}{K_i + \frac{1-V}{V}} \qquad (3)$$

where V is the fraction of the feed vaporized. $0 < V < 1$

$$V_i = y_i V \qquad (4)$$

where y_i is the mole fraction in the vapor of the $i^{\underline{th}}$ component.

The fraction vaporized is found by using the Newton-Raphson iteration process.

$$V^{(N+1)} = V^{(N)} - \frac{g(V^{(N)})}{g'(V^{(N)})} \tag{5}$$

where
$$g(V) = \sum_i \frac{K_i Z_i}{V(K_i - 1)+1} - 1 \tag{6}$$

and
$$g'(V) = \sum_i \frac{-K_i Z_i (K_i-1)}{\left[V(K_i-1) +1\right]^2} \tag{7}$$

If the initial value used for V is zero, this method should always converge. Justification of this solution is given in Holland (5). A graphical representation of the function is shown in Figure 4 .

The format of the input data for this equipment (NE) is as follows:

EN(NE,1)= PACER equipment number. NE is also the PACER equipment number.

EN(NE,2)= PACER external equipment flag.

EN(NE,3)= blank

EN(NE,J)= Equilibrium value of each component, 1 to NOCOMP, at the conditions of the flash. J=4,····,NOCOMP+3.

ENC(NE,1)= PACER equipment number.

ENC(NE,2)= PACER external equipment flag.

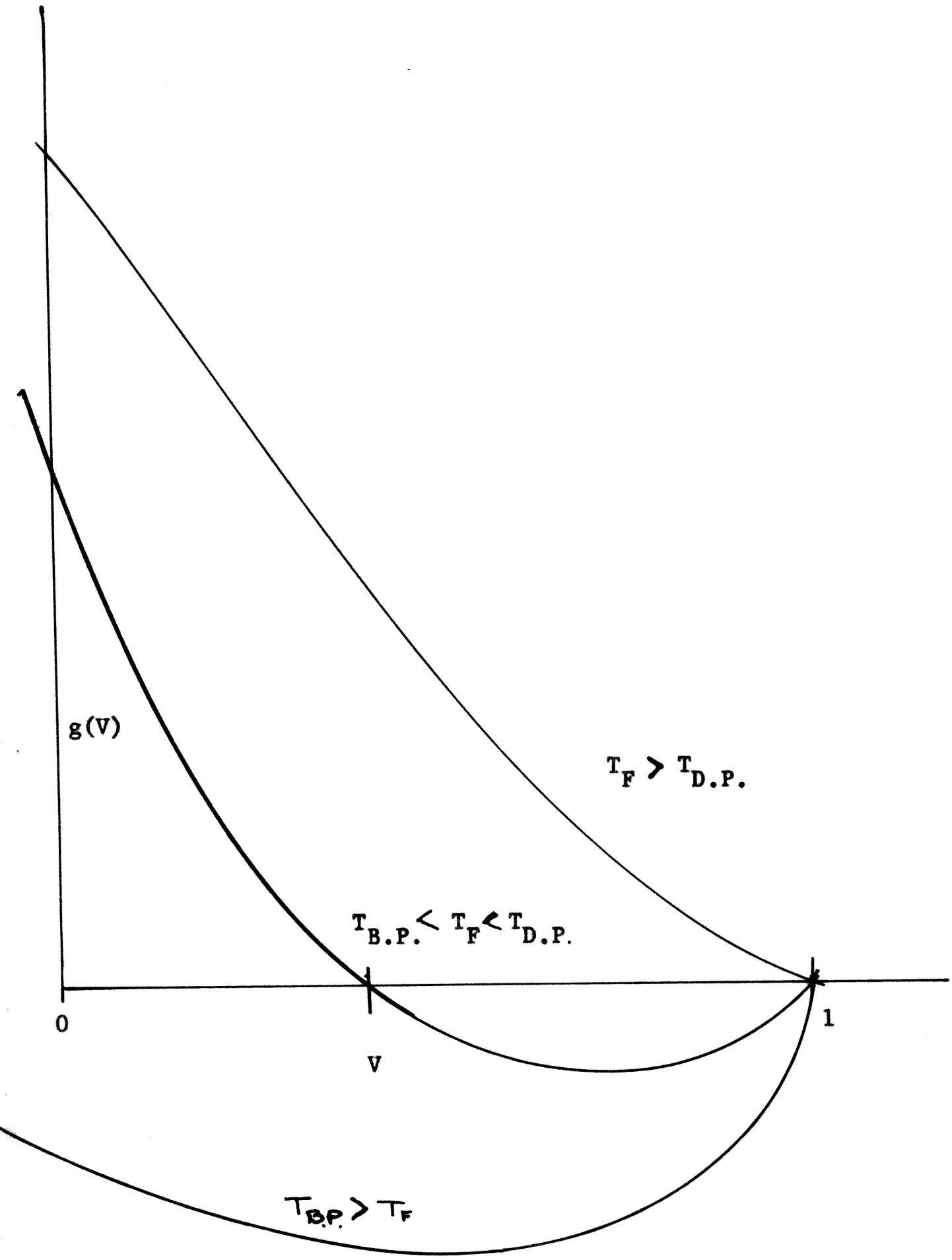

FIGURE 4

Graphical Representation of the Flash Function

FUTURE WORK

PACER is now a working executive system with the ability to set up and solve process design problems. There are three areas in which additional work seems necessary. First is the need for additional equipment subroutines to simulation processes in industry today. Second, very little has been said about the problem of convergence. Finally, a method of optimization should be built into PACER.

Equipment Subroutines

More equipment subroutines will be written as they are needed. When the PACER program becomes available, it is hoped that outside sources will contribute to the equipment subroutine library. Few restrictions have been placed on the equipment subroutine. As these subroutines become more sophisticated, a larger amount of input data will become necessary.

A subroutine should be written for the determination of equilibrium values of components given the operating temperature and pressure. To accurately interpolate equilibrium values at given conditions, a vast amount of data must be available at various conditions. It is conceivable that the data pack for such a program might become very large. Another welcome addition to the equipment subroutine library would be the simulation of a reactor.

Out of necessity the subroutines used for equipment in a recycle calculation cannot be too complex. A large number of iteration loops might be necessary before convergence is even roughly obtained. Then

after the problem begins to converge, more sophisticated, time consuming subroutines might be introduced in solving for the final convergent values. In example 4, run 18, 37 iterations loops were needed before the predetermined convergence requirements were satisfied. 23 equipment units needed to be calculated for a single iteration loop. This would seem to indicate a time problem even on a large computer.

Convergence

PACER solves recycle problems by using a trial and error iterative solution until the calculations converge to within the values of fluctuation specified in the input data. Will this type of solution always converge if enough iterative loops are calculated? The five example problems illustrated in Appendix B converged. In real plants, a process eventually converges if the operating conditions are held constant over a long enough period of time. This does not prove, however, that PACER will always converge.

All that can be said is if convergence is obtained, the results will be a satisfactory solution to the input data and equipment subroutine restraints. It seems, therefore, that more work is needed in the area of convergence.

It is sometimes the case in first order and higher order recycle problems, that there are several sets unknown streams which when assumed known will give a satisfactory list of equipment on which an iterative solution can be performed. It may be possible that the solution to this type of problem is not unique.

Optimization

PACER is unable to optimize a proposed process design. For a given

set of operating conditions PACER can evaluate the process design. Only by the solution of different sets of operating conditions might an insight be gained as to where the optimum set of conditions lie.

The profit function is usually optimized in this type of calculation since economics determine whether the proposed plant should be built. It would seem necessary to introduce this profit function into the PACER calculations. Once the above is written, an optimizing scheme must be found.

Shell Oil is presently using a system called CHEOPS (12). Problems are optimized by the method of steepest ascent. Several other optimization techniques have been published. This would at least indicate a starting point on the problem of optimization.

SUMMARY

A working executive system for process simulation and design is now available. Given a proposed design which is encoded in the Process Matrix and the necessary equipment subroutines, PACER can solve for the steady-state solution. In the past, an input-output bottleneck has existed in that the engineer could only solve one case at a time. PACER relieves this bottleneck with its executive program. This should enable the design engineer to make greater use of the digital computer.

PACER minimizes the changes that must be made with the program when a new case is to be evaluated. By performing all bookkeeping within the executive program, only two pieces of external information are required, the input data for the proposed design and the equipment subroutines simulating the operation of units in the design. PACER will function on any consistent set of streams and equipment.

When a recycle problem necessitates the use of a trial and error iterative solution, the executive system finds a set of unknown streams to assume known such that the iterative solution can be performed. PACER then determines the shortest list of equipment which must be calculated to obtain convergence. Only after the iterative problem has converged are the branch equipment calculated. Finding this shortest list minimizes the number of calculations necessary in completely performing one iterative loop. Because recycle calculations appear to require the bulk of computing time in evaluating a process design, minimizing these calculations is definitely an asset.

The example problems in Appendix B illustrate the minimum effort required in programming problems which are completely different in design. The key to the PACER method of solution is the one to one correspondence between the actual process flow diagram and the Process Matrix (KPM). Through the Process Matrix, PACER can "see" the process flow diagram in the same manner as the engineer.

Future work appears to be necessary in three areas, increasing the equipment subroutine library, studying in more detail the problem of convergence and finally, introducing into PACER a method of optimizing a given design.

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APPENDICES

APPENDIX A

PACER FORTRAN Statements

The following pages contain the actual printout of PACER program. The IBM 7090 computer at Purdue uses "READ INPUT TAPE 5" for a read statement and "WRITE OUTPUT TAPE 6" for a write statement.

```
C                              P A C E R
C
C                   PROCESS ASSEMBLY CASE EVALUATOR ROUTINE
C
C       WRITTEN BY PAUL T SHANNON  JULY-AUGUST, 1961
C       THAYER SCHOOL OF ENGINEERING
C             DARTMOUTH COLLEGE
C          HANOVER, NEW HAMPSHIRE
C                    HENRY MOSLER  MAY-SEPTEMBER, 1963
C          SCHOOL OF CHEMICAL ENGINEERING
C                    PURDUE UNIVERSITY
C                    LAFAYETTE, INDIANA
C
C       MAXIMUM NO. OF EQUIPMENT IS 30
C       MAXIMUM LENGTH OF EQUIPMENT PARAMETER VECTOR EQUALS 10
C       MAXIMUM LENGTH OF EQUIPMENT CONTROL VECTOR EQUALS 10
C       MAXIMUM NO. OF STREAMS IS   60
C       MAXIMUM LENGTH OF STREAM PROPERTIES VECTOR EQUALS 20
C       MAXIMUM LENGTH OF STREAM CONTROL VECTOR EQUAL 10
C       MAXIMUM NO. OF STREAMS PER EQUIPMENT IS 9
C       MAXIMUM NUMBER OF SPECIFIED PREFERRED STREAMS EQUALS  10
C       SN(NSN,NSL) IS STREAM MATRIX.  ONE COLUMN FOR EACH STREAM.
C          NSN CORRESPONDS TO STREAM NUMBER  1 TO NSMAX
C          NSL CORRESPONDS TO STREAM VARIABLES  1 TO  NSLMAX
C       SNC(NSNC,NSCL) IS STREAM CONTROL MATRIX. ONE COL. FOR EACH STREAM
C          NSNC CORRES. STREAM NUMBER  1  TO NSMAX
C          NSCL CORRES. TO STREAM CONTROL VARIABLES 1 TO NSCLMX
C       EN(NEN,NEL) IS EQUIPMENT VARIABLES MATRIX.  ONE COL. EACH EQUIP.
C          NEN CORRES. TO EQUIP. NUMBER  1 TO NEMAX
C          NEL CORRES. TO EQUIP. VARIABLE  1 TO  NELMAX
C       ENC(NENC,NECL) IS EQUIPMENT CONTROL MATRIX
C            NENC  CORRES. TO EQUIPMENT NUMBER  1 TO NEMAX
C            NECL CORRES. TO EQUIPMENT CONTROL VARIABLE 1 TO NECLMX
C       FEED STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 1.0
C       PRODUCT STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 2.0
C       SPECIAL FEED STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR
C                                                       EQUAL TO 3.0
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
```

```
      COMMON   J2,J3,J4,J5
      COMMON   J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1         J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2         J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON   J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON   K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1         K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON   L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
C
C
C
C
C
C     MAIN CALLING PROGRAM
C
    1 CALL   DREAD
      IF(KSETS+1)  2,3,2
    2 CALL   DCHECK
    3 CALL   DPRINT
    4 CALL   MAJOR
      IF(KSETS+1)  5,10,5
    5 CALL   RPRINT
   10 CALL SPRINT
      IF(KSETS+1)  6,9,6
    6 CALL   CHANGE
    7 IF(KBACK) 9,9,1
    9 GO TO 1
      END
```

```
      SUBROUTINE  DREAD
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON  NDELS, NDELSC
      COMMON NEXEQN, NAME,KSEM
      EQUIVALENCE (A,I)
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
      READS IN INPUT DATA FOR PACER
      READ ALPHABETIC HEADER CARD  12A6 FIELD
      READ TITLE CARD
    1 READ INPUT TAPE 5,  200, TITLE
      READ CONTROL VARIABLES
    2 READ INPUT TAPE 5,  201, KRUN, KSETS, KCLEAN, LOOPS, NOGO
      READ DIMENSION VALUES
    3 READ INPUT TAPE 5,  202, N2MAX,N3MAX,NEMAX,NSMAX,NELMAX,NECLMX,NSL
     1MAX,NSCLMX,NPSMAX
      DO 777 N4=1,30
      DO 777 N5=1,12
  777 KPMR(N4,N5) = 0
      IF KCLEAN IS ZERO, SET MATRICIES TO ZERO OTHERWISE READ KPM
    4 IF(KCLEAN)                  36,5,36
    5 DO  7   N4 = 1, 30
    6 DO  7   N5 = 1,10
```

```
      KPM(N4,N5) = 0
      DO 12    N4 = 1,60
      DO 12    N5 = 1,20
      SN(N4,N5) = 0.
      STPM(N4,N5) = 0.
      DELS(N5) = 0.
      DO 17    N4 = 1,60
      DO 17    N5 = 1,10
      SNC(N4,N5) = 0.
      STRMC(N4,N5) = 0.
      KUS(N4) = 0
      KPS(N5) = 0
      DELSC(N5) = 0.
      DO 20    N4 = 1,30
      DO 20    N5 = 1,10
      EN(N4,N5) = 0.
      ENC(N4,N5) = 0.
      DO 27    N4 = 1,8
      DO 27    N5 = 1,20
      STRMI(N4,N5) = 0.
      STRMO(N4,N5) = 0.
      DO 31    N4 = 1,8
      DO 31    N5 = 1,10
      STRMCI(N4,N5) = 0.
      STRMCO(N4,N5) = 0.
C     READ KPM ROWS INTO KPMR
      READ INPUT TAPE 5,  203,  NOKPM
      IF(NOKPM)          44,44,38
C     NOKPM IS NUMBER OF ROWS OF KPM TO BE READ. IF ZERO, GO TO READ EN
      READ INPUT TAPE 5,   204, ((KPMR(K2,K3),K3=1,12),K2=1,NOKPM)
C     SET UP EQUIPMENT SUBROUTINE CALLING NUMBER FROM EQUIPMENT NAME
      KVOID=0
      DO  243 K2 = 1,  NOKPM
      NEX =  KPMR(K2,1)
C     NEX IS EQUIPMENT NUMBER OF PROCESS MATRIX ROW KPMR(K2,1)
C     KPMR(K2,2) IS THE EQUIPMENT SUBROUTINE NAME
      I = KPMR(K2,2)
      DO 243 K3 = 1,15
      GO TO      (401,402,403,404,405,406,407,408,409,410,411,412,413,
     1 414,415),K3
C     ONLY THE NAMES USED IN THE IF STATEMENTS GIVEN BELOW ARE
C     ACCEPTABLE EQUIPMENT SUBROUTINE NAMES FOR PACER
  401 IF(A- (+6HUNAME1)) 243,143,243
  402 IF(A- (+6HUNAME2)) 243,143,243
  403 IF(A- (+6HUNAME3)) 243,143,243
  404 IF(A- (+6HUNAME4)) 243,143,243
  405 IF(A- (+6HUNAME5)) 243,143,243
  406 IF(A- (+6HUNAME6)) 243,143,243
  407 IF(A- (+6HUNAME7)) 243,143,243
  408 IF(A- (+6HUNAME8)) 243,143,243
  409 IF(A- (+6HUNAME9)) 243,143,243
  410 IF(A- (+6HDEBUG )) 243,143,243
  411 IF(A- (+6HEXTRA1)) 243,143,243
  412 IF(A- (+6HEXTRA2)) 243,143,243
  413 IF(A- (+6HEXTRA3)) 243,143,243
  414 IF(A- (+6HEXTRA4)) 243,143,243
  415 IF(A- (+6HEXTRA5)) 416,143,416
  416 IF(NECALL(NEX)) 418,418,243
  418 WRITE OUTPUT TAPE 6,417,NEX,A,KRUN
  417 FORMAT(26H1IN DREAD EQUIPMENT NUMBER,I3,32H HAS BEEN GIVEN THE ILL
     1EGAL NAME,A6,1H.///35H THIS DATA IS NOT BEING CALCULATED.///11H RU
```

```
     2N NUMBER,I4,17H HAS BEEN VOIDED.)
        KVOID=1
   143 NECALL(NEX) = K3
        NAME(NEX)=KPMR(K2,2)
        NEXEQN(NEX)=KPMR(K2,3)
   243 CONTINUE
C       A DUMMY SUBROUTINE MUST BE SUPPLIED FOR EACH OF THE ABOVE NAMES
C       NOT SUPPLIED BY THE USER OF PACER
C       IF THE NUMBER OF EQUIPMENT SUBROUTINE NAMES OR AN EQUIPMENT
C       SUBROUTINE NAME ITSELF IS CHANGED, MUST MODIFY SUBROUTINE EQCALL
C       AND CHANGE DREAD SUBROUTINE STATEMENT 140
C       STORE FROM KPMR INTO KPM
    39 DO  43  K2 = 1, NOKPM
    40 N2 = KPMR(K2,1)
       KPM(N2,1)=N2
    41 DO 43 N3 = 2, N3MAX
    42 KPM(N2,N3) = KPMR(K2,N3+2)
    43 CONTINUE
C       READ EN INTO STRM, THEN INTO EN
    44 READ INPUT TAPE 5, 205, NOEN
   144 IF(NOEN)          57,57,45
    45 DO 51 K2=1,NOEN
    51 READ INPUT TAPE 5,   206,  (STRM(K2,K3),K3=1,NELMAX)
C       STORE STRM VALUES OF EN INTO EN
    52 DO 56   K2 = 1, NOEN
    53 NEN = STRM(K2,1)
    54 DO 56   K3 = 1, NELMAX
    55 EN(NEN,K3) = STRM(K2,K3)
    56 CONTINUE
C       READ ENC INTO STRM AND THEN INTO ENC
    57 READ INPUT TAPE 5,   205,  NOENC
   157 IF(NOENC)              70,70,58
    58 DO 64 K2=1,NOENC
    64 READ INPUT TAPE 5,   206,   (STRM(K2,K3),K3=1,NECLMX)
C       STORE STRM VALUES INTO ENC
    65 DO 69   K2 = 1, NOENC
    66 NENC = STRM(K2,1)
    67 DO 69   K3 = 1, NECLMX
    68 ENC(NENC,K3) = STRM(K2,K3)
    69 CONTINUE
       READ SN INTO STRM THEN INTO SN
    70 READ INPUT TAPE 5,  205, NOSN
    71 IF(NOSN)          85,85,72
    72 DO 79 K2=1,NOSN
    79 READ INPUT TAPE 5,   206,   (STRM(K2,K3),K3=1,NSLMAX)
       STORE STRM VALUES OF SN INTO SN
    80 DO 84   K2 = 1, NOSN
    81 NSN = STRM(K2,1)
    82 DO 84   K3 = 1, NSLMAX
    83 SN(NSN,K3) = STRM(K2,K3)
    84 CONTINUE
C       READ SNC INTO STRM AND THEN INTO SNC
    85 READ INPUT TAPE 5,   205, NOSNC
    86 IF(NOSNC)              300,300,87
    87 DO 94 K2=1,NOSNC
    94 READ INPUT TAPE 5,   206,   (STRM(K2,K3),K3=1,NSCLMX)
C       STORE STRM VALUES OF SNC INTO SNC
    95 DO 99   K2 = 1, NOSNC
    96 NSNC = STRM(K2,1)
    97 DO 99   K3 = 1, NSCLMX
    98 SNC(NSNC,K3) = STRM(K2,K3)
```

```
   99 CONTINUE
C       READ STREAM VARIABLES TEST VECTOR
  300 READ INPUT TAPE 5,    205,  NDELS
  301 IF(NDELS)                   302,302,100
  100 READ INPUT TAPE 5,    206,    (DELS(K2),K2=1,NSLMAX)
C       READ STREAM CONTROL TEST VECTOR
  302 READ INPUT TAPE 5,    205,  NDELSC
  303 IF(NDELSC)                  102,102,101
  101 READ INPUT TAPE 5,    206,    (DELSC(K2),K2=1,NSCLMX)
C       READ PREFERRED STREAM LIST
  102 READ INPUT TAPE 5,    205, NOKPS
  103 IF(NOKPS)                   105,105,104
  104 READ INPUT TAPE 5, 207, (KPS(K2),K2=1,NOKPS)
C       SET STRM MATRIX TO ZERO
  105 DO 108  K2 = 1,60
  106 DO 108  K3 = 1,20
  107 STRM(K2,K3) = 0.
  108 CONTINUE
      DO 514 K2=1,60
      DO 514 K3=1,3
  514 KSEM(K2,K3)=0
      IF(KVOID)510,510,1
  510 DO 507 M1=1,N2MAX
      IF(KPM(M1,1))507,507,502
  502 M2=KPM(M1,1)
      DO 506 M3=2,N3MAX
      IF(KPM(M1,M3)) 504,507,503
  503 M4=(KPM(M1,M3))
      KSEM(M4,3)=M2
      GO TO 505
  504 M4=-(KPM(M1,M3))
      KSEM(M4,2)=M2
  505 KSEM(M4,1)=M4
  506 CONTINUE
  507 CONTINUE
  109 RETURN
  200 FORMAT   (12A6)
  201 FORMAT   (5I5)
  202 FORMAT   (9I5)
  203 FORMAT   (1I5)
  204 FORMAT   (I5, A6, 4X, I5, 5X, 9I5)
C       IF CHANGE FORMAT 204 FOR KPM, MUST ALSO CHANGE STATEMENT NUMBER 38
  205 FORMAT    (1I5)
  206 FORMAT   (5F14.4)
C       IF CHANGE FORMAT 206, MUST CHANGE NUMBER OF WORDS PER CARD IN
C       STATEMENTS 45,46,48,50, 58,59,61,63,72,73,75,77, 87,88,90,92
  207 FORMAT    (10I5)
      END
```

```
      SUBROUTINE   DCHECK
C     DUMMY SUBROUTINE.  WHEN WRITTEN, SUBROUTINE WILL CHECK AND ARRANGE
C     INPUT DATA.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KPET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2;L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30),ENAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6,3,  KRUN
    2 RETURN
    3 FORMAT(1H1 28X18HSUBROUTINE  DCHECK///63H WHEN WRITTEN, WILL CHECK
     1 INPUT DATA FOR ERRORS AND MAY ALSO DO//23H SOME NUMBER SHUFFLING.
     2///19H THIS IS RUN NUMBER I12)
      END
```

```
      SUBROUTINE  DPRINT
C     THIS SUBROUTINE PRINTS THE INPUT DATA,  CONTROL CARDS, DIMENSIONS,
C     KPM, EN, ENC, SN, SNC, DELS, DELSC, AND KPS.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON  NDELS, NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6, 100,  KRUN
C     PRINT TITLE CARD
    2 WRITE OUTPUT TAPE 6, 101,  (TITLE(K2),K2=1,12)
C     PRINT PROCESS MATRIX HEADING, THEN PROCESS MATRIX
    3 WRITE OUTPUT TAPE 6, 102
    4 DO 5     K2 = 1, N2MAX
   44 IF(KPM(K2,1))              5,5,47
   47 WRITE OUTPUT TAPE 6, 103, KPM(K2,1), NAME(K2),NEXEQN(K2),
     1(KPM(K2,K3),K3=2,N3MAX)
    5 CONTINUE
      WRITE OUTPUT TAPE 6,511
      DO 508 M1=1,NSMAX
  508 WRITE OUTPUT TAPE 6,509,(KSEM(M1,M2),M2=1,3)
      WRITE OUTPUT TAPE 6,512,KSETS,KCLEAN,LOOPS,NOGO
C     PRINT LIST OF PREFERRED STREAMS USED IN THE CALCULATIONS IF GIVEN
```

```
    6 IF(NOKPS)          8,8,7
    7 WRITE OUTPUT TAPE 6, 104,  (KPS(K2),K2=1,NOKPS)
    8 WRITE OUTPUT TAPE 6, 105, N2MAX, N3MAX, NEMAX, NSMAX, NSLMAX,
     1NSCLMX, NELMAX, NECLMX
      WRITE OUTPUT TAPE 6,25
   25 FORMAT(////35H STREAM VARIABLES TEST VECTOR, DELS//)
      WRITE OUTPUT TAPE 6,107,(DELS(K2),K2=1,NSLMAX)
      WRITE OUTPUT TAPE 6,26
   26 FORMAT(////34H STREAM CONTROL TEST VECTOR, DELSC//)
      WRITE OUTPUT TAPE 6,107,(DELSC(K2),K2=1,NSCLMX)
C     PRINT OUT STREAM VARIABLES MATRIX
    9 WRITE OUTPUT TAPE 6, 106, KRUN
   10 DO 11   K2 = 1, NSMAX
   11 WRITE OUTPUT TAPE 6, 107,    (SN(K2,K3),K3=1,NSLMAX)
C     PRINT OUT STREAM CONTROL PARAMETERS MATRIX
   12 WRITE OUTPUT TAPE 6, 108, KRUN
   13 DO 14   K2 = 1, NSMAX
   14 WRITE OUTPUT TAPE 6, 107,   (SNC(K2,K3),K3=1,NSCLMX)
C     PRINT OUT EQUIPMENT PARAMETERS MATRIX
   15 WRITE OUTPUT TAPE 6, 109, KRUN
   16 DO 17   K2 = 1, NEMAX
   17 WRITE OUTPUT TAPE 6, 107,    (EN(K2,K3),K3=1,NELMAX)
C     PRINT OUT EQUIPMENT CONTROL PARAMETERS MATRIX
   18 WRITE OUTPUT TAPE 6, 110, KRUN
   19 DO 20   K2 = 1, NEMAX
   20 WRITE OUTPUT TAPE 6, 107,   (ENC(K2,K3),K3=1,NECLMX)
   21 RETURN
  100 FORMAT(1H1,22X5HPACER//8X39HPROCESS ASSEMBLY CASE EVALUATOR ROUTIN
     1E ///14X 18HDATA OF RUN NUMBER I5//)
  101 FORMAT(12A6)
  102 FORMAT(//13X14HPROCESS MATRIX//11H  EQUIPMENT,14X14HSTREAM NUMBERS
     1/14H NUMBER   NAME)
  103 FORMAT(/1XI4,4XA6,1X10I5)
  104 FORMAT(///60H PREFERRED STREAMS USED IN RECYCLE CALCULATIONS  ARE
     1NUMBERS // 1X 10I5)
  105 FORMAT(//9X25HDIMENSIONS OF THE PROBLEM//33H NUMBER OF ROWS IN PRO
     1CESS MATRIX,I12/36H NUMBER OF COLUMNS IN PROCESS MATRIX,I9//25H MA
     2XIMUM EQUIPMENT NUMBER,10XI10/22H MAXIMUM STREAM NUMBER,13XI10//
     335H LENGTH OF STREAM PARAMETERS VECTOR  I10/ 32H LENGTH OF STREAM
     4CONTROL VECTOR  I13// 38H LENGTH OF EQUIPMENT PARAMETERS VECTOR
     5 I7/ 35H LENGTH OF EQUIPMENT CONTROL VECTOR I10)
  106 FORMAT(1H1,18X30HSTREAM VARIABLES MATRIX AT RUN,I5//)
  107 FORMAT(/(1X5F15.4))
  108 FORMAT(1H1,15X39HSTREAM CONTROL PARAMETERS MATRIX AT RUN,I5//)
  109 FORMAT(1H1,18X34HEQUIPMENT PARAMETERS MATRIX AT RUN,I5//)
  110 FORMAT(1H1,14X42HEQUIPMENT CONTROL PARAMETERS MATRIX AT RUN,I5//)
  509 FORMAT(/1X,3I6)
  511 FORMAT(1H1,7X,4HKSEM //20H STREAM    FROM     TO)
  512 FORMAT(1H1,18H CONTROL VARIABLES//8H KSETS= ,I5//8H KCLEAN=,I5//
     18H LOOPS= ,I5//8H NOGO=  ,I5)
      END
```

```
      SUBROUTINE  RPRINT
C     THIS SUBROUTINE PRINTS RESULTS AFTER EACH COMPLETE SET OF CALC
C     CALCULATIONS IN MAJOR
C     RPRINT IS CALLED BY PACER
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6, 100,  KRUN
C     PRINT TITLE CARD
    2 WRITE OUTPUT TAPE 6, 101,  (TITLE(K2),K2=1,12)
C     PRINT PROCESS MATRIX HEADING, THEN PROCESS MATRIX
    3 WRITE OUTPUT TAPE 6, 102
    4 DO 5     K2 = 1, N2MAX
   44 IF(KPM(K2,1))                5,5,47
   47 WRITE OUTPUT TAPE 6, 103, KPM(K2,1), NAME(K2),NEXEQN(K2),
     1(KPM(K2,K3),K3=2,N3MAX)
    5 CONTINUE
C     PRINT LIST OF PREFERRED STREAMS USED IN THE CALCULATIONS IF GIVEN
    6 IF(NOKPS)              8,8,7
    7 WRITE OUTPUT TAPE 6, 104,  (KPS(K2),K2=1,NOKPS)
C     PRINT OUT FOR EACH EQUIPMENT IN KPM, EN,ENC, SN,SNC
```

```
    8 DO 21    K2 = 1, N2MAX
    9 NE = KPM(K2,1)
   10 IF(NE)                21,21,11
   11 WRITE OUTPUT TAPE 6, 105, NE, KRUN, (EN(NE,K4),K4=1,NELMAX)
   12 WRITE OUTPUT TAPE 6, 106,              (ENC(NE,K4),K4=1,NECLMX)
   13 DO 20    K3 = 2, N3MAX
   14 NS = KPM(K2,K3)
   15 IF(NS)                     19,20,16
C       PRINT INPUT STREAM RESULTS
   16 WRITE OUTPUT TAPE 6, 107, NS,NE,(SN(NS,K4),K4=1,NSLMAX)
   17 WRITE OUTPUT TAPE 6, 108,        (SNC(NS,K4),K4=1,NSCLMX)
   18 GO TO 20
C       PRINT OUTPUT STREAM RESULTS
   19 NS = - NS
  119 WRITE OUTPUT TAPE 6, 109, NS,NE,(SN(NS,K4),K4=1,NSLMAX)
  219 WRITE OUTPUT TAPE 6, 110,        (SNC(NS,K4),K4=1,NSCLMX)
   20 CONTINUE
   21 CONTINUE
   22 RETURN
  100 FORMAT(1H1,23X5HPACER//8X39HPROCESS ASSEMBLY CASE EVALUATOR ROUTIN
     1E ///11X 21HRESULTS OF RUN NUMBER I5//)
  101 FORMAT(12A6)
  102 FORMAT(//13X14HPROCESS MATRIX//11H  EQUIPMENT,14X14HSTREAM NUMBERS
     1/14H NUMBER    NAME)
  103 FORMAT(/1XI4,4XA6,1X10I5)
  104 FORMAT(///60H PREFERRED STREAMS USED IN RECYCLE CALCULATIONS WERE
     1NUMBERS//1X10I5)
  105 FORMAT(1H1 8X 34HFINAL RESULTS FOR EQUIPMENT NUMBER I4,14H OF RUN
     1NUMBER I5///25H EQUIPMENT PARAMETERS ARE//(1X 5F15.4))
  106 FORMAT(//33H EQUIPMENT CONTROL PARAMETERS ARE//(1X 5F15.4))
  107 FORMAT(//20H INPUT STREAM NUMBER I4,20H OF EQUIPMENT NUMBER I4//
     127H FINAL STREAM VARIABLES ARE//(1X 5F15.4))
  108 FORMAT(//36H FINAL STREAM CONTROL PARAMETERS ARE//(1X 5F15.4))
  109 FORMAT(//21H OUTPUT STREAM NUMBER I4,20H OF EQUIPMENT NUMBER I4//
     1 27H FINAL STREAM VARIABLES ARE // (1X 5F15.4))
  110 FORMAT(//36H FINAL STREAM CONTROL PARAMETERS ARE//(1X 5F15.4))
      END
```

```
      SUBROUTINE SPRINT
      COMMON   KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON   KEFLAG, NE, NEMAX
      COMMON   KSFLAG, NS, NSMAX
      COMMON   EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON   SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON   KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON   NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON   KEUS, NEUS,   NENUM
      COMMON   KECV1, KECV2, KECV3, KECV4
      COMMON   NECV1, NECV2, NECV3, NECV4
      COMMON   KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON   KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON   KES, KES2, KES3
      COMMON   KRET, KRET2, KRET3
      COMMON   STRM, STRMC
      COMMON   STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON   DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON   KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON   TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON   NLX, NLY, NLXS, NLXSC
      COMMON   ICONV, NEX, NECALL, NEQUIP
      COMMON   ND2, ND3, ND4, ND5, ND6, ND7
      COMMON   EXX, D15, D16
      COMMON   I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON   J2,J3,J4,J5
      COMMON   J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1         J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2         J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON   J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON   K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1         K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON   L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON   NDELS, NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    8 WRITE OUTPUT TAPE 6, 105, N2MAX, N3MAX, NEMAX, NSMAX, NSLMAX,
     1NSCLMX, NELMAX, NECLMX
      IF(20-NSLMAX)1,9,9
    1 WRITE OUTPUT TAPE 6,3
      NSLMAX=20
C     PRINT OUT STREAM VARIABLES MATRIX
    9 WRITE OUTPUT TAPE 6, 106, KRUN
   10 DO 11    K2 = 1, NSMAX
   11 WRITE OUTPUT TAPE 6, 107,    (SN(K2,K3),K3=1,NSLMAX)
      RETURN
  105 FORMAT(//9X25HDIMENSIONS OF THE PROBLEM//33H NUMBER OF ROWS IN PRO
     1CESS MATRIX,I12/36H NUMBER OF COLUMNS IN PROCESS MATRIX,I9//25H MA
     2XIMUM EQUIPMENT NUMBER,10XI10/22H MAXIMUM STREAM NUMBER,13XI10//
     335H LENGTH OF STREAM PARAMETERS VECTOR  I10/ 32H LENGTH OF STREAM
     4CONTROL VECTOR  I13// 38H LENGTH OF EQUIPMENT PARAMETERS VECTOR
     5 I7/ 35H LENGTH OF EQUIPMENT CONTROL VECTOR I10)
  106 FORMAT(1H1////18X30HSTREAM VARIABLES MATRIX AT RUN,I5//)
```

```
107 FORMAT(/(1X5F15.4))
  3 FORMAT(1H6,76H NOTE THAT NSLMAX, THE LENGTH OF THE STREAM PARAMETE
    1RS VECTOR, IS TOO LARGE.//27H IT IS BEING CHANGED TO 20.)
      END
```

```
      SUBROUTINE  CHANGE
C     USED FOR INTERNAL VARIATION OF PROBLEM PARAMETERS
C     DUMMY SUBROUTINE AT THIS TIME
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6,3,  KRUN
    4 KBACK = 0
    2 RETURN
    3 FORMAT(1H1 25X18HSUBROUTINE  CHANGE///60H NO INTERNAL CHANGES IN TH
     1HE PROBLEM PARAMETERS ARE REQUIRED.//23H THIS IS PROBLEM NUMBER
     2I10)
      END
```

```
      SUBROUTINE MAJOR
C     SUBROUTINE MAJOR - PERFORMS MATERIAL AND ENERGY BALANCES FOR GIVEN
C         PROCESS. MAXIMUM OF THIRD ORDER TRIAL AND ERROR STREAM RECYCLE
C         CALCULATIONS. UPON COMPLETING MAJOR ALL PROCESS STREAMS HAVE
C         BEEN CALCULATED.
C     CALLING SEQUENCE OF MAJOR
C                                SCAN
C                                  SSUB(N)
C                                EQUIP
C                                  EQCALL
C                                GUESS1
C                                  GUESS2
C                                    GUESS3
C                                    SETUP
C                                      EQCALL
C                                      TEST
C                                GOOF
C     KPM(N2,N3) IS THE PROCESS MATRIX SEARCH VARIABLE. N2=ROW NO. 1 TO
C         N2MAX.  N3= COLUMN NO. 1 TO N3MAX.
C     KEFLAG = EQUIPMENT FLAG
C     KSFLAG = STREAM FLAG
C     SN(NSN,NSL) IS STREAM MATRIX.  ONE COLUMN FOR EACH STREAM.
C        NSN CORRESPONDS TO STREAM NUMBER  1 TO NSMAX
C        NSL CORRESPONDS TO STREAM VARIABLES  1 TO  NSLMAX
C     SNC(NSNC,NSNCL) IS STREAM CONTROL MATRIX. ONE COL. FOR EACH STREAM
C        NSNC CORRES. STREAM NUMBER  1  TO NSMAX
C        NSNCL  CORRES. TO STREAM CONTROL VARIABLES  1 TO NSCMAX
C     EN(NEN,NEL) IS EQUIPMENT VARIABLES MATRIX.  ONE COL. EACH EQUIP.
C        NEN CORRES. TO EQUIP. NUMBER  1 TO NEMAX
C        NEL CORRES. TO EQUIP. VARIABLE  1 TO  NELMAX
C     ENC(NENC,NENCL) IS EQUIPMENT CONTROL MATRIX
C         NENC  CORRES. TO EQUIPMENT NUMBER  1 TO NEMAX
C         NENCL  CORRES. TO EQUIPMENT CONTROL VARIABLE  1 TO  NECMAX
C     FEED STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 1.0
C     PRODUCT STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 2.0
C     SPECIAL STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 3.0
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
```

```
      COMMON   J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1         J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2         J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON   J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON   K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1         K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON   L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
C     SET EQUIP. AND STREAM FLAGS TO ZERO
    2 DO 3 NE=1,NEMAX
    3 KEFLAG(NE)= 0
    4 DO 5 NS=1,NSMAX
    5 KSFLAG(NS)= 0
C     SET FEED STREAM FLAGS ON (+1). CALCULATE SPECIAL STREAMS AND FLAG
    6 CALL SCAN
    7 KES=0
C     KES IS +1 IF ANY EQUIPMENT IS CALCULATED IN KPM SCAN IN MAJOR
C     START PROCESS MATRIX SCAN ROW BY ROW
    8 DO 32 I2=1, N2MAX
    9 NE=KPM(I2,1)
  119 IF(NE)                32,32,10
   10 IF(KEFLAG(NE)) 32,11,32
C     START SCAN OF ROW OF UNCALC. EQUIP. LOOK AT SIGN OF S. IS IT KNOWN
   11 I3=2
C     CHECK SIGN OF STREAM IN PROCESS MATRIX
   12 IF(KPM(I2,I3)) 25,13,21
C     IF FIRST STREAM IS ZERO, FLAG KNOWN AS +9. SET ASSOCIATED S FLAG=9
   13 WRITE OUTPUT TAPE 6,41, NE
   14 KEFLAG(NE)= 9
C     FLAG INPUT AND OUTPUT STREAMS AS 9
   15 DO 19 I4=2,N3MAX
   16 I5 = XABSF(KPM(I2,I4))
   17 IF(I5) 19,19,18
   18 KSFLAG(I5)= 9
   19 CONTINUE
   20 GO TO 32
C     IF STREAM IS INPUT SEE IF IT IS KNOWN. IF NOT GO TO NEXT EQUIP.
   21 I6=KPM(I2,I3)
   22 IF(KSFLAG(I6)) 32,32,23
   23 I3=I3+1
   24 GO TO 12
C     OUTPUT STREAM. CALL OUT SUBROUTINE OF EQUIPMENT NE. CALC. OUTPUT S
   25 IF(KSETS)                225,225,125
  125 WRITE OUTPUT TAPE 6,42, NE,(KPM(I2,I10),I10=2,N3MAX)
  225 CALL EQUIP
   26 KES= 1
   27 KEFLAG(NE)= 1
C     SET OUTPUT STREAM FLAGS FROM 0 TO 1
   28 DO 132 I7=2, N3MAX
   29 I8=KPM(I2,I7)
   30 IF(I8)                   31,132,132
```

```
   31 I8 = - I8
  130 IF(KSFLAG(I8))          132,131,132
  131 KSFLAG(I8) = 1
  132 CONTINUE
   32 CONTINUE
C       THIS COMPLETES A SCAN OF PROCESS MATRIX ROW BY ROW
C       IF KES IS 1 RESCAN PROCESS MATRIX
C       IF KES IS 0, SEE IF ALL EQUIP. HAS BEEN CALC. IF NOT GO TO GUESS1
   33 IF(KES)          34,34,7
C       ARE ALL EQUIP. KNOWN  IF NOT GO TO GUESS1
   34 DO38 I9=1,NEMAX
   35 IF(KEFLAG(I9)) 38,36,38
   36 IF(KSETS)                    136,136,336
  336 WRITE OUTPUT TAPE 6,43, (KEFLAG(I11),I11=1,NEMAX)
  436 WRITE OUTPUT TAPE 6,44, (KSFLAG(I12),I12=1,NSMAX)
C       START TRIAL AND ERROR CALCULATION OF RECYCLE STREAMS.
  136 CALL GUESS1
C       IF KRET2 = 2, CALCULATION WAS NOT POSSIBLE.  PRESENT STATUS OF
C       MATRICIES HAS BEEN PRINTED OUT BY SUBROUTINE GOOF.
  236 IF(KRET2 - 2) 7,39,7
   38 CONTINUE
   39 RETURN
C       MAJOR LOOP COMPLETE.
   41 FORMAT(17H1EQUIPMENT NUMBER I5,39H IS NOT BEING USED IN THIS CALCU
     1LATION.//52H FIRST INPUT DESIGNATION IN PROCESS MATRIX WAS ZERO.//
     242H EQUIPMENT AND OUTPUT STREAM FLAGGED AS +9)
   42 FORMAT(17H1EQUIPMENT NUMBER I5,30H IS BEING CALCULATED DIRECTLY.//
     123H ASSOCIATED STREAMS ARE 3X 5(I4,1H,)/26X 5(I4,1H,))
   43 FORMAT(97H1ENTER TRIAL AND ERROR CALCULATIONS. CURRENT STATUS OF E
     1QUIPMENT AND STREAM FLAGS ARE AS FOLLOWS.//12H KEFLAGS ARE//11H E0
     21 TO E10 10(1H, I3)//11H E11 TO E20 10(1H, I3)//11H E21 TO E30 10(
     31H, I3))
   44 FORMAT(12H0KSFLAGS ARE//11H S01 TO S10 10(1H,I3)//11H S11 TO S20
     110(1H, I3)//11H S21 TO S30  10(1H, I3)//11H S31 TO S40 10(1H, I3)/
     2/11H S41 TO S50 10(1H, I3)//11H S51 TO S60 10(1H, I3))
      END
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```
      SUBROUTINE  SCAN
C     THIS SUBROUTINE FLAGS FEED STREAMS TO 7, FLAGS PRODUCT STREAMS
C     TO 10, AND CALCULATES AND FLAGS SPECIAL PRODUCT STREAMS USING
C     SUBROUTINE SSUB(N) TO 8 BEFORE BEGINNING THE MATERIAL AND ENERGY
C     BALANCE CALCULATIONS.
C     FEED STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 1.0
C     PRODUCT STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 2.0
C     SPECIAL STREAMS HAVE SECOND COLUMN WORD OF SN VECTOR EQUAL TO 3.0
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,  NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 DO 16  I12 = 1, NSMAX
    2 IF(SN(I12,1))                3,6,3
    3 IF(SN(I12,2) - 1.)           4,8,4
    4 IF(SN(I12,2) - 2.)           5,10,5
    5 IF(SN(I12,2) - 3.)          16,12,16
C     FLAG UNUSED STREAM NUMBERS AS 99
    6 KSFLAG(I12) = 99
    7 GO TO 16
C     FLAG FEED STREAMS EQUAL TO 7
    8 KSFLAG(I12) = 7
```

```
    9 GO TO 16
C      FLAG PRODUCT STREAMS EQUAL TO 10
   10 KSFLAG(I12) = 10
   11 GO TO 16
C      CALCULATE SPECIAL STREAMS.  FLAG EQUAL TO 8
   12 KSFLAG(I12) = 8
   13 CALL SSUB(I12)
   14 WRITE OUTPUT TAPE 6,30,KRUN,I12,  (SN(I12,I14),I14=1,NSLMAX)
   15 WRITE OUTPUT TAPE 6,31,           (SNC(I12,I15),I15=1,NSCLMX)
   16 CONTINUE
   17 DO  20 I13 = 1, NEMAX
C      FLAG UNUSED EQUIPMENT NUMBERS AS 99
   18 IF(EN(I13,1))              19,19,20
   19 KEFLAG(I13) = 99
   20 CONTINUE
   21 RETURN
   30 FORMAT(1H1 28X 10HRUN NUMBER I5/14H STREAM NUMBER I4,70H CALCULATE
     1D BY STREAM SUBROUTINE.  NEW STREAM VALUES ARE LISTED BELOW.//28X
     213HSTREAM VECTOR//(1X 5F15.4))
   31 FORMAT(/24X 21HSTREAM CONTROL VECTOR//(1X 5F15.4))
      END
```

```
      SUBROUTINE  SSUB (N)
C     THIS TO BE USED ONLY TO DEBUG
C     SSUB MEANS STREAM SUBROUTINE,  USED TO CALC. STREAM PRIOR TO START
C     OF MAJOR MATERIAL AND ENERGY BALANCES.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,  NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NEXEQN, NAME,KSEM
      COMMON NDELS,NDELSC
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6,3, (N)
    2 RETURN
    3 FORMAT(///23H VALUE OF STREAM NUMBER I5,60H IS BEING CHANGED BEFOR
     1E DOING MATERIAL AND ENERGY BALANCES.)
      END
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```
      SUBROUTINE  EQUIP
C     THIS SUBROUTINE DOES THE BOOK KEEPING BEFORE CALLING THE EQUIPMENT
C     SUBROUTINE CORRESPONDING TO EQUIPMENT NUMBER NEX.  AFTER THE
C     OUTPUT STREAMS OF NEX ARE CALCULATED, IT STORES THE ANSWERS IN
C     THE SN AND SNC MATRICIES.
C     EQUIP IS CALLED BY MAJOR
C     EQUIP CALLS SUBROUTINE EQCALL WHICH CALLS THE PROPER
C     EQUIPMENT SUBROUTINE
C     I2 IS ROW NUMBER OF KPM OF EQUIPMENT NUMBER NE
C     SET INPUT AND OUTPUT STREAMS OF EQUIP. NUMBER NE OF KPM ROW I2
C     FROM SN AND SNC TO STRMI, STRMCI, STRMO, AND STRMCO.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON   ICONV, NEX, NECALL, NEQUIP
      COMMON   ND2, ND3, ND4, ND5, ND6, ND7
      COMMON   EXX, D15, D16
      COMMON   I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON   J2,J3,J4,J5
      COMMON   J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1         J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2         J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON   J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON   K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1         K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON   L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 L4 = 1
    2 L6 = 1
    3 DO 21 L2 = 2, N3MAX
    4 L3 = KPM(I2,L2)
C     L3 IS THE STREAM NUMBER OF KPM ROW I2
    5 IF(L3)                          14,22,8
C     SET INPUT STREAMS IN STRMI AND STRMCI
```

```
    8 DO 9  L5 = 1,  NSLMAX
    9 STRMI(L4,L5)    = SN(L3,L5)
   10 DO 11 L5 = 1, NSCLMX
   11 STRMCI(L4,L5)   = SNC(L3,L5)
   12 L4 = L4 + 1
   13 GO TO 21
C     SET OUTPUT STREAMS IN STRMO AND STRMCO
   14 NIN = L4 - 1
C     NIN = NUMBER OF INPUT STREAMS
   15 L3 = - L3
   17 DO 18   L7 = 1, NSLMAX
   18 STRMO(L6,L7)    = SN(L3,L7)
   19 DO 20   L7 = 1, NSCLMX
   20 STRMCO(L6,L7)   = SNC(L3,L7)
  120 L6 = L6 + 1
   21 CONTINUE
   22 NOUT = L6 - 1
   23 CALL EQCALL
C     SET CALC. OUTPUT STREAM VALUES IN SN AND SNC
   24 DO 30   L8 = 1, NOUT
   25 L9 = STRMO(L8,1)
   26 DO 27  L10 = 1, NSLMAX
   27 SN(L9,L10) = STRMO(L8,L10)
   28 DO 29  L10 = 1, NSCLMX
   29 SNC(L9,L10) = STRMCO(L8,L10)
   30 CONTINUE
  130 IF(KSETS)                  42,42,31
   31 WRITE OUTPUT TAPE 6,99,  NE
   32 DO 33  L12 = 1,  NOUT
   33 WRITE OUTPUT TAPE 6,98,  STRMO(L12,1),  (STRMO(L12,L13),L13=1,NSLM
     1AX)
   34 DO 35  L12 = 1, NOUT
   35 WRITE OUTPUT TAPE 6,97,  STRMCO(L12,1),  (STRMCO(L12,L13),L13=1,NS
     1CLMX)
   36 DO 37  L12 = 1, NIN
   37 WRITE OUTPUT TAPE 6,96,  STRMI(L12,1),  (STRMI(L12,L13),L13=1,NSLM
     1AX)
   38 DO 39  L12 = 1, NIN
   39 WRITE OUTPUT TAPE 6,95,  STRMCI(L12,1),  (STRMCI(L12,L13),L13=1,NS
     1CLMX)
   40 WRITE OUTPUT TAPE 6,94,  EN(NE,1),  (EN(NE,L13),L13=1,NELMAX)
   41 WRITE OUTPUT TAPE 6,93,  ENC(NE,1),  (ENC(NE,L13),L13=1,NECLMX)
   42 RETURN
   99 FORMAT(21H1FOR EQUIPMENT NUMBER I3,38H, THE OUTPUT STREAM VALUES G
     1IVEN BELOW/62HOWERE CALCULATED DIRECTLY USING THE INPUT STREAM AND
     2 EQUIPMENT/24HOPARAMETER VALUES SHOWN.)
   98 FORMAT(/21HOOUTPUT STREAM NUMBER F6.0,18H PARAMETERS VECTOR//(1X
     15F15.4))
   97 FORMAT(/21HOOUTPUT STREAM NUMBER F6.0,15H CONTROL VECTOR//(1X 5F15
     1.4))
   96 FORMAT(/20HOINPUT STREAM NUMBER F6.0, 18H PARAMETERS VECTOR//(1X 5
     1F15.4))
   95 FORMAT(/20HOINPUT STREAM NUMBER F6.0, 15H CONTROL VECTOR//(1X 5F15
     1.4))
   94 FORMAT(/17HOEQUIPMENT NUMBERF6.0,17H PARAMETER VECTOR//(1X5F15.4))
   93 FORMAT(/17HOEQUIPMENT NUMBER F6.0, 15H CONTROL VECTOR//(1X5F15.4))
      END
```

```
      SUBROUTINE  EQCALL
      COMMON   KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON   KEFLAG, NE, NEMAX
      COMMON   KSFLAG, NS, NSMAX
      COMMON   EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON   SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON   KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON   NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON   KEUS, NEUS,   NENUM
      COMMON   KECV1, KECV2, KECV3, KECV4
      COMMON   NECV1, NECV2, NECV3, NECV4
      COMMON   KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON   KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON   KES, KES2, KES3
      COMMON   KRET, KRET2, KRET3
      COMMON   STRM, STRMC
      COMMON   STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON   DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON   KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON   TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON   NLX, NLY, NLXS, NLXSC
      COMMON   ICONV, NEX, NECALL, NEQUIP
      COMMON   ND2, ND3, ND4, ND5, ND6, ND7
      COMMON   EXX, D15, D16
      COMMON   I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON   J2,J3,J4,J5
      COMMON   J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1         J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2         J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON   J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON   K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1         K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON   L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
  100 NEQUIP = NECALL(NE)
  101 GO TO (1,2,3,4,5,6,7,8,9,10,11,12,13,14,15),  NEQUIP
    1 CALL  UNAME1
      RETURN
    2 CALL  UNAME2
      RETURN
    3 CALL  UNAME3
      RETURN
    4 CALL  UNAME4
      RETURN
    5 CALL  UNAME5
      RETURN
    6 CALL  UNAME6
      RETURN
    7 CALL  UNAME7
      RETURN
    8 CALL  UNAME8
```

```
      RETURN
    9 CALL   UNAME9
      RETURN
   10 CALL   DEBUG
      RETURN
   11 CALL   EXTRA1
      RETURN
   12 CALL   EXTRA2
      RETURN
   13 CALL   EXTRA3
      RETURN
   14 CALL   EXTRA4
      RETURN
   15 CALL   EXTRA5
      RETURN
      END
```

```
      SUBROUTINE GUESS1
C     THIS SUBROUTINE FORMS AN ORDERED LIST OF UNKNOWN INPUT STREAMS AND
C     THEN GENERATES THE UNKNOWN STREAM CHECK VECTOR.  THE CHECK VECTOR
C     CONTAINS THE STREAM NUMBERS WHICH HAVE BEEN ASSUMED TEMPORARILY
C     KNOWN IN DOING TRIAL AND ERROR RECYCLE CALCULATIONS.
C     GUESS1 IS CALLED BY SUBROUTINE MAJOR
C     GUESS1 CALLS SUBROUTINES  GUESS2  AND  GOOF
C     KUS(NUS) IS LIST OF UNKNOWN INPUT STREAMS.  1 TO NUSMAX
C     KPS(NPS) IS LIST OF PREFERRED UNKNOWN INPUT STREAMS TO BE USED
C     FIRST IN TRYING TO ESTABLISH A SEQUENCE OF EQUIPMENT FOR TRIAL
C     AND ERROR RECYCLE CALCULATIONS.
C     KCV1, KCV2, KCV3 RECORD INPUT STREAMS ASSUMED KNOWN.
C     J2, J3, J4, J5  ARE DO STATEMENT INDICIES.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
      MAKE LIST OF UNKNOWN INPUT STREAMS.
    1 NUS = 1
    2 DO 5  J2=1, NSMAX
  102 IF(KSFLAG(J2)) 5,3,5
    3 KUS(NUS) = J2
```

```
    4 NUS=NUS + 1
    5 CONTINUE
    6 NUSMAX = NUS - 1
C       IS A PREFRED STREAM LIST GIVEN  (IS NOKPS GREATER THAN ZERO )
    7 IF(NOKPS) 114,114,8
C       SORT LIST SO PREFERRED STREAMS HEAD UNKNOWN STREAM LIST.
    8 DO 14  J3=1,  NOKPS
    9 DO 13  J4=1, NUSMAX
   10 IF(KUS(J4)-KPS(J3)) 13,11,13
   11 KUS(J4) = KUS(J3)
   12 KUS(J3) = KPS(J3)
   13 CONTINUE
   14 CONTINUE
  114 IF(KSETS) 15,15,115
  115 WRITE OUTPUT TAPE 6,99 , (KUS(J5),J5=1,NUSMAX)
   15 NUS2 = 0
   16 NUS3 = 0
   17 KCV3 = 0
   18 KCV2 = 0
   19 GO TO 28
C       FLAG INPUT STREAM ASSUMED KNOWN AS 2.  RECORD STREAM NO. IN KCV3
   20 NUS3 = 0
   21 NUS3 = NUS3 + 1
   22 KCV3 = KUS(NUS3)
C       FLAG INPUT STREAM ASSUMED KNOWN AS 2.  RECORD STREAM NO. IN KCV2
   24 NUS2 = 0
   25 NUS2 = NUS2 +1
  125 ND3 = NUS3 + NUS2
   26 KCV2 = KUS(ND3)
C       FLAG INPUT STREAM ASSUMED KNOWN AS 2.  RECORD STREAM NO. IN KCV1
   28 NUS1 = 0
   29 NUS1 = NUS1 + 1
  129 ND4 = NUS3 + NUS2 + NUS1
   30 KCV1 = KUS(ND4)
   31 KSFLAG(KCV1) = 2
       IF(KCV2) 32,32,27
   27 KSFLAG(KCV2) = 2
       IF(KCV3) 32,32,23
   23 KSFLAG(KCV3) = 2
       CALL GUESS2 TO SEE IF ASSUMED INPUTS CAN BE FOUND AS CALC. OUTPUTS
        IF POSSIBLE, DO TRIAL AND ERROR RECYCLE CALCULATIONS.
   32 KRET2 = 0
   33 KRET  = 0
  133 IF(KSETS)                    34,34,134
       PRINT KCV1, KCV2, KCV3 BEFORE CALLING GUESS2
  134 WRITE OUTPUT TAPE 6,98, KCV1, KCV2, KCV3
   34 CALL GUESS2
C       IF KRET IS 2 ON RETURN, UNABLE TO CALCULATE AS OUTPUT STREAMS THE
C       ASSUMED INPUTS KCV1, KCV2, KCV3,  TRY ANOTHER KCVS.
C       IF KRET IS 0 ON RETURN, ASSUMED KNOWN INPUT STREAMS HAVE BEEN
C       CALCULATED AS OUTPUT STREAMS.  FINAL VALUES  STORED IN SN MATRIX
   35 IF(KRET) 36,36,40
   36 RETURN
   40 IF(KCV3) 42,42,41
   41 KSFLAG(KCV3)=0
   42 IF(KCV2) 44,44,43
   43 KSFLAG(KCV2)=0
   44 IF(KCV1) 46,46,45
   45 KSFLAG(KCV1)=0
   46 IF(NUS1 - (NUSMAX-NUS3 - NUS2)) 29,47,61
   47 IF(NUS2) 48,51,48
```

```
48 IF(NUS2 - (NUSMAX - 1 - NUS3)) 25,49,61
49 IF(NUS3) 50,52,50
50 IF(NUS3 - (NUSMAX - 2)) 21,61,61
51 IF(NUSMAX-2) 61,24,24
52 IF(NUSMAX-3) 61,20,20
61 CALL GOOF
62 KRET2 = 2
63 RETURN
99 FORMAT(80H1ORDERED LIST OF UNKNOWN INPUT STREAMS.  PREFERRED STREA
   1MS, IF ANY, GIVEN FIRST./65H0BEGIN CHOOSING STREAMS FOR TRIAL AND
   2ERROR RECYCLE CALCULATIONS.///10(1X,I4,1H,))
98 FORMAT(73H1INPUT STREAMS ASSUMED KNOWN FOR TRIAL AND ERROR RECYCLE
   1 CALCULATIONS ARE // 319  //66H NOTE ONLY NON-ZERO NUMBERS ARE ASS
   2UMED INPUT STREAMS BEING TRIED.  )
    END
```

```
      SUBROUTINE GUESS2
C     THIS SUBROUTINE FIRST SEES IF IT IS POSSIBLE TO FIND A LIST OF
C     EQUIPMENT WHICH HAS AS OUTPUT STREAMS THE ASSUMED KNOWN INPUT
C     STREAMS.  IF POSSIBLE, IT CALLS FOR TRIAL AND ERROR RECYCLE CALC.
C     TO BE DONE AFTER FIRST ESTABLISHING THE SHORTEST LIST OF EQUIPMENT
C     REQUIRED.
C     GUESS2 IS CALLED BY SUBROUTINE GUESS1
C     GUESS2 CALLS SUBROUTINES  GUESS3  AND  SETUP
C     KE1 IS SHORTEST LIST OF EQUIPMENT WITH ASSUMED INPUTS AS OUTPUTS
C     KE2 IS FIRST LIST OF EQUIP. MADE TO SEE IF ASSUMED INPUTS CAN BE
C      FOUND AS OUTPUTS.
C     KE3 IS LIST OF EQUIP. OBTAINED BY LETTING ONLY SELLECTD OUTPUT
C      STREAMS OF EQUIP. PLACED IN KE1 BE TAKEN AS TEMPORARILY KNOWN.
C     KE4 IS SHORTEST LIST OF EQUIP. HAVING AS OUTPUTS THE ASSUMED
C      INPUT STREAMS OBTAINED BY LETTING ONLY SELECTED OUTPUT STREAMS
C      OF EQUIP. PLACED IN KE1 BE TAKEN AS TEMPORARILY KNOWN.
C     KEUS IS A LIST OF OUTPUT STREAMS OF A GIVEN EQUIP. OF LIST KE1
C     WHICH ARE NOT PRODUCT STREAMS
C     NEUS IS THE NUMBER OF OUTPUT STREAMS OF A GIVEN EQUIP. OF LIST KE1
C     WHICH ARE NOT PRODUCT STREAMS
C     NENUM IS THE EQUIPMENT NUMBER WHOSE OUTPUT STREAMS WILL BE CHECKED
C     TO DETERMINE THE SHORTEST EQUIPMENT LIST.
C     NENUM IS EQUIP. NUMBER IN LIST KE1(NE1) CORRES. TO ROW J27 IN KPM
C     KECV1, KECV2, KECV3, KECV4 ARE OUTPUT STREAM NUMBER CHECK VECTOR
C     NECV1, NECV2, NECV3, NECV4 ARE INDICIES USED TO FORM THE STREAM
C     NUMBER CHECK VECTOR KECVS
C     KES2 AND KES3 ARE SEARCH FLAGS
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NF, NFMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN,  NAME,KSEM
```

```
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
C     SET LISTS KE1 AND KE2 EQUAL TO ZERO
    1 DO 2  J6=1, NEMAX
      KE1(J6) = 0
      KE2(J6) = 0
    2 KE3(J6) = 0
      NE1MAX = 0
      NE2MAX = 0
      NE3MAX = 0
    5 NE2 = 1
    6 KES2 = 0
C     START SCAN OF PROCESS MATRIX.  J9 IS EQUIP. NUMBER J8 IS ROW NO.
    7 DO 25 J8 = 1, N2MAX
    8 J9=KPM(J8,1)
  118 IF(J9)                25,25,9
    9 IF(KEFLAG(J9))  25,10,25
C     IF KEFLAG IS ZERO, EQUIP. IS UNKNOWN. START SCAN OF ROW.
   10 J10 = 2
   11 IF(KPM(J8,J10)) 16,25,12
C     IF INPUT STREAM, SEE IF IT IS KNOWN.  IF UNKNOWN, GO TO NEXT ROW.
C     IF OUTPUT STREAM, ALL INPUTS ARE KNOWN. SET EQUIP. IN KE2.
C     FLAG EQUIP. AND OUTPUT STREAMS EQUAL TO 3
   12 J11 = KPM(J8,J10)
   13 IF(KSFLAG(J11)) 25,25,14
   14 J10 = J10 + 1
   15 GO TO 11
   16 KE2(NE2) = J9
   17 NE2 = NE2 + 1
   18 KEFLAG(J9) = 3
   19 KES2 = 2
      FLAG KES2 EQUAL TO 2 EVERY TIME AN EQUIP. NO. IS PLACED IN KE2
      FLAG STREAMS FROM 0 TO 3 AND FROM 2 TO 3
   20 DO 24 J13 = 2, N3MAX
   21 J14 = KPM(J8,J13)
   22 IF(J14)  23,24,24
   23 J14 = - J14
  123 IF(KSFLAG(J14))          124,125,124
  124 IF(KSFLAG(J14) - 2)      24,125,24
  125 KSFLAG(J14) = 3
   24 CONTINUE
   25 CONTINUE
C     IF ANY EQUIP. FLAGGED IN SCAN, RESCAN PROCESS MATRIX.
   26 IF(KES2) 28,28,6
   28 NE2MAX = NE2 - 1
C     EQUIP. LIST KE2 HAS BEEN FORMED OF LENGTH NE2MAX.  MAY BE ZERO
C     IF LIST KE2 IS ZERO, PRINT KE2, KEFLAG AND KSFLAG AND RETURN
   29 IF(NE2MAX)  30,30,33
   30 IF(KSETS)                31,31,130
  130 WRITE OUTPUT TAPE 6,99, NE2MAX, (KEFLAG(J12),J12=1,NEMAX)
  131 WRITE OUTPUT TAPE 6,199,  (KSFLAG(J19),J19=1,NSMAX)
   31 KRET = 2
   32 RETURN
C     IF ANY EQUIP. IN LIST KE2, SEE IF ASSUMED INPUTS ARE CALC. OUTPUTS!
```

```
   33 DO 42 J15 = 1, NSMAX
   34 IF(KSFLAG(J15) - 2)  42,35,42
C     SOME ASSUMED INPUT STREAM NOT A CALC. OUTPUT STREAM (KSFLAG IS 2)
C     REFLAG INPUT STREAMS FLAGGED 3 TO 0 AND RETURN
   35 IF(KSETS)                36,36,135
  135 WRITE OUTPUT TAPE6,98,NE2MAX,(KE2(J20),J20=1,NE2MAX)
  235 WRITE OUTPUT TAPE 6,198, (KEFLAG(J21),J21=1,NEMAX)
  335 WRITE OUTPUT TAPE 6,298, (KSFLAG(J22),J22=1,NSMAX)
C     REFLAG STREAMS FLAGGED 3 TO FLAG 0
   36 DO 39 J18 = 1, NSMAX
   37 IF(KSFLAG(J18) - 3)  39,38,39
   38 KSFLAG(J18) = 0
   39 CONTINUE
C     REFLAG EQUIPMENT FROM 3 TO 0
  244 DO247 J16 = 1, NEMAX
  245 IF(KEFLAG(J16) - 3)  247,246,247
  246 KEFLAG(J16) = 0
  247 CONTINUE
   40 KRET = 2
   41 RETURN
   42 CONTINUE
C     KCVS ARE IN LIST KE2 AT THIS POINT IN PROGRAM
   43 IF(KSETS)                44,44,143
  143 WRITE OUTPUT TAPE 6, 97, NE2MAX, (KE2(J23),J23=1,NE2MAX)
  243 WRITE OUTPUT TAPE 6,197, (KEFLAG(J24),J24=1,NEMAX)
      WRITE OUTPUT TAPE 6,1000
  343 WRITE OUTPUT TAPE 6,297, (KSFLAG(J25),J25=1,NSMAX)
C     REFLAG ALL EQUIP. AND STREAMS WITH FLAGS EQUAL TO 3 TO 0
   44 DO 47 J16 = 1, NEMAX
   45 IF(KEFLAG(J16) - 3)  47,46,47
   46 KEFLAG(J16) = 0
   47 CONTINUE
   48 DO 51 J17 = 1, NSMAX
   49 IF(KSFLAG(J17) - 3)  51,50,51
   50 KSFLAG(J17) = 0
   51 CONTINUE
C     NOW FIND SHORTEST LIST OF EQUIPMENT WITH ASSUMED INPUTS AS OUTPUTS
      WRITE OUTPUT TAPE 6,800,KCV1,KCV2,KCV3
  800 FORMAT(1H1,6H KCV1=,I3///6H KCV2=,I3///6H KCV3=,I3)
      M1=1
      IF(KCV3) 702,702,701
  701 M2=KSEM(KCV3,2)
      IF(KEFLAG(M2)) 726,725,726
  725 KEFLAG(M2)=4
      KE3(M1)=M2
      M1=M1+1
  726 KSFLAG(KCV3)=4
  702 IF(KCV2) 704,704,703
  703 M2=KSEM(KCV2,2)
      IF(KEFLAG(M2)) 728,727,728
  727 KEFLAG(M2)=4
      KE3(M1)=M2
      M1=M1+1
  728 KSFLAG(KCV2)=4
  704 IF(KCV1) 706,706,705
  705 M2=KSEM(KCV1,2)
      IF(KEFLAG(M2)) 723,724,723
  724 KEFLAG(M2)=4
      KE3(M1)=M2
      M1=M1+1
  723 KSFLAG(KCV1)=4
```

```
706 DO 715 M2=1,NE2MAX
    M3=KE3(M2)
    IF(M3) 716,716,707
707 DO 714 M4=2,N3MAX
    IF(KPM(M3,M4))715,715,708
708 M5=KPM(M3,M4)
    IF(KSFLAG(M5)) 714,711,714
711 KSFLAG(M5)=4
712 M6=KSEM(M5,2)
    IF(KEFLAG(M6)) 714,713,714
713 KEFLAG(M6)=4
    KE3(M1)=KSEM(M5,2)
    M1=M1+1
714 CONTINUE
715 CONTINUE
716 NE3MAX=M1-1
    M3=1
    DO 731 M1=1,NE2MAX
    DO 730 M2=1,NE2MAX
    IF(KE3(M2)-KE2(M1)) 730,729,730
729 KE1(M3)=KE2(M1)
    M3=M3+1
730 CONTINUE
731 CONTINUE
    NE1MAX=M3-1
    WRITE OUTPUT TAPE 6,732,(KE2(M1),M1=1,NE2MAX)
    WRITE OUTPUT TAPE 6,735,NE2MAX
    WRITE OUTPUT TAPE 6,733,(KE3(M1),M1=1,NE3MAX)
    WRITE OUTPUT TAPE 6,736,NE3MAX
    WRITE OUTPUT TAPE 6,734,(KE1(M1),M1=1,NE1MAX)
    WRITE OUTPUT TAPE 6,737,NE1MAX
732 FORMAT(///////9H KE2 LIST,30(I2,1H,))
733 FORMAT(9H KE3 LIST,30(I2,1H,))
734 FORMAT(9H KE1 LIST,30(I2,1H,))
735 FORMAT(8H NE2MAX=I2)
736 FORMAT(8H NE3MAX=I2)
737 FORMAT(8H NE1MAX=I2)
738 DO 742 M1=1,NE1MAX
    M2=KE1(M1)
    DO 741 M3=2,N3MAX
    IF(KPM(M2,M3)) 739,742,741
739 M4=-KPM(M2,M3)
    IF(KSFLAG(M4))741,740,741
740 KSFLAG(M4)=4
741 CONTINUE
742 CONTINUE
 73 CALL SETUP
 74 IF(KRET3)   76,78,75
C       IF KRET3 = - 2, HAVE TAKEN NON-CONVERGED ANSWERS AS CORRECT.
C       KRET = 0 BY SETUP STATEMENT 123 AND S AND E FLAGGED AS 13
C       IF KRET3 = 0, CONVERGED
C       IF KRET3 IS 2, ITERATIVE CALC. DID NOT CONVERGE.  PRINT REMARK.
C       SET KRET TO 2 AND RETURN TO TRY NEW KCVS.  SAME EXIT AS IF KCVS
C       NOT IN LIST KE2.
 75 KRET = 2
 76 IF(KSETS)                    77,77,176
176 WRITE OUTPUT TAPE 6,94 , KRET, KRET2 , KRET3
 77 RETURN
 78 KRET = 0
 79 IF(KSETS)                   80,80,179
179 WRITE OUTPUT TAPE 6,93   , KRET, KRET2, KRET3
```

```
C     FLAG EQUIP. AND STREAMS FROM 4 TO 1 AS TRIAL AND ERROR CALCULATIONI
C     CONVERGED TO LIMIT
   80 DO 83 J35 = 1, NEMAX
   81 IF(KEFLAG(J35) - 4)  83,82,83
   82 KEFLAG(J35) = 1
   83 CONTINUE
   84 DO 87 J36 = 1, NSMAX
   85 IF(KSFLAG(J36) - 4)  87,86,87
   86 KSFLAG(J36) = 1
   87 CONTINUE
   88 RETURN
   99 FORMAT(57H1NO EQUIPMENT CAN BE CALCULATED WITH INPUTS ASSUMED KNOWI
     1N ///21H EQUIPMENT LIST IS OF ,I5, 8H LENGTH ///13H KEFLAGS ARE  /I
     2/11H E01 TO E10,10(1H,I3)//11H E11 TO E20,10(1H,I3) //11H E21 TO EI
     330,10(1H,I3))
  199 FORMAT(// 13H KSFLAGS ARE //11H S01 TO S10,10(1H,I3)//11H S11 TO SI
     120,10(1H,I3) //11H S21 TO S30,10(1H,I3) //11H S31 TO S40,10(1H,I3)I
     2//11H S41 TO S50,10(1H,I3)//11H S51 TO S60,10(1H,I3) // )
   98 FORMAT(60H1ALL ASSUMED INPUTS WERE NOT CONTAINED AS OUTPUTS IN LISI
     1T OF I5/24H EQUIPMENTS GIVEN BELOW.///  19X 16HEQUIPMENT NUMBER/
     2(14X I10))
  198 FORMAT(/12HOKEFLAGS ARE/11HOE01 TO E10 10(1H, I3)/11HOE11 TO E20
     110(1H, I3)/11HOE21 TO E30 10(1H, I3))
   97 FORMAT(59H1ASSUMED INPUT STREAMS ARE CALCULATED OUTPUT STREAMS OF
     1THE / 18H FOLLOWING LIST OF I4, 12H EQUIPMENTS.///19X 16HEQUIPMENTI
     2 NUMBER /(14X I10))
 1000 FORMAT(/46HOSHORTEST LIST OF EQUIPMENT WILL NOW BE FOUND./14HOAT TI
     1HIS POINT)
   94 FORMAT(45H1TRIAL AND ERROR ITERATIONS DID NOT CONVERGE.//59H WILL I
     1TRY ANOTHER SET OF ASSUMED INPUT STREAMS IF POSSIBLE.//34H PRESENTI
     2 STATUS OF RETURN FLAGS IS//10X 7HKRET  =  I4/10X  7HKRET2 =  I4/ I
     310X  7HKRET3 =  I4)
   93 FORMAT(51H1TRIAL AND ERROR ITERATIVE CALCULATIONS SUCCESSFUL.//47HI
     1 WILL SEE IF ALL EQUIPMENT HAS BEEN CALCULATED.//34H PRESENT STATUI
     2S OF RETURN FLAGS IS//10X 7HKRET  = I4/10X 7HKRET2 = I4/10X 7HKRETI
     33 = I4)
 1197  FORMAT(/12HOKEFLAGS ARE/11HOE01 TO E10 10(1H, I3)/11HOE11 TO E20  I
     110(1H, I3)/11HOE21 TO E30 10(1H, I3))
 S298  FORMAT(// 13H KSFLAGS ARE //11H S01 TO S10,10(1H,I3)//11H S11 TO SI
     120,10(1H,I3) //11H S21 TO S30,10(1H,I3) //11H S31 TO S40,10(1H,I3)I
     2//11H S41 TO S50,10(1H,I3)//11H S51 TO S60,10(1H,I3) // )
 S297  FORMAT(// 13H KSFLAGS ARE //11H S01 TO S10,10(1H,I3)//11H S11 TO SI
     120,10(1H,I3) //11H S21 TO S30,10(1H,I3) //11H S31 TO S40,10(1H,I3)I
     2//11H S41 TO S50,10(1H,I3)//11H S51 TO S60,10(1H,I3) // )
      END
```

```
      SUBROUTINE  SETUP
C     THIS SUBROUTINE DOES THE BOOKING FOR THE TRIAL AND ERROR ITERATIVE
C     CALCULATIONS.
C     STORE SN AND SNC FOR EQUIP. IN KE1 IN STRM AND STRMC
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 K5 = 1
    2 DO 15   K2 = 1, NE1MAX
    3 DO 415  K3 = 1, N2MAX
    4 IF(KPM(K3,1) - KE1(K2))          415,5,415
    5 DO 315  K4 = 2, N3MAX
    6 IF(KPM(K3,K4))                   9,315,7
C     K3 IS ROW NUMBER OF EQUIP. KE1(K2)
C     IF STREAM NUMBER IS NOT ZERO, STORE STREAM IN STRM AND STRMC
C     K6 IS STREAM NUMBER
    7 K6 = KPM(K3,K4)
    8 GO TO 10
    9 K6 = - KPM(K3,K4)
   10 DO 11   K7 = 1, NSLMAX
   11 STRM(K6,K7) = SN(K6,K7)
```

```
   12 DO 13    K7 = 1, NSCLMX
   13 STRMC(K6,K7) = SNC(K6,K7)
   14 K5 = K5 + 1
  315 CONTINUE
  415 CONTINUE
   15 CONTINUE
   16 NSTRM = K5 - 1
C     NSTRM IS THE NUMBER STREAMS ASSOCIATED WITH LIST KE1
   17 LOOP = 1
C     FOR EACH EQUIP. IN KE1, STORE STRM AND STRMC IN STRMI, STRMCI
C     AND STRMO, STRMCO.
   18 LIMIT2 = 0
   19 K8 = 1
C     K8 IS COUNTING INDEX FOR KE1
   20 K9 = 1
C     K9 IS COUNTING INDEX FOR STRMI AND STRMCI
   21 K10 = 1
C     K10 IS COUNTING INDEX FOR STRMO AND STRMCO
   22 DO 45    K3 = 1, N2MAX
   23 IF(KPM(K3,1) - KE1(K8))                    45,24,45
   24 K11 = K3
C     K3 IS ROW NUMBER IN KPM OF KE1 AS IS K11
   25 DO 345   K4 = 2, N3MAX
   26 IF(KPM(K3,K4))                             27,44,36
   27 NIN = K9 - 1
   28 K6 =-KPM(K3,K4)
   31 DO 32    K7 = 1, NSLMAX
   32 STRMO(K10,K7) = STRM(K6,K7)
   33 DO 34    K7 = 1, NSCLMX
   34 STRMCO(K10,K7) = STRMC(K6,K7)
   35 K10 = K10 + 1
  335 GO TO 345
   36 K6 = KPM(K3,K4)
   39 DO 40    K7 = 1, NSLMAX
   40 STRMI(K9,K7) = STRM(K6,K7)
   41 DO 42    K7 = 1, NSCLMX
   42 STRMCI(K9,K7) = STRMC(K6,K7)
   43 K9 = K9 + 1
  343 GO TO 345
   44 NOUT = K10 - 1
  345 CONTINUE
   45 CONTINUE
   46 NE = KE1(K8)
   47 CALL EQCALL
   48 CALL  TEST
   49 K13 = 1
   50 K14 = 1
C     IF LIMIT IS ZERO, CONVERGENCE OBTAINED FOR EQUIP. KE1(K8)
C     PLACE CALCULATED STREAMS BACK IN STRM AND STRMC
   51 IF(LIMIT)                                  53,53,52
C     NUMBER OF ITEMS EXCEEDING DELS IS VALUE OF LIMIT
   52 LIMIT2 = LIMIT2 + 1
   53 DO 67    K4 = 2, N3MAX
   54 IF(KPM(K11,K4))                            55,67,61
   55 K6 =-KPM(K11,K4)
   56 DO 57    K7 = 1, NSLMAX
   57 STRM(K6,K7) = STRMO(K13,K7)
   58 DO 59    K7 = 1, NSCLMX
   59 STRMC(K6,K7) = STRMCO(K13,K7)
   60 K13 = K13 + 1
  360 GO TO 67
```

```
   61 K6 = KPM(K11,K4)
   62 DO 63   K7 = 1, NSLMAX
   63 STRM(K6,K7) = STRMI(K14,K7)
   64 DO 65   K7 = 1, NSCLMX
   65 STRMC(K6,K7) = STRMCI(K14,K7)
   66 K14 = K14 + 1
   67 CONTINUE
   68 IF(NE1MAX - K8)          71,71,69
   69 K8 = K8 + 1
   70 GO TO 20
C          GET TO STATEMENT 71, AFTER EACH LOOP THRU KE1
   71 IF(LIMIT2)                          72,72,85
C     CONVERGENCE OBTAINED IF LIMIT2 IS ZERO
   72 WRITE OUTPUT TAPE 6,200,   LOOP
   73 K15 = 1
C     K15 IS USED TO CONTROL RETURN FROM PRINT CYCLE
   74 GO TO  126
C     STORE STRM AND STRMC IN SN AND SNC AND RETURN TO GUESS2
   76 DO 82   K2 = 1, NE1MAX
  500 DO 515  K3 = 1, N2MAX
  501 IF(KPM(K3,1) - KE1(K2))          515,505,515
  505 DO 615  K4 = 2, N3MAX
  506 IF(KPM(K3,K4))                   509,615,507
C     K3 IS ROW NUMBER OF EQUIP. KE1(K2)
C     IF STREAM NUMBER IS NOT ZERO, STORE STRM AND STRMC IN SN AND SNC
C     K6 IS STREAM NUMBER
  507 K6 = KPM(K3,K4)
  508 GO TO 510
  509 K6 = - KPM(K3,K4)
  510 DO 511  K7 = 1, NSLMAX
  511 SN(K6,K7) = STRM(K6,K7)
  512 DO 513  K7 = 1, NSCLMX
  513 SNC(K6,K7) = STRMC(K6,K7)
  615 CONTINUE
  515 CONTINUE
   82 CONTINUE
   83 KRET3 = 0
C     CONVERGED, RETURNED TO GUESS2
   84 RETURN
C     RETURNING TO GUESS2 WITH KRET3 = 0 WILL CAUSE GUESS2 TO REFLAG
C     EQUIPMENT AND STREAMS FROM 4 TO 1 AND RETURN TO MAJOR.
   85 IF(KSETS)                    86,86,185
  185 WRITE OUTPUT TAPE 6,201,  LOOP, LIMIT2
C     HAVE NOT YET CONVERGED.
C     CHECK NUMBER OF LOOPS THRU KE1
   86 IF(LOOPS - LOOP)             89,89,87
   87 LOOP = LOOP + 1
   88 GO TO 18
C     GO THRU KE1 LIST AGAIN
   89 WRITE OUTPUT TAPE 6,202
C     MAXIMUM NUMBER OF LOOPS TRIED WITHOUT CONVERGING.
C     IF NOGO IS ZERO, STORE LAST LOOP ANSWERS IN SN AND SNC AND GO ON
C     WITH THE CALCULATION. FLAG E AND S FROM 4 TO 13
C     IF NOGO IS NOT ZERO, REFLAG EQUIP. AND STREAMS FROM 4 TO 0 AND
C     RETURN TO GUESS2 TO TRY AND FIND ANOTHER EQUIPMENT LIST WHICH WILL
C     CONVERGE.
   90 IF(NOGO)                        91,104,91
   91 WRITE OUTPUT TAPE 6,203
   92 K15 = 2
   93 GO TO 126
C     REFLAG E AND S FROM 4 TO 0
```

```
   94 DO  97  K8 = 1, NEMAX
   95 IF(KEFLAG(K8) - 4)                97,96,97
   96 KEFLAG(K8) = 0
   97 CONTINUE
   98 DO 101  K8 = 1, NSMAX
   99 IF(KSFLAG(K8) - 4)               101,100,101
  100 KSFLAG(K8) = 0
  101 CONTINUE
  102 KRET3  = 2
  103 RETURN
  104 WRITE OUTPUT TAPE 6, 204
  105 K15 = 3
  106 GO TO  126
C     STORE STRM AND STRMC IN SN AND SNC
  108 DO 114  K2 = 1, NE1MAX
  600 DO 815  K3 = 1, N2MAX
  601 IF(KPM(K3,1) - KE1(K2))                815,605,815
  605 DO 715  K4 = 2, N3MAX
C     K3 IS ROW NUMBER OF EQUIP. KE1(K2)
C     IF STREAM NUMBER IS NOT ZERO, STORE STRM AND STRMC IN SN AND SNC
C     K6 IS STREAM NUMBER
  606 IF(KPM(K3,K4))                     609,715,607
  607 K6 = KPM(K3,K4)
  608 GO TO 610
  609 K6 = - KPM(K3,K4)
  610 DO 611  K7 = 1, NSLMAX
  611 SN(K6,K7) = STRM(K6,K7)
  612 DO 613  K7 = 1, NSCLMX
  613 SNC(K6,K7) = STRMC(K6,K7)
  715 CONTINUE
  815 CONTINUE
  114 CONTINUE
      REFLAG EQUIP. AND STREAMS FROM 4 TO 13
  115 DO 118  K8 = 1, NEMAX
  116 IF(KEFLAG(K8) - 4)              118,117,118
  117 KEFLAG(K8) = 13
  118 CONTINUE
  119 DO 122  K8 = 1, NSMAX
  120 IF(KSFLAG(K8) - 4)              122,121,122
  121 KSFLAG(K8) = 13
  122 CONTINUE
  123 KRET3 = (-2)
  124 KRET  =  0
  125 RETURN
      PRINT SEQUENCE
  126 IF(KSETS)                      153,153,326
  326 GO TO (127,129,131), K15
  127 WRITE OUTPUT TAPE 6,205, KCV1,KCV2,KCV3, NE1MAX, (KE1(K16),K16=1,NE
     1E1MAX)
  128 GO TO 133
  129 WRITE OUTPUT TAPE 6,206, KCV1,KCV2,KCV3,NE1MAX,LOOPS,(KE1(K16),K16=
     1=1,NE1MAX)
  130 GO TO 133
  131 WRITE OUTPUT TAPE 6,207,KCV1,KCV2,KCV3,NE1MAX, (KE1(K16),K16=1,NE1
     1MAX)
  132 GO TO 133
      K21 IS KPM ROW OF EQUIPMENT NUMBER KE1(K20)
      K24 IS EQUIPMENT NUMBER KE1(K20)
  133 DO 452 K20 = 1, NE1MAX
  134 DO 352 K21 = 1, N2MAX
  135 IF(KPM(K21,1) - KE1(K20))              352,136,352
```

```
      K24 = KE1(K20)
      WRITE OUTPUT TAPE 6, 208, K24, (EN(K24,K25),K25=1,NELMAX)
      WRITE OUTPUT TAPE 6, 209,      (ENC(K24,K25),K25=1,NECLMX)
      DO 152 K22 = 2, N3MAX
      IF(KPM(K21,K22))           147,152,141
C     INPUT STREAM PRINT OUT
  141 K23 = KPM(K21,K22)
      WRITE OUTPUT TAPE 6,210,  K23, (STRM(K23,K25),K25=1,NSLMAX)
      WRITE OUTPUT TAPE 6,211,         (SN(K23,K25),K25=1,NSLMAX)
      WRITE OUTPUT TAPE 6,212,       (STRMC(K23,K25),K25=1,NSCLMX)
      WRITE OUTPUT TAPE 6,213,        (SNC(K23,K25),K25=1,NSCLMX)
      GO TO 152
C     OUTPUT STREAM PRINT OUT
  147 K23 =-KPM(K21,K22)
      WRITE OUTPUT TAPE 6, 214, K23, (STRM(K23,K25),K25=1,NSLMAX)
      WRITE OUTPUT TAPE 6, 215,        (SN(K23,K25),K25=1,NSLMAX)
      WRITE OUTPUT TAPE 6, 216,      (STRMC(K23,K25),K25=1,NSCLMX)
      WRITE OUTPUT TAPE 6, 217,       (SNC(K23,K25),K25=1,NSCLMX)
  152 CONTINUE
  352 CONTINUE
  452 CONTINUE
      GO TO (76,94,108), K15
  200 FORMAT(////10X 38HCONVERGENCE OBTAINED AFTER LOOP NUMBER I4///59H
     1THE ANSWERS AND THE INITIAL STREAM VALUES ARE GIVEN BELOW.)
  201 FORMAT(1H1 10X 30HCONVERGENCE NOT OBTAINED AFTER I5, 7H LOOPS.///
     131H STREAMS HAVE NOT CONVERGED FOR I4, 13H EQUIPMENT(S) ///63H ANO
     2THER LOOP OF CALCULATIONS WILL BE TRIED IF ALLOWABLE NUMBER //29H
     30F LOOPS HAS NOT BEEN TRIED.)
  202 FORMAT(///67H ALLOWABLE NUMBER OF LOOPS HAS BEEN TRIED.  THIS WAS
     1THE LAST LOOP.)
  203 FORMAT(///67H ANOTHER LIST OF ASSUMED KNOWN INPUT STREAMS WILL BE
     1TRIED IN ORDER //59H TO TRY AND FIND A PATH OF CALCULATION WHICH W
     2ILL CONVERGE. )
  204 FORMAT(///72H WILL PROCEED WITH CALCULATION EVEN THOUGH CONVERGENC
     1E WAS NOT OBTAINED. ///59H THE RESULTS OF THE LAST LOOP OF CALCULA
     2TIONS WILL BE USED. ///50H STREAMS AND EQUIPMENT WILL BE FLAGGED E
     3QUAL TO 13 ///54H THE ANSWERS AND INITIAL STREAM VALUES ARE GIVEN
     4BELOW )
  205 FORMAT(27H1THE ASSUMED INPUT STREAMS  2( I4, 1H,),I4, 21H HAVE BEE
     1N CALCULATED //19H AS OUTPUT STREAMS. //12H THE LIST OF I4, 38H EQ
     2UIPMENTS IN ORDER OF CALCULATION IS / (15X I10))
  206 FORMAT(27H1THE ASSUMED INPUT STREAMS  2(I4, 1H,),I4, 24H COULD NO
     1T BE CALCULATED //43H AS OUTPUT STREAMS OF THE FOLLOWING LIST OF
     2I4, 18H EQUIPMENTS AS THE //34H CALCULATION WOULD NOT CONVERGE IN
     6 I4,
     37H LOOPS. ///51H THE FINAL VALUES OF THE VARIABLES ARE SHOWN BELOW
     4. ///50H THE LIST OF EQUIPMENT IN ORDER OF CALCULATION WAS/(15X I1
     50))
  207 FORMAT(46H1THE CALCULATION OF THE ASSUMED INPUT STREAMS  2(I4, 1H,
     1), I4//64H DID NOT CONVERGE.  HOWEVER, ANSWERS OBTAINED FROM THE L
     2AST LOOP //58H WILL BE TAKEN AS CORRECT FOR ANY ADDITIONAL CALCULA
     3TIONS. ///12H THE LIST OF I4, 38H EQUIPMENTS IN ORDER OF CALCULATI
     40N IS/(15X I10))
  208 FORMAT(1H1 19X 20HFOR EQUIPMENT NUMBER I4///20X 24HEQUIPMENT PARAM
     1ETERS ARE //(5F15.4))
  209 FORMAT(/16X 32HEQUIPMENT CONTROL PARAMETERS ARE /(5F15.4))
  210 FORMAT(//19X 19HINPUT STREAM NUMBER I4//16X 31HCALCULATED STREAM VA
     1ARIABLES ARE //(5F15.4))
  211 FORMAT(/18X 29HINITIAL STREAM VARIABLES WERE//( 5F15.4))
  212 FORMAT(/13X 40HCALCULATED STREAM CONTROL PARAMETERS ARE//(5F15.4))
  213 FORMAT(/14X 38HINITIAL STREAM CONTROL PARAMETERS WERE //(5F15.4))
```

```
  214 FORMAT(//19X 20HOUTPUT STREAM NUMBER I4//16X 31HCALCULATED STREAM
     1VARIABLES ARE //(5F15.4))
  215 FORMAT(/18X 29HINITIAL STREAM VARIABLES WERE //(5F15.4))
  216 FORMAT(/13X 40HCALCULATED STREAM CONTROL PARAMETERS ARE//(5F15.4))
  217 FORMAT(/14X 38HINITIAL STREAM CONTROL PARAMETERS WERE //(5F15.4))
      END
```

```
      SUBROUTINE  TEST
C     THIS SUBROUTINE TESTS FOR CONVERGENCE BY SEEING IF CALCULATED
C     OUTPUT STREAM VALUES STRMO, STRMCO ARE THE SAME AS LAST LOOP
C     VALUES STRM, STRMC BY USING THE CONVERGENCE TEST VECTORS
C     DELS(NSLMAX), DELSC(NSCLMX).  THE ONE CHECK VECTOR IS USED FOR ALL
C     STREAMS.
C     TEST IS CALLED BY SUBROUTINE SETUP
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 LIMIT = 0
    2 LIMIT3 = 0
    3 DO 21  K26 = 1, NOUT
    4 K27 = STRMO(K26,1)
      K27 IS OUTPUT STREAM NUMBER OF EQUIP. NUMBER NE
      CHECK STREAM VALUES STRMO AGAINST STRM USING DELS
    5 DO  8  K28 = 1, NSLMAX
    6 IF(  ABSF(STRM(K27,K28)-STRMO(K26,K28)) - DELS(K28))      8,8,7
    7 LIMIT3 = LIMIT3 + 1
    8 CONTINUE
    9 IF(LIMIT3)              13,13,10
```

```
 10 IF(KSETS)                    11,11,100
100 WRITE OUTPUT TAPE 6, 25, LOOP, K27, NE, LIMIT3
 11 LIMIT3 = 0
 12 LIMIT = LIMIT + 1
C     CHECK STREAM VALUES STRMCO AGAINST STRMC USING DELSC
 13 DO 16  K28 = 1, NSCLMX
 14 IF(  ABSF(STRMC(K27,K28)-STRMCO(K26,K28)) - DELSC(K28))  16,16,15
 15 LIMIT3 = LIMIT3 + 1
 16 CONTINUE
 17 IF(LIMIT3)           21,21,18
 18 IF(KSETS)                 19,19,118
118 WRITE OUTPUT TAPE 6, 26, LOOP, K27, NE, LIMIT3
 19 LIMIT3 = 0
 20 LIMIT = LIMIT + 1
 21 CONTINUE
 22 RETURN
 25 FORMAT(15H1IN LOOP NUMBER I4, 47H OF THE TRIAL AND ERROR ITERATIVES
    1 CALCULATIONS, ///21H OUTPUT STREAM NUMBER I5, 20H OF EQUIPMENT NU
    2MBER I4/// 4H HAS I4, 51H STREAM VALUES WHICH EXCEED THE CONVERGEN
    3CE LIMITS.)
 26 FORMAT(15H1IN LOOP NUMBER I4, 47H OF THE TRIAL AND ERROR ITERATIVES
    1 CALCULATIONS, ///21H OUTPUT STREAM NUMBER I4, 20H OF EQUIPMENT NU
    2MBER I4///4H HAS I4, 63H STREAM CONTROL PARAMETERS WHICH EXCEED TH
    3E CONVERGENCE LIMITS.)
     END
```

```
      SUBROUTINE  GOOF
C     THIS SUBROUTINE PRINTS OUT KUS, KPS, KEFLAG, KSFLAG, KPM, SN, SNC
C     EN, AND ENC WHEN PROCESS IS GREATER THAN THIRD ORDER IN RECYCLE
C     CALCULATIONS
C     GOOF IS CALLE BY SUBROUTINE GUESS1
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6, 100, KRUN, (KUS(K2),K2=1,NUSMAX)
C     IF PREFERRED STREAMS GIVEN, LIST THEM
    2 IF(NPSMAX)                  4,4,3
    3 WRITE OUTPUT TAPE 6, 101,  (KPS(K2),K2=1,NPSMAX)
C     PRINT EQUIPMENT AND STREAM FLAGS
    4 WRITE OUTPUT TAPE 6, 102,  (KEFLAG(K2),K2=1,NEMAX)
    5 WRITE OUTPUT TAPE 6, 103,  (KSFLAG(K2),K2=1,NSMAX)
C     PRINT OUT PROCESS MATRIX ON NEW PAGE
    6 WRITE OUTPUT TAPE 6, 104, KRUN
    7 DO 49   K2 = 1, N2MAX
      IF(KPM(K2,1))    49,49,47
   47 WRITE OUTPUT TAPE 6,105,KPM(K2,1),NAME(K2),NEXEQN(K2),
     1(KPM(K2,K3),K3=2,N3MAX)
```

```
   49 CONTINUE
  108 IF(NOKPS)                    9,9,109
  109 WRITE OUTPUT TAPE 6, 101,  (KPS(K2),K2=1,NOKPS)
C     PRINT OUT STREAM VARIABLES MATRIX
    9 WRITE OUTPUT TAPE 6, 106, KRUN
   10 DO 11    K2 = 1, NSMAX
   11 WRITE OUTPUT TAPE 6, 107,   (SN(K2,K3),K3=1,NSLMAX)
C     PRINT OUT STREAM CONTROL PARAMETERS MATRIX
   12 WRITE OUTPUT TAPE 6, 208, KRUN
   13 DO 14    K2 = 1, NSMAX
   14 WRITE OUTPUT TAPE 6, 107,  (SNC(K2,K3),K3=1,NSCLMX)
C     PRINT OUT EQUIPMENT PARAMETERS MATRIX
   15 WRITE OUTPUT TAPE 6, 209, KRUN
   16 DO 17    K2 = 1, NEMAX
   17 WRITE OUTPUT TAPE 6, 107,   (EN(K2,K3),K3=1,NELMAX)
C     PRINT OUT EQUIPMENT CONTROL PARAMETERS MATRIX
   18 WRITE OUTPUT TAPE 6, 110, KRUN
   19 DO 20    K2 = 1, NEMAX
   20 WRITE OUTPUT TAPE 6, 107,  (ENC(K2,K3),K3=1,NECLMX)
   21 RETURN
  100 FORMAT(1H1 8X 27HCALCULATIONS FOR RUN NUMBER I5, 11H IMPOSSIBLE //
     1/67H ALL POSSIBLE COMBINATIONS OF ALL UNKNOWN INPUT STREAMS TAKEN
     2, 2, /67H AND 3 AT A TIME HAVE BEEN TRIED.  NOT POSSIBLE TO FIND
     3A SERIES OF / 71H EQUIPMENT WHICH HAS AS OUTPUT STREAMS THE ASSUME
     4D KNOWN INPUT STREAMS. ///61H THE FOLLOWING LIST OF INPUT STREAMS
     5COULD NOT BE CALCULATED. // 9(I4, 1H,), I4 /)
  101 FORMAT(//31H PREFERRED STREAM NUMBERS WERE 515/ (31X 515))
  102 FORMAT(//37H PRESENT STATUS OF EQUIPMENT FLAGS IS //11H E01 TO E10
     1 10(1H, I3)//11H E11 TO E20 10(1H, I3)//11H E21 TO E30 10(1H, I3))
  103 FORMAT(//34H PRESENT STATUS OF STREAM FLAGS IS //11H S01 TO S10 10
     1(1H, I3)//11H S11 TO S20 10(1H, I3)//11H S21 TO S30 10(1H, I3)//
     2          11H S31 TO S40 10(1H, I3)//11H S41 TO S50 10(1H, I3)//
     311H S51 TO S60 10(1H, I3))
  104 FORMAT(1H1 23X 21HPROCESS MATRIX AT RUN I5///10H EQUIPMENT 14X 14H
     1STREAM NUMBERS /8H  NUMBER //)
  105 FORMAT(/1X14,4XA6,1X10I5)
  106 FORMAT(1H1 18X 30HSTREAM VARIABLES MATRIX AT RUN I5//)
  107 FORMAT(/(5F15.4))
  208 FORMAT(1H1 15X 39HSTREAM CONTROL PARAMETERS MATRIX AT RUN I5//)
  209 FORMAT(1H1 18X 34HEQUIPMENT PARAMETERS MATRIX AT RUN I5//)
  110 FORMAT(1H1 13X 42HEQUIPMENT CONTROL PARAMETERS MATRIX AT RUN I5//)
   48 FORMAT(/1X14,4XA6,1X10I5)
      END
```

```
      SUBROUTINE  DEBUG
C     THIS IS USED TO DEBUG LOGIC OF PACE.  IT SERVES IN PLACE OF
C     EQUIPMENT SUBROUTINES.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
    1 WRITE OUTPUT TAPE 6,99,  NE, (STRMI(K2,1), K2=1, NIN)
    2 WRITE OUTPUT TAPE 6,98,        (STRMO(K6,1), K6=1, NOUT)
    3 EXX = NE
    4 DO 8  K15 = 1, NOUT
    5 DO 8  K16 = 3, NSLMAX
    6 D15 = K15
    7 D16 = K16
    8 STRMO(K15,K16) = D15 + D16 + 0.99
    9 DO 13 K15 = 1, NOUT
   10 DO 13 K16 = 3, NSCLMX
   11 D15 = K15
   12 D16 = K16
   13 STRMCO(K15,K16) = D15 + D16 + 0.66
   14 DO 15 K16 = 3, NELMAX
   15 EN(NE,K16) = EXX
```

```
16 DO 17 K16 = 3, NECLMX
17 ENC(NE,K16) = EXX + 100.
18 RETURN
99 FORMAT(1H1 18X 35HPACER DEBUGGER EQUIPMENT SUBROUTINE ///17H EQUIP
   1MENT NUMBER I4,25H IS NOW BEING CALCULATED. //33H THE ASSOCIATED I
   2NPUT STREAMS ARE I6/(33X I6))
98 FORMAT(//34H THE CALCULATED OUTPUT STREAMS ARE I11/(34X I11))
   END
```

```
      SUBROUTINE UNAME1
C     UNAMI IS A MIXER SUBROUTINE
C     THIS SUBROUTINE TAKE NIN INPUT STREAMS AND PRODUCES NOUT OUTPUT
C     STREAMS FOR EACH VARIABLE STRMI(I,K), K=3 TO NSLMAX, THE I INPUT
C     VALUES ARE ADDED TOGETHER AND THEN SPLIT AMOUNG THE OUTPUT STREAMS
C     IN FRACTIONS DETERMINED BY THE EN VECTOR FOR THE EQUIPMENT BEING
C     CALCULATED.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
      DIMENSION  TOTAL(18)
    1 NDD = NSLMAX - 2
    2 DO  3  JE2 = 1, NDD
    3 TOTAL(JE2) = 0.
C     ADD VALUES FOR EACH ROW OF NIN STREAMS
    4 DO 6  JE3 = 1, NDD
    5 DO  6  JE4 = 1, NIN
    6 TOTAL(JE3) = TOTAL(JE3) + STRMI(JE4,JE3+2)
C     STORE TOTALS FOR EACH ROW DIVIDED BY THE NUMBER OF OUTPUT STREAMS
C     IN THE OUTPUT STREAM VECTORS STRMO
    7 OUT = NOUT
```

```
      DO 10   JE5 = 1, NOUT
      DO 10   JE6 = 1,  NDD
   10 STRMO(JE5,JE6 + 2) = TOTAL(JE6)/OUT
      RETURN
      END
```

```
      SUBROUTINE  UNAME2
C     DISTILLATION SIMULATION        SHANNON 8/24/61
C     5 COMPONENT SYSTEM.  1 FEED IS SPLIT INTO TWO STREAMS, OVERHEADS
C     AND BOTTOMS.  THIS SUBROUTINE CAN NOW WORK WITH NOCOMP COMPONENTS
C     FIRST OUTPUT STREAM IS OVERHEADS
C     SECOND OUTPUT STREAM IS BOTTOMS
C     SUBROUTINE MAY ALSO BE USED TO SIMULATE A REACTOR
C     STREAM PARAMETERS VECTOR FORMAT          STREAM CONTROL VECTOR FORMAT
C          ROW                VARIABLE                 VARIABLE
C           1          STREAM NUMBER            STREAM NUMBER
C           2          STREAM FLAG              STREAM FLAG
C           3          TOTAL STREAM QUANTITY    LIGHT KEY COMPONENT NUMBER
C           4-8        PERCENT OF COMPONENTS    NOT USED
C                      1 TO 5
C           9-13       AMOUNT OF COMPONENTS     NOT USED
C                      1 TO 5
C     EQUIPMENT PARAMETERS VECTOR
C          ROW                VARIABLE
C           1          EQUIPMENT NUMBER
C           2          EQUIPMENT FLAG
C           3          FICTICIOUS HEAT LOAD
C           4-8        FRACTION OF AMOUNT OF COMPONENT 1 TO 5 IN FEED
C                      APPEARING IN OVERHEAD
C     EQUIPMENT CONTROL VECTOR
C          ROW                VARIABLE
C           1          EQUIPMENT NUMBER
C           2          EQUIPMENT FLAG
C           3          FIRST CONSTANT IN HEAT EQUATION
C           4          SECOND CONSTANT IN HEAT EQUATION
C           5          THIRD CONSTANT IN HEAT EQUATION
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,       ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,       SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,           KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,           KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,           KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,           KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,         STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
```

```
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
C     CALCULATE THE AMOUNTS OF EACH COMPONENT IN OVERHEADS AND BOTTOMS
      COMPNO=STRMCI(1,4)
      NOCOMP=COMPNO+0.5
      J59=NOCOMP+3
    1 DO  3  NSN = 1, NOCOMP
      J58=J59+NSN
    2 STRMO(1, J58 ) = STRMI(1, J58 )*EN(NE,NSN+3)
    3 STRMO(2, J58 ) = STRMI(1, J58 ) *(1. - EN(NE,NSN+3))
C     CALCULATE AMOUNT OF OVERHEADS AND BOTTOMS
    4 STRMO(1,3) = 0.
    5 STRMO(2,3) = 0.
    6 DO 8   NRN = 1, NOCOMP
      J58=J59+NRN
    7 STRMO(1,3) = STRMO(1,3) + STRMO(1, J58 )
    8 STRMO(2,3) = STRMO(2,3) + STRMO(2, J58 )
C     CALCULATE PERCENTAGE COMPOSITION OF COMPONENTS
    9 DO  11  NSN = 1, NOCOMP
      J58=J59+NSN
   10 STRMO(1,NSN+3) = STRMO(1, J58 )/STRMO(1,3)
   11 STRMO(2,NSN+3) = STRMO(2, J58 )/STRMO(2,3)
C     CALCULATE FICTICIOUS HEAT LOAD
   12 NX = STRMCI(1,3)
   13 EN(NE,3)=ENC(NE,3)*STRMO(1,3)+ENC(NE,4)*(((ABSF(EN(NE,NX+3) -
     1  EN(NE,NX+4)))*100.)**ENC(NE,5))
   14 RETURN
      END
```

```
      SUBROUTINE  UNAME3
C     THIS MIXER TAKES NIN INPUT STREAMS AND SPLITS THE TOTAL INPUT
C     EQUALLY AMOUNG THE NOUT OUTPUT STREAMS. NOCOMP IS THE NO. OF COMP.
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,        ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,        SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,            KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,            KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,            KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,            KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,          STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION  KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION  SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION  KUS(60), KPS(10), KEUS(8)
      DIMENSION  KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION  STRM(60,20), STRMC(60,10)
      DIMENSION  STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION  DELS(20), DELSC(10)
      DIMENSION  TITLE(12), ICONV(15), NECALL(30)
      DIMENSION TOTAL(6)
    1 OUT = NOUT
    3 DO 4  NSN = 1, 6
    4 TOTAL(NSN) = 0.
C     TOTAL AMOUNTS FOR EACH OF THE COMPONENTS
    5 DO  7  NRN = 1,5
    6 DO  7  NSN = 1, NIN
    7 TOTAL(NRN) = TOTAL(NRN) + STRMI(NSN,NRN+8)
C     TOTAL AMOUNTS OF TOTAL QUANTITY FOR THE INPUT STREAMS
    8 DO  9  NSN = 1, NIN
    9 TOTAL(6) = TOTAL(6) + STRMI(NSN,3)
C     SPLIT TOTAL QUANTITY AMOUNG OUTPUT STREAMS
   10 DO 11 NSN = 1, NOUT
   11 STRMO(NSN,3) = TOTAL(6)/OUT
C     SPLIT AMOUNTS OF EACH COMPONENT AMOUNG OUTPUT STREAMS AND
```

```
C       CALCULATE PERCENTAGES
   12 DO 15  NSN = 1, NOUT
   13 DO 15  NRN = 1, 5
   14 STRMO(NSN,NRN+8) = TOTAL(NRN)/OUT
   15 STRMO(NSN,NRN+3) = STRMO(NSN,NRN+8)/STRMO(NSN,3)
   16 RETURN
      END
```

```
      SUBROUTINE UNAME4
C               FLASH  DISTILLATION
C     NUMBER OF COMPONENTS EQUALLS NOCOMP (2-10)
C     EQIULIBRIUM VALUES FOR THE COMPONENTS AT THE OPERATING CONDITIONS
C     MUST BE GIVEN IN THE EN VECTOR FOR THE EQUIPMENT BEING CALCULATED.
C     K VALUES MUST BE IN EN(NE,K2)          K2=4,NOCOMP+3
      COMMON  KPM, N2, N3, N2MAX, N3MAX, KPMR
      COMMON  KEFLAG, NE, NEMAX
      COMMON  KSFLAG, NS, NSMAX
      COMMON  EN, NEN, NEL, NELMAX,          ENC, NENC, NECL, NECLMX
      COMMON  SN, NSN, NSL, NSLMAX,          SNC, NSNC, NSCL, NSCLMX
      COMMON  KUS, NUS, NUSMAX,              KPS, NPS, NPSMAX
      COMMON  NUS1, NUS2, NUS3,              KCV1, KCV2, KCV3
      COMMON  KEUS, NEUS,   NENUM
      COMMON  KECV1, KECV2, KECV3, KECV4
      COMMON  NECV1, NECV2, NECV3, NECV4
      COMMON  KE1, NE1, NE1MAX,              KE2, NE2, NE2MAX
      COMMON  KE3, NE3, NE3MAX,              KE4, NE4, NE4MAX
      COMMON  KES, KES2, KES3
      COMMON  KRET, KRET2, KRET3
      COMMON  STRM, STRMC
      COMMON  STRMI, STRMCI, NIN,            STRMO, STRMCO, NOUT
      COMMON  DELS, DELSC, LIMIT, LIMIT2, LIMIT3, LOOP, KBACK
      COMMON  KRUN, KSETS, KCLEAN, LOOPS, NOGO
      COMMON  TITLE, NOKPM, NOEN, NOENC, NOSN, NOSNC, NOKPS, NSTRM
      COMMON  NLX, NLY, NLXS, NLXSC
      COMMON  ICONV, NEX, NECALL, NEQUIP
      COMMON  ND2, ND3, ND4, ND5, ND6, ND7
      COMMON  EXX, D15, D16
      COMMON  I2,I3,I4,I5,I6,I7,I8,I9,I10,I11,I12,I13,I14,I15
      COMMON  J2,J3,J4,J5
      COMMON  J6, J7,J8,J9,J10,J11,J12,J13,J14,J15,J16,J17,J18,J19,J20,
     1        J21,J22,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,
     2        J35,J36,J37,J38,J39,J40,J41,J42,J43,J44,J45,J46
      COMMON  J47,J48,J49,J50,J51,J52,J53,J54,J55,J56,J57,J58,J59,J60
      COMMON  K2,K3,K4,K5,K6,K7,K8,K9,K10,K11,K12,K13,K14,K15,K16,K17,
     1        K18,K19,K20,K21,K22,K23,K24,K25,K26,K27,K28,K29,K30
      COMMON  L2,L3,L4,L5,L6,L7,L8,L9,L10,L11,L12,L13
      COMMON NDELS,NDELSC
      COMMON NEXEQN, NAME,KSEM
      DIMENSION NEXEQN(30), NAME(30),KSEM(60,3)
      DIMENSION KPM(30,10), KEFLAG(30), KSFLAG(60), KPMR(30,12)
      DIMENSION SN(60,20) ,SNC(60,10), EN(30,10), ENC(30,10)
      DIMENSION KUS(60), KPS(10), KEUS(8)
      DIMENSION KE1(30), KE2(30), KE3(30), KE4(30)
      DIMENSION STRM(60,20), STRMC(60,10)
      DIMENSION STRMI(8,20), STRMCI(8,10), STRMO(8,20), STRMCO(8,10)
      DIMENSION DELS(20), DELSC(10)
      DIMENSION TITLE(12), ICONV(15), NECALL(30)
      DIMENSION Y(10),YPRIME(10),V(10)
      SUMY=0.
      SUMX=0.
      COMPNO=STRMCI(1,4)
C     COMPNO IS THE NUMBER OF COMPONENTS IN THE INPUT STREAM.
      X=COMPNO+0.5
      NOCOMP=X
      DO 1 K1=1,NOCOMP
      SUMY=SUMY+STRMI(1,K1+3)*EN(NE,K1+3)
      SUMX=SUMX+STRMI(1,K1+3)/EN(NE,K1+3)
    1 CONTINUE
      WRITE OUTPUT TAPE 6,900,SUMX,SUMY
```

```
  900 FORMAT(1H1,6H SUMX=,E20.8,5X,6H SUMY=,E20.8)
      WRITE OUTPUT TAPE 6,901,COMPNO,NOCOMP
  901 FORMAT(//8H COMPNO=,F5.2,5X,7HNOCOMP=,I3)
      IF(SUMX-1.) 2,2,3
    2 WRITE OUTPUT TAPE 6,100,NE,STRMI(1,1)
  100 FORMAT(1H1,17H EQUIPMENT NUMBER,I2,1X,28H, A FLASH DISTILLATION UN
     1IT,/42H HAS ENTIRELY VAPORIZED FEED STREAM NUMBER,F5.2)
      DO 4 K1=3,NSLMAX
      STRMO(1,K1)=STRMI(1,K1)
      STRMO(2,K1)=0.
    4 CONTINUE
      GO TO 14
    3 IF(SUMY-1.) 5,5,6
    5 WRITE OUTPUT TAPE 6,101,NE,STRMI(1,1)
  101 FORMAT(1H1,17H EQUIPMENT NUMBERI2,1X,28H, A FLASH DISTILLATION UNI
     1T,/44H HAS NOT VAPORIZED ANY OF FEED STREAM NUMBER,F5.2)
      DO 20 K1=3,NSLMAX
      STRMO(1,K1)=0.
      STRMO(2,K1)=STRMI(1,K1)
   20 CONTINUE
      GO TO 14
C     BEGIN FLASH TRIAL AND ERROR. ASSUME V IS 0.5
    6 KLOOP=0
    7 CALCV=0.
   10 VAPOR=CALCV
      SUMY=0.
      SUMYP=0.
      DO 11 I=1,NOCOMP
      K2=I+3
      TOP=EN(NE,K2)*STRMI(1,K2)
      XKM1=EN(NE,K2)-1.
      BOT=VAPOR*XKM1+1.
      Y(I)=TOP/BOT
      SUMY=SUMY+Y(I)
      YPRIME(I)=-TOP*XKM1/BOT**2
   11 SUMYP=SUMYP+YPRIME(I)
      KLOOP=KLOOP+1
      WRITE OUTPUT TAPE 6,998,KLOOP,VAPOR,SUMY,(Y(I),I=1,NOCOMP)
  998 FORMAT(////7H KLOOP=,I3//7H VAPOR=,F10.7//6H SUMY=,F15.7//6H Y(1)=
     1,5(F10.7,1H,))
      WRITE OUTPUT TAPE 6,997,SUMYP,(YPRIME(I),I=1,NOCOMP)
  997 FORMAT(/7H SUMYP=,F15.7//11H YPRIME(I)=,5(F10.7,1H,))
      CALCV=VAPOR-(SUMY-1.)/SUMYP
      WRITE OUTPUT TAPE 6,996,CALCV
  996 FORMAT(/7H CALCV=,F10.7)
      IF(CALCV-VAPOR) 21,12,21
   21 IF(15-KLOOP) 12,12,10
C     CONVERGENCE HAS BEEN OBTAINED.
   12 STRMO(1,3)=VAPOR*STRMI(1,3)
      STRMO(2,3)=(1.-VAPOR)*STRMI(1,3)
      DO 13 K1=1,NOCOMP
      K2=K1+NOCOMP+3
      STRMO(1,K2)=Y(K1)*STRMO(1,3)
      STRMO(2,K2)=STRMI(1,K2)-STRMO(1,K2)
      K3=K1+3
      STRMO(1,K3)=STRMO(1,K2)/STRMO(1,3)
      STRMO(2,K3)=STRMO(2,K2)/STRMO(2,3)
   13 CONTINUE
   14 J=3+2*NOCOMP
      WRITE OUTPUT TAPE 6,15,(V(I),I=1,NOCOMP)
   15 FORMAT(1H1,17H FINAL V(I) VALUE/(5F15.8))
```

```
      WRITE OUTPUT TAPE 6,16,(STRMI(1,I),I=1,J)
   16 FORMAT(///18H STRMI(1,I) VALUES/(5F15.8))
      DO 18 K=1,2
      WRITE OUTPUT TAPE 6,17,K,(STRMO(K,I),I=1,J)
   17 FORMAT(///7H STRMO(,I1,10H,I) VALUES/(5F15.8))
   18 CONTINUE
      IF(CALCV-0.001) 22,23,23
   22 WRITE OUTPUT TAPE 6,24,CALCV
   24 FORMAT(1H1,22H IN SUBROUTINE UNAME4,,47H THE FLASH CALCULATIONS HA
     1VE CONVERGED ON ZERO.////7H CALCV=,F20.8)
   23 IF(0.999-CALCV)25,19,19
   25 WRITE OUTPUT TAPE 6,26,CALCV
   26 FORMAT(1H1,22H IN SUBROUTINE UNAME4,,46H THE FLASH CALCULATIONS HA
     1VE CONVERGED ON ONE.////7H CALCV=,F20.8)
   19 RETURN
      END
```

```
*       LABEL
C5UNAM
        SUBROUTINE  UNAME5
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C6UNAM
        SUBROUTINE  UNAME6
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C7UNAM
        SUBROUTINE  UNAME7
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C8UNAM
        SUBROUTINE  UNAME8
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C9UNAM
        SUBROUTINE  UNAME9
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C1EXTR
        SUBROUTINE  EXTRA1
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C2EXTR
        SUBROUTINE  EXTRA2
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
*       LABEL
C3EXTR
        SUBROUTINE  EXTRA3
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
        RETURN
        END
C4EXTR
*       LABEL
        SUBROUTINE  EXTRA4
C       DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C       IS SUPPLIED
```

```
      RETURN
      END
*     LABEL
CSEXTR
      SUBROUTINE  EXTRA5
C     DUMMY SUBROUTINE    REPLACE IF ANOTHER SUBROUTINE WITH THIS NAME
C     IS SUPPLIED
      RETURN
      END
```

APPENDIX B

Example Problems

This section illustrates several sample problems to show how PACER sets up and solves a problem. Data input format and results are also given.

Example 1

A certain process is to be designed to separate five components. Figure 5 shows the process flow diagram. The process consists of 5 pieces of equipment and 10 streams. The poly plant reaction is to be tested at two sets of operating conditions which will of course give two different degrees of separation on input stream 4. Three different feed streams are to be tested for each set of operating conditions resulting in 6 runs.

The input data statements and the results are shown on the following pages.

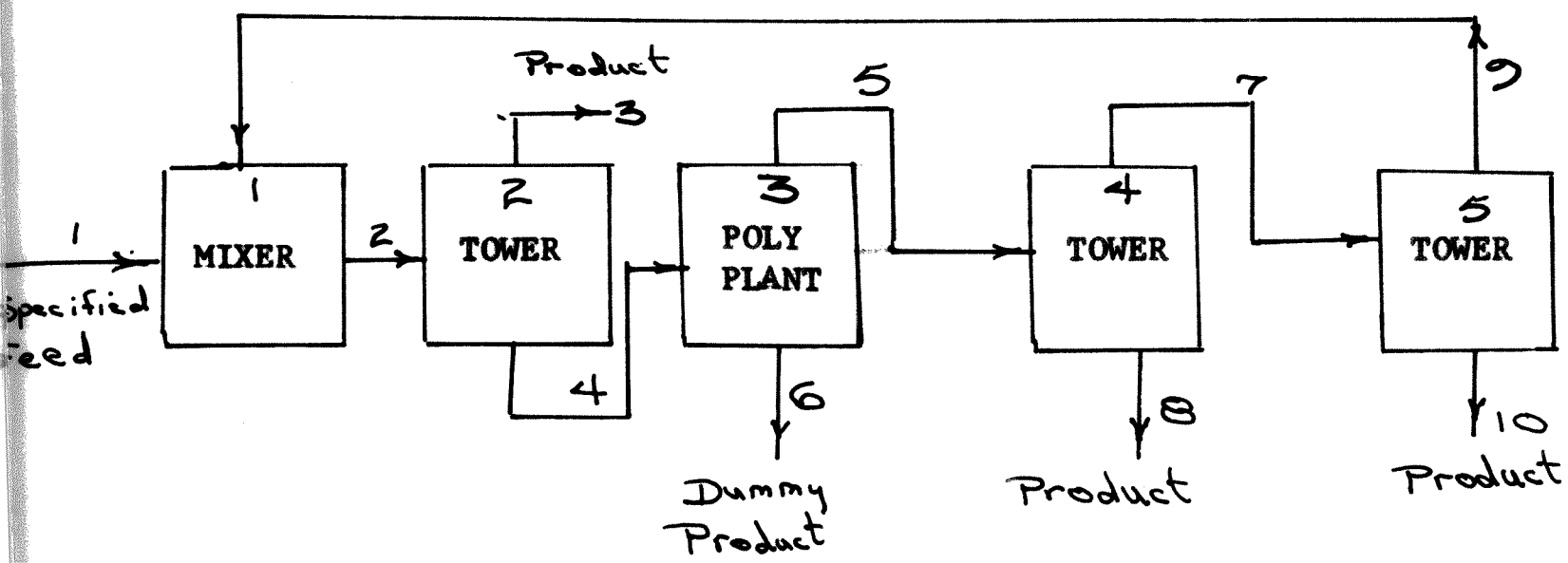

FIGURE 5

Flow Diagram for Example 1

Process Matrix (13)

Equipment

Number	Name (Subroutine)	Associate Streams		
1	Mixer (UNAME3)	1	9	-2
2	Tower (UNAME2)	2	-3	-4
3	Reactor (UNAME2)	4	-5	-6
4	Tower (UNAME2)	5	-7	-8
5	Tower (UNAME2)	7	-9	-10

Data Input for Example 1

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEEDI REACTI
 700    -1     0    20     1
  10     5    10    15     8     5    13     5     5
   5
   1UNAME3        501            1     9    -2
   2UNAME2        502            2    -3    -4
   3UNAME2        503            4    -5    -6
   4UNAME2        504            5    -7    -8
   5UNAME2        505            7    -9   -10
   5
        1.             99.
        0.
        2.             66.            0.          0.800          0.70
        0.001           0.001         0.001
        3.             66.            0.          0.50           1.0
        0.07            1.00          2.43
        4.             66.            0.          1.00           1.00
        0.95            0.90          0.02
        5.             66.            0.          0.90           0.80
        0.35            0.25          0.0
   5
        1.             99.            0.          0.             0.
        2.             99.          100.        100.             2.
        3.             99.           10.         10.             2.
        4.             99.          100.        100.             2.
        5.             99.          100.        100.             2.
  10
        1.             1.          1000.          0.10           0.10
        0.45           0.30           0.05       100.           100.  0
      450.           300.            50.
        2.             0.
        0.
        0.
        3.             2.
        0.
        0.
        4.             0.
        0.
        0.
        5.             0.
        0.
        0.
        6.             2.
        0.
        0.
        7.             0.
        0.
        0.
        8.             2.
        0.
        0.
        9.             0.          100.          0.20           0.20
        0.20           0.20          0.20       20.            20.
       20.            20.           20.
       10.             2.
        0.
        0.
```

```
10
      1.          1.          0.          5.
      2.          0.          2.          5.
      3.          2.          0.          5.
      4.          0.          3.          5.
      5.          0.          4.          5.
      6.          2.          0.          5.
      7.          0.          4.          5.
      8.          2.          0.          5.
      9.          0.          0.          5.
     10.          2.          0.          5.
13
    100.        100.         20.       0.01        0.01
   0.01        0.01        0.01        10.         10.
     10.         10.         10.
 4
    100.        100.        100.        100.
 1
 9
```

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED2 REACT1
701    -1     1    20     1
 10     5    10    15     8     5    13     5     5
  0
  0
  0
  1
      1.          1.       1000.       0.40        0.40
   0.15        0.04        0.01        400.        400.
  150.          40.         10.
  0
  0
  0
  1
  9
```

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED3 REACT1
702    -1     1    20     1
 10     5    10    15     8     5    13     5     5
  0
  0
  0
  1
      1.          1.       1000.       0.01        0.04
   0.15        0.40        0.40        10.         40.
  150.         400.        400.
  0
  0
  0
  1
  9
```

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED1 REACT2
 710    -1     1    20     1
  10     5    10    15     8     5    13     5     5
   0
   1
         3.          66.          0.          0.30          1.00
        0.15         1.00         2.55
   0
   1
         1.          1.         1000.         0.10          0.10
        0.45         0.30         0.05         100.          100.  0
       450.        300.         50.
   0
   0
   0
   1
   9
```

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED2 REACT2
 711    -1     1    20     1
  10     5    10    15     8     5    13     5     5
   0
   0
   0
   1
         1.          1.         1000.         0.40          0.40
        0.15         0.04         0.01         400.          400.
       150.         40.          10.
   0
   0
   0
   1
   9
```

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED3 REACT2
 712    -1     1    20     1
  10     5    10    15     8     5    13     5     5
   0
   0
   0
   1
         1.          1.         1000.         0.01          0.04
        0.15         0.40         0.40         10.           40.
       150.         400.         400.
   0
   0
   0
   1
   9
```

Results for Example 1

STREAM VARIABLES MATRIX AT RUN 700

1.0000	1.0000	1000.0000	0.1000	0.1000
0.4500	0.3000	0.0500	100.0000	100.0000
450.0000	300.0000	50.0000		
2.0000	0.	1138.2525	0.0965	0.1155
0.4048	0.3393	0.0439	109.8975	131.4189
460.7125	386.2237	50.0000		
3.0000	2.0000	180.8081	0.4863	0.5088
0.0025	0.0021	0.0003	87.9180	91.9932
0.4607	0.3862	0.0500		
4.0000	0.	957.4444	0.0230	0.0412
0.4807	0.4030	0.0522	21.9795	39.4257
460.2518	385.8375	49.9500		
5.0000	0.	589.8490	0.0186	0.0668
0.0546	0.6541	0.2058	10.9897	39.4257
32.2176	385.8375	121.3785		
6.0000	2.0000	367.5954	0.0299	0.
1.1644	0.	-0.1943	10.9897	0.
428.0341	0.	-71.4285		
7.0000	0.	430.7035	0.0255	0.0915
0.0711	0.8062	0.0056	10.9897	39.4257
30.6067	347.2537	2.4276		
8.0000	2.0000	159.1456	0.	0.
0.0101	0.2424	0.7474	0.	0.
1.6109	38.5837	118.9509		
9.0000	0.	138.9571	0.0712	0.2270
0.0771	0.6247	0.	9.8908	31.5405
10.7124	86.8134	0.		
10.0000	2.0000	291.7464	0.0038	0.0270
0.0682	0.8927	0.0083	1.0990	7.8851
19.8944	260.4403	2.4276		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

STREAM VARIABLES MATRIX AT RUN 701

1.0000	1.0000	1000.0000	0.4000	0.4000
0.1500	0.0400	0.0100	400.0000	400.0000
150.0000	40.0000	10.0000		
2.0000	0.	1180.9205	0.3722	0.4454
0.1300	0.0439	0.0085	439.5585	526.0013
153.5708	51.7899	10.0000		
3.0000	2.0000	720.0631	0.4884	0.5113
0.0002	0.0001	0.0000	351.6468	368.2009
0.1536	0.0518	0.0100		
4.0000	0.	460.8574	0.1908	0.3424
0.3329	0.1123	0.0217	87.9117	157.8004
153.4172	51.7381	9.9900		
5.0000	0.	288.5093	0.1524	0.5470
0.0372	0.1793	0.0841	43.9558	157.8004
10.7392	51.7381	24.2757		
6.0000	2.0000	172.3482	0.2550	0.
0.8278	0.	-0.0829	43.9558	0.
142.6780	0.	-14.2857		
7.0000	0.	259.0083	0.1697	0.6092
0.0394	0.1798	0.0019	43.9558	157.8004
10.2022	46.5643	0.4855		
8.0000	2.0000	29.5010	0.	0.
0.0182	0.1754	0.8064	0.	0.
0.5370	5.1738	23.7902		
9.0000	0.	181.0125	0.2185	0.6974
0.0197	0.0643	0.	39.5603	126.2403
3.5708	11.6411	0.		
10.0000	2.0000	77.9959	0.0564	0.4046
0.0850	0.4478	0.0062	4.3956	31.5601
6.6315	34.9232	0.4855		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

STREAM VARIABLES MATRIX AT RUN 702

1.0000	1.0000	1000.0000	0.0100	0.0400
0.1500	0.4000	0.4000	10.0000	40.0000
150.0000	400.0000	400.0000		
2.0000	0.	1133.2837	0.0097	0.0468
0.1355	0.4551	0.3530	10.9915	53.0085
153.5708	515.7129	400.0000		
3.0000	2.0000	46.9685	0.1872	0.7900
0.0033	0.0110	0.0085	8.7932	37.1060
0.1536	0.5157	0.4000		
4.0000	0.	1086.3152	0.0020	0.0146
0.1412	0.4743	0.3678	2.1983	15.9026
153.4172	515.1972	399.6000		
5.0000	0.	1513.9660	0.0007	0.0105
0.0071	0.3403	0.6414	1.0992	15.9026
10.7392	515.1972	971.0280		
6.0000	2.0000	-427.6508	-0.0026	-0.
-0.3336	-0.	1.3362	1.0992	0.
142.6780	0.	-571.4280		
7.0000	0.	510.3020	0.0022	0.0312
0.0200	0.9086	0.0381	1.0992	15.9026
10.2022	463.6774	19.4206		
8.0000	2.0000	1003.6641	0.	0.
0.0005	0.0513	0.9481	0.	0.
0.5370	51.5197	951.6074		
9.0000	0.	133.2014	0.0074	0.0955
0.0268	0.8703	0.	0.9892	12.7220
3.5708	115.9194	0.		
10.0000	2.0000	377.1005	0.0003	0.0084
0.0176	0.9222	0.0515	0.1099	3.1805
6.6315	347.7581	19.4206		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

STREAM VARIABLES MATRIX AT RUN 710

1.0000	1.0000	1000.0000	0.1000	0.1000
0.4500	0.3000	0.0500	100.0000	100.0000
450.0000	300.0000	50.0000		
2.0000	0.	1148.1809	0.0921	0.1137
0.4124	0.3383	0.0435	105.6945	130.4928
473.5473	388.4463	50.0000		
3.0000	2.0000	176.8125	0.4782	0.5166
0.0027	0.0022	0.0003	84.5556	91.3450
0.4735	0.3884	0.0500		
4.0000	0.	971.3684	0.0218	0.0403
0.4870	0.3995	0.0514	21.1389	39.1478
473.0738	388.0579	49.9500		
5.0000	0.	631.8809	0.0100	0.0620
0.1123	0.6141	0.2016	6.3417	39.1478
70.9611	388.0579	127.3725		
6.0000	2.0000	339.4874	0.0436	0.
1.1845	0.	-0.2281	14.7972	0.
402.1127	0.	-77.4225		
7.0000	0.	464.7020	0.0136	0.0842
0.1451	0.7516	0.0055	6.3417	39.1478
67.4130	349.2521	2.5474		
8.0000	2.0000	167.1789	0.	0.
0.0212	0.2321	0.7467	0.	0.
3.5481	38.8058	124.8250		
9.0000	0.	147.9333	0.0386	0.2117
0.1595	0.5902	0.	5.7075	31.3183
23.5946	87.3130	0.		
10.0000	2.0000	316.7687	0.0020	0.0247
0.1383	0.8269	0.0080	0.6342	7.8296
43.8185	261.9391	2.5474		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

STREAM VARIABLES MATRIX AT RUN 711

1.0000	1.0000	1000.0000	0.4000	0.4000
0.1500	0.0400	0.0100	400.0000	400.0000
150.0000	40.0000	10.0000		
2.0000	0.	1168.4904	0.3619	0.4502
0.1351	0.0443	0.0086	422.8328	526.0006
157.8658	51.7912	10.0000		
3.0000	2.0000	706.6863	0.4787	0.5210
0.0002	0.0001	0.0000	338.2663	368.2004
0.1579	0.0518	0.0100		
4.0000	0.	461.8041	0.1831	0.3417
0.3415	0.1120	0.0216	84.5666	157.8002
157.7079	51.7394	9.9900		
5.0000	0.	284.0402	0.0893	0.5556
0.0833	0.1822	0.0897	25.3700	157.8002
23.6562	51.7394	25.4745		
6.0000	2.0000	177.7638	0.3330	0.
0.7541	0.	-0.0871	59.1966	0.
134.0517	0.	-15.4845		
7.0000	0.	252.7185	0.1004	0.6244
0.0889	0.1843	0.0020	25.3700	157.8002
22.4734	46.5655	0.5095		
8.0000	2.0000	31.3218	0.	0.
0.0378	0.1652	0.7971	0.	0.
1.1828	5.1739	24.9650		
9.0000	0.	168.5802	0.1354	0.7488
0.0467	0.0691	0.	22.8330	126.2401
7.8657	11.6414	0.		
10.0000	2.0000	84.1383	0.0302	0.3751
0.1736	0.4151	0.0061	2.5370	31.5600
14.6077	34.9241	0.5095		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

STREAM VARIABLES MATRIX AT RUN 712

1.0000	1.0000	1000.0000	0.0100	0.0400
0.1500	0.4000	0.4000	10.0000	40.0000
150.0000	400.0000	400.0000		
2.0000	0.	1137.1581	0.0093	0.0466
0.1388	0.4535	0.3518	10.5710	53.0085
157.8657	515.7129	400.0000		
3.0000	2.0000	46.6363	0.1813	0.7956
0.0034	0.0111	0.0086	8.4568	37.1060
0.1579	0.5157	0.4000		
4.0000	0.	1090.5217	0.0019	0.0146
0.1446	0.4724	0.3664	2.1142	15.9026
157.7078	515.1972	399.6000		
5.0000	0.	1574.3701	0.0004	0.0101
0.0150	0.3272	0.6472	0.6343	15.9026
23.6562	515.1972	1018.9800		
6.0000	2.0000	-483.8484	-0.0031	-0.
-0.2771	-0.	1.2801	1.4799	0.
134.0516	0.	-619.3800		
7.0000	0.	523.0672	0.0012	0.0304
0.0430	0.8865	0.0390	0.6343	15.9026
22.4734	463.6774	20.3796		
8.0000	2.0000	1051.3029	0.	0.
0.0011	0.0490	0.9499	0.	0.
1.1828	51.5197	998.6004		
9.0000	0.	137.0779	0.0042	0.0928
0.0574	0.8456	0.	0.5708	12.7220
7.8657	115.9194	0.		
10.0000	2.0000	385.9893	0.0002	0.0082
0.0378	0.9010	0.0528	0.0634	3.1805
14.6077	347.7581	20.3796		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.

Example 2

The process being designed in Example 1 is to be modified by the addition of three more pieces of equipment as shown in Figure 6. It is desired to see how these modifications will affect the results of feed 1, reaction 1 in Example 1. The first set of data is for the initial problem. The second set makes the modifications for Example 2. The input data statements and the results are shown on the following pages. Note that the second set is not complete but only contains the modifications to the first set. This is permissible when KCLEAN≠0.

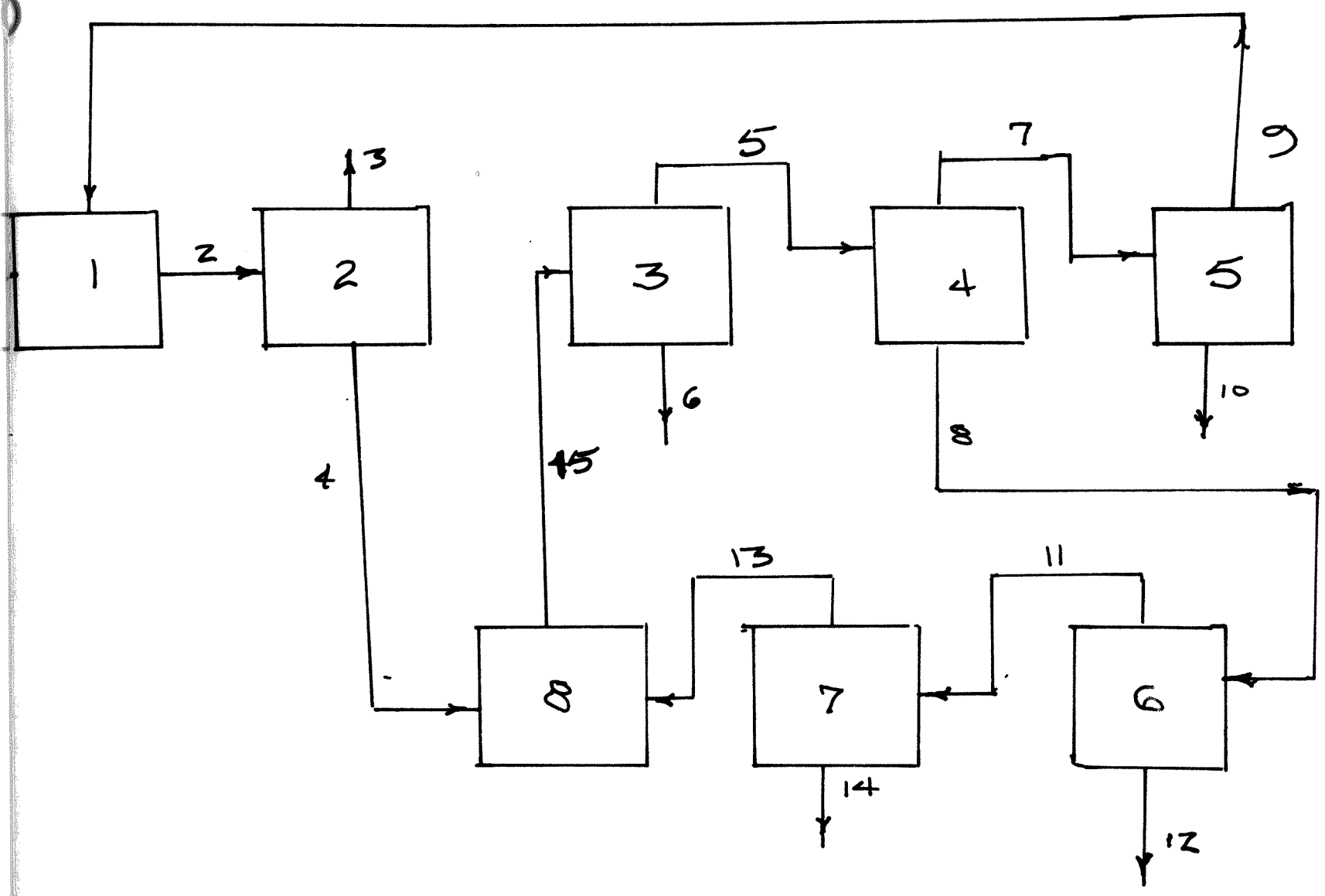

FIGURE 6

Flow Diagram for Example 2

Process Matrix (13)

Equipment Number	Name (Subroutines)	Associated Streams		
1	Mixer (UNAME3)	1	9	-2
2	Tower (UNAME2)	2	-3	-4
3	Reactor (UNAME2)	15	-5	-6
4	Tower (UNAME2)	5	-7	-8
5	Tower (UNAME2)	7	-9	-10
6	Tower (UNAME2)	8	-11	-12
7	Splitter (UNAME2)	11	-13	-14
8	Mixer (UNAME3)	4	13	-15

Data Input for Example 2

```
LIGHT ENDS SIMULATION  SHANNON  8/24/61  SIMPLE CASE  FEED1 REACT1
 700    -1      0    20     1
  10     5     10    15     8     5    13     5     5
   5
    1UNAME3          501          1     9    -2
    2UNAME2          502          2    -3    -4
    3UNAME2          503          4    -5    -6
    4UNAME2          504          5    -7    -8
    5UNAME2          505          7    -9   -10
    5
         1.              99.
         0.
         2.              66.              0.          0.800          0.70
         0.001            0.001           0.001
         3.              66.              0.          0.50           1.0
         0.07             1.00            2.43
         4.              66.              0.          1.00           1.00
         0.95             0.90            0.02
         5.              66.              0.          0.90           0.80
         0.35             0.25            0.0
    5
         1.              99.              0.          0.             0.
         2.              99.            100.        100.             2.
         3.              99.             10.         10.             2.
         4.              99.            100.        100.             2.
         5.              99.            100.        100.             2.
   10
         1.              1.            1000.          0.10           0.10
         0.45            0.30             0.05      100.          100.  0
       450.            300.              50.
         2.              0.
         0.
         0.
         3.              2.
         0.
         0.
         4.              0.
         0.
         0.
         5.              0.
         0.
         0.
         6.              2.
         0.
         0.
         7.              0.
         0.
         0.
         8.              2.
         0.
         0.
         9.              0.            100.          0.20           0.20
         0.20            0.20            0.20       20.           20.
        20.             20.             20.
        10.              2.
         0.
         0.
```

```
10
        1.          1.          0.          5.
        2.          0.          2.          5.
        3.          2.          0.          5.
        4.          0.          3.          5.
        5.          0.          4.          5.
        6.          2.          0.          5.
        7.          0.          4.          5.
        8.          2.          0.          5.
        9.          0.          0.          5.
       10.          2.          0.          5.
13
     100.        100.         20.        0.01        0.01
     0.01        0.01        0.01         10.         10.
     10.         10.         10.
 4
     100.        100.        100.        100.
 1
 9
```

```
LIGHT ENDS SIMULATION   ALL COMPONENTS SEPARATED
 699   -1    1   20    1
   8    5    8   20    8    5   13    5    5
   8
   1UNAME3       101          1    9   -2
   2UNAME2       102          2   -3   -4
   3UNAME2       103         15   -5   -6
   4UNAME2       104          5   -7   -8
   5UNAME2       105          7   -9  -10
   6UNAME2       106          8  -11  -12
   7UNAME2       107         11  -13  -14
   8UNAME3       108          4   13  -15
   3
6.             66.                          0.95        0.95
0.95           0.95        0.03
7.             66.                          0.95        0.95
0.95           0.02        0.003
8.             99.
0.
   3
6.             66.         100.        100.        2.
7.             66.         100.        100.        2.
8.             99.
   6
8.
0.
0.
11.
0.
0.
12.            2.
0.
0.
13.
0.
0.
```

10

```
14.          2.
 0.
 0.
15.
 0.
 0.
     6
 8.                                    5.
11.                                    5.
12.          2.                        5.
13.                                    5.
14.          2.                        5.
15.                                    5.
     0
     0
     0
```

Results for Example 2

STREAM VARIABLES MATRIX AT RUN 699

1.0000	1.0000	1000.0000	0.1000	0.1000
0.4500	0.3000	0.0500	100.0000	100.0000
450.0000	300.0000	50.0000		
2.0000	0.	1139.3170	0.0965	0.1155
0.4044	0.3398	0.0439	109.8902	131.5697
460.7462	387.1109	50.0000		
3.0000	2.0000	180.9088	0.4859	0.5091
0.0025	0.0021	0.0003	87.9121	92.0988
0.4607	0.3871	0.0500		
4.0000	0.	958.4082	0.0229	0.0412
0.4803	0.4035	0.0521	21.9780	39.4709
460.2854	386.7238	49.9500		
5.0000	0.	591.6459	0.0186	0.0667
0.0546	0.6549	0.2052	10.9890	39.4709
32.3221	387.4594	121.4045		
6.0000	2.0000	368.9670	0.0298	0.
1.1638	0.	-0.1936	10.9890	0.
429.4218	0.	-71.4438		
7.0000	0.	432.0265	0.0254	0.0913
0.0711	0.8065	0.0056	10.9891	39.4622
30.7034	348.4437	2.4281		
8.0000	-0.	159.3084	0.	0.
0.0101	0.2430	0.7468	0.	0.
1.6160	38.7160	118.9764		
9.0000	0.	139.3170	0.0710	0.2266
0.0771	0.6253	0.	9.8902	31.5697
10.7462	87.1109	0.		
10.0000	2.0000	292.7094	0.0038	0.0270
0.0682	0.8928	0.0083	1.0989	7.8924
19.9572	261.3328	2.4281		
11.0000	-0.	41.8846	0.	0.
0.0367	0.8781	0.0852	0.	0.
1.5352	36.7802	3.5693		
12.0000	2.0000	117.4237	0.	0.
0.0007	0.0165	0.9828	0.	0.
0.0808	1.9358	115.4071		
13.0000	-0.	2.2047	0.	0.
0.6615	0.3336	0.0049	0.	0.
1.4584	0.7356	0.0107		
14.0000	2.0000	39.6799	0.	0.
0.0019	0.9084	0.0897	0.	0.

0.0768	36.0446	3.5586		
15.0000	-0.	960.6129	0.0229	0.0411
0.4807	0.4033	0.0520	21.9780	39.4709
461.7439	387.4594	49.9607		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		
0.	0.	0.	0.	0.
0.	0.	0.	0.	0.
0.	0.	0.		

Example 3

This problem gives an insight as to how PACER performs recycle calculations. The problem illustrated in Figure 7 appears to be of second order magnitude such that when streams 8 and 9 are assumed known, the problem may be solved. This is true, however the problem is not second order. If the correct stream, in this case stream 3, is assumed known, an iterative solution is possible. Thus, the problem is really only a first order recycle.

Figure 8 illustrates a problem which appears to be of third order magnitude. Once again by choosing the proper stream, in this case stream 22, the problem is shown to be of only first order magnitude. The input data statements and results are shown on the following pages.

The above discussion does not mean that all problems are of only first order magnitude. It would seem to imply that only an extremely complicated problem will be of greater magnitude than third order recycle, and thus be unsolvable by PACER in its present form.

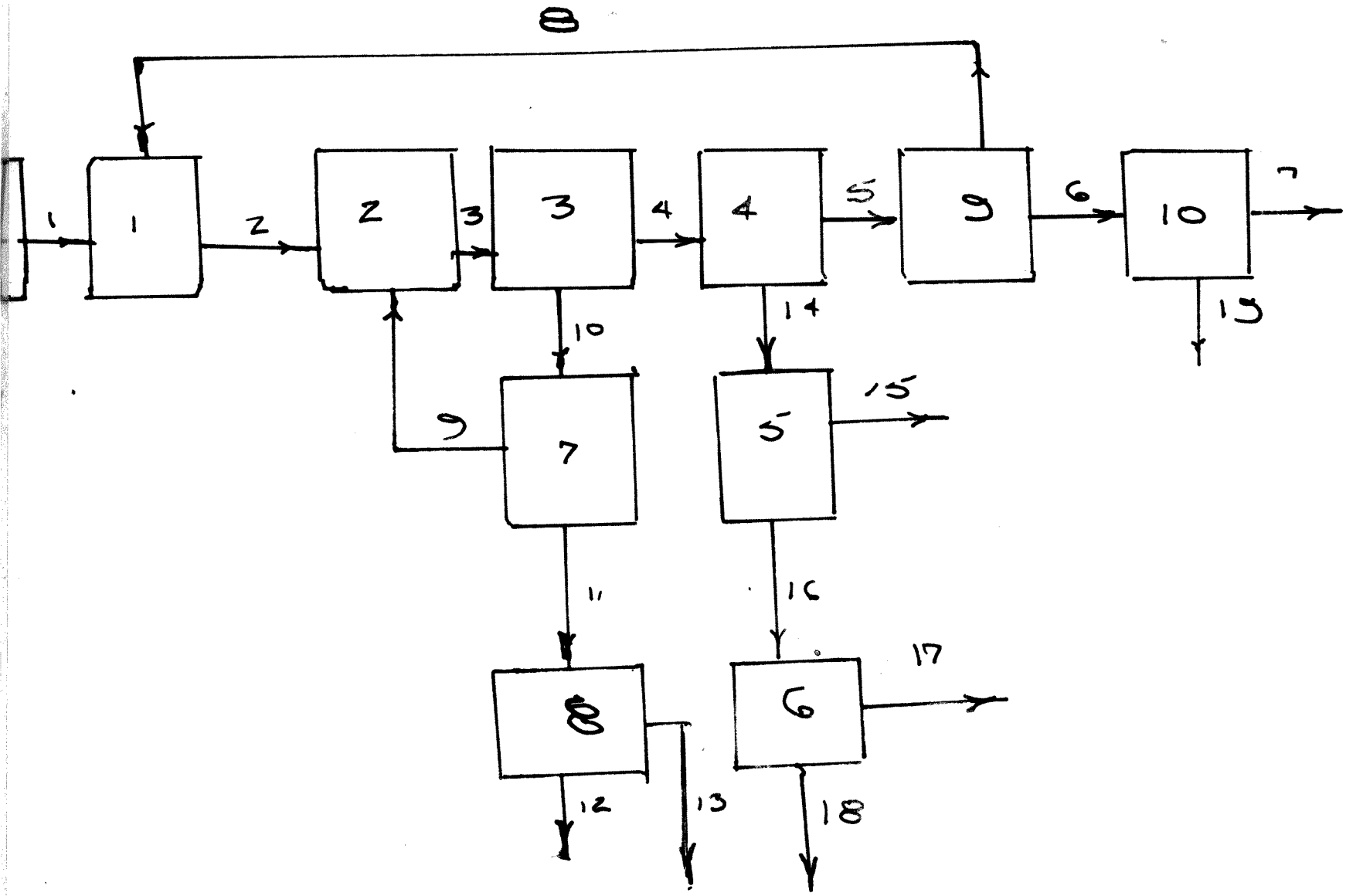

FIGURE 7

Flow Diagram for Example 3, Run 1

Process Matrix

Equipment

Number	Name	Associated Streams		
1	UNAME 3	1	8	-2
2	" 3	2	9	-3
3	" 2	3	-4	-10
4	" 2	4	-5	-14
5	" 2	14	-15	-16
6	" 2	16	-17	-18
7	" 2	10	-9	-11
8	" 2	11	-13	-12
9	" 2	5	-8	-6
10	" 2	6	-7	-19
11	" 2	20	-1	-21

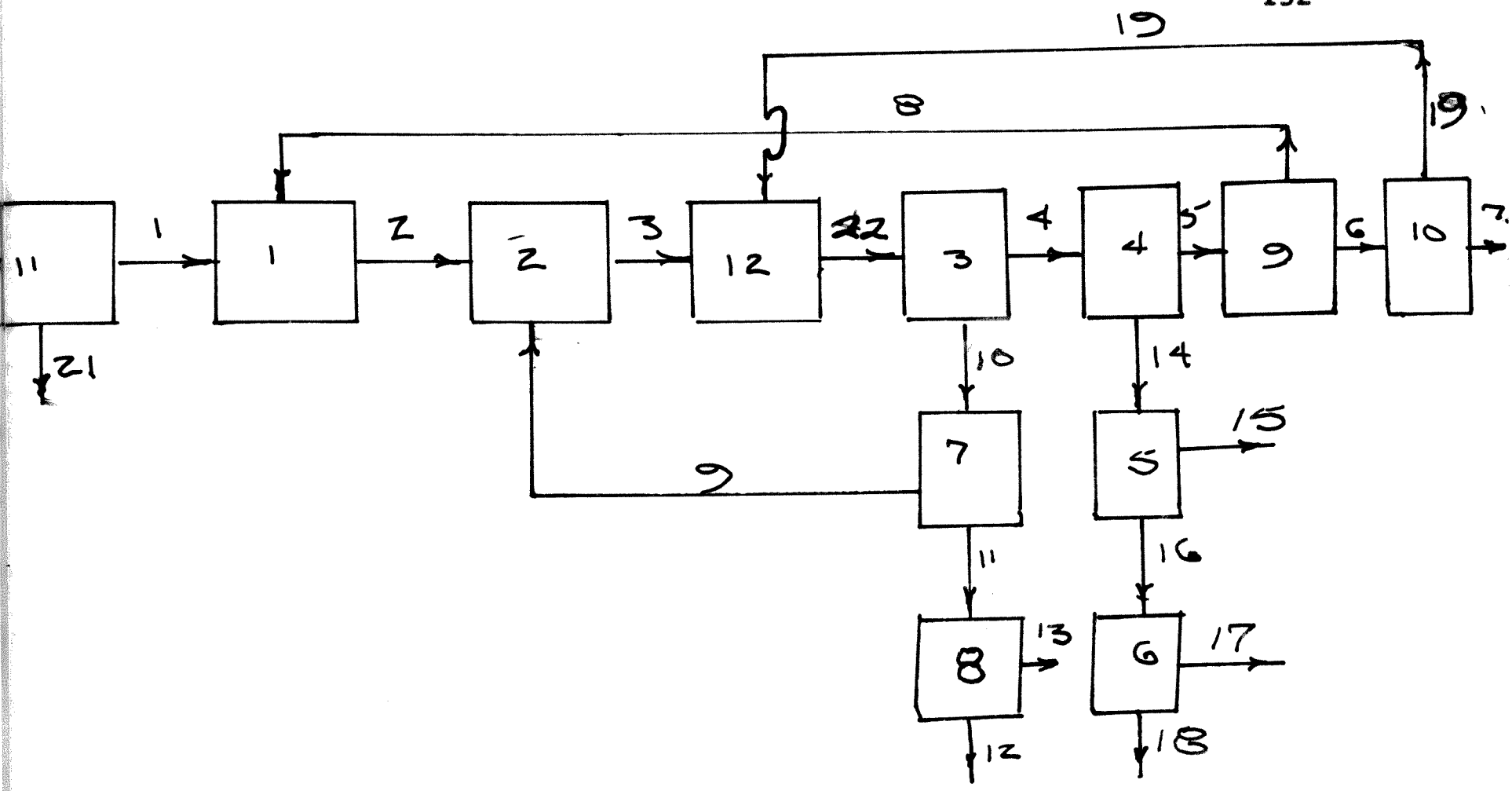

FIGURE 8

Flow Diagram for Example 3, Run 2

Process Matrix

Equipment

Number	Name
1	UNAME 3
2	" 3
3	" 2
4	" 2
5	" 2
6	" 2
7	" 2
8	" 2
9	" 2
10	" 2
11	" 2
12	" 3

Associated Streams		
1	8	-2
2	9	-3
22	-4	-10
4	-5	-14
14	-15	-16
16	-17	-18
10	-9	-11
11	-13	-12
5	-8	-6
6	-7	-19
20	-1	-21
3	19	-22

Data Input for Example 3

```
SECOND ORDER RECYCLE          NOT REALLY
     1    -1     0    20     1
    11     5    11    21     8     5    13     5     2
    11
     1UNAME3        103              1     8    -2
     2UNAME3        107              2     9    -3
     3UNAME2        436              3    -4   -10
     4UNAME2        126              4    -5   -14
     5UNAME2        228             14   -15   -16
     6UNAME2        416             16   -17   -18
     7UNAME2         31             10    -9   -11
     8UNAME2        135             11   -13   -12
     9UNAME2        155              5    -8    -6
    10UNAME2         22              6    -7   -19
    11UNAME2        444             20    -1   -21
    11
1.          23.
0.
2.          23.
0.
3.          33.                                .8          .7
.2          .15          .7
4.          33.                                .8          .7
.2          .2           .75
5.          33.                                .2          .3
.75         .82          .3
6.          33.                                .333        .25
.8          .75          .25
7.          33.                                .4          .333
.8          .75          .3
8.          33.                                .8          .75
.25         .2           .75
9.          33.                                .65         .8
.5          .6           .8
10.         33.                                .45         .55
.7          .8           .2
11.         33.                                .75         .8
.5          .45          .25
    11
1.          23.
2.          23.
3.          33.          100.          100.          2.
4.          33.          100.          100.          2.
5.          33.          100.          100.          2.
6.          33.          100.          100.          2.
7.          33.          100.          100.          2.
8.          33.          100.          100.          2.
9.          33.          100.          100.          2.
10.         33.          100.          100.          2.
11.         33.          100.          100.          2.
    21
1.
0.
0.
2.
0.
0.
```

3.
0.
0.
4.
0.
0.
5.
0.
0.
6.
0.
0.
7. 2.
0.
0.
8.
0.
0.
9.
0.
0.
10.
0.
0.
11.
0.
0.
12. 2.
0.
0.
13. 2.
0.
0.
14.
0.
0.
15. 2.
0.
0.
16.
0.
0.
17. 2.
0.
0.
18. 2.
0.
0.
19. 2.
0.
0.
20. 1. 1000. .1 .1
.4 .25 .15 100. 100.
400. 250. 150.
21. 2.
0.
0.

```
   21
1.          0.          2.          5.
2.          0.          2.          5.
3.          0.          2.          5.
4.          0.          2.          5.
5.          0.          2.          5.
6.          0.          2.          5.
7.          2.                      5.
8.          0.          2.          5.
9.          0.          2.          5.
10.         0.          2.          5.
11.         0.          2.          5.
12.         2.                      5.
13.         2.                      5.
14.         0.          2.          5.
15.         2.                      5.
16.         0.          2.          5.
17.         2.                      5.
18.         2.                      5.
19.         2.                      5.
20.         1.                      5.
21.         2.                      5.
   13
100.        100.        20.         .01         .01
.01         .01         .01         10.         10.
10.         10.         10.
   4
100.        100.        100.        100.
   2
   8   9
```

THIRD ORDER RECYCLE NOT REALLY

```
    2   -1    1   20    1
   12    5   12   22    8    5   13    5    3
    2
   3UNAME2      436        22   -4   -10
  12UNAME3      555         3   19   -22
    1
12.             23.
0.
    1
12.             23.
    2
19.
0.
0.
22.
0.
0.
    2
19.         0.          2.          5.
22.         0.          2.          5.
    0
    0
    3
    8   9   19
```

Example 4

A feed stream is available with a flow rate of 1000 moles/hr equally split between 5 components. A product stream is desired with a much greater purity of component 1. To separate the components, 11 separators are available. These separators are simulated by equipment subroutine UNAME2. The equipment units are not very efficient. Each separator takes the input stream and places the following fractions of components 1 through 5 respectively in the overhead:

0.7, 0.6, 0.5, 0.4, 0.3.

The question is, how will recycle of the "waste" bottoms from each of the separators affect the flow rate and purity of the final product stream, in this case stream 34. Figure 9 shows the planned process design. Dotted lines signify possible recycle. Table 4 shows the effect of recycle on stream 34.

This problem illustrates the advantage of PACER over complete programming of individual problems. 13 different designs have been analyzed with one pass at the computer.

The complexity of the recycled streams in runs 17 and 18 have made them greater than third order recycle. By purposely misflagging stream 28 in run 17 and streams 7 and 28 in run 18, the problem is reduced to third order magnitude. The results are correct because all required unknown equipments are still in the KE1 list used in making trial and error iterative calculations.

With no recycle the final product stream has a flow rate of 3.2559 moles/hr and contains 85.02% component 1. With recycle of all "waste" bottoms except that from the first equipment, stream 34 has a flow rate of 73.2198 moles/hr and contains 86.38% component 1. This conclusively shows the effect of recycle.

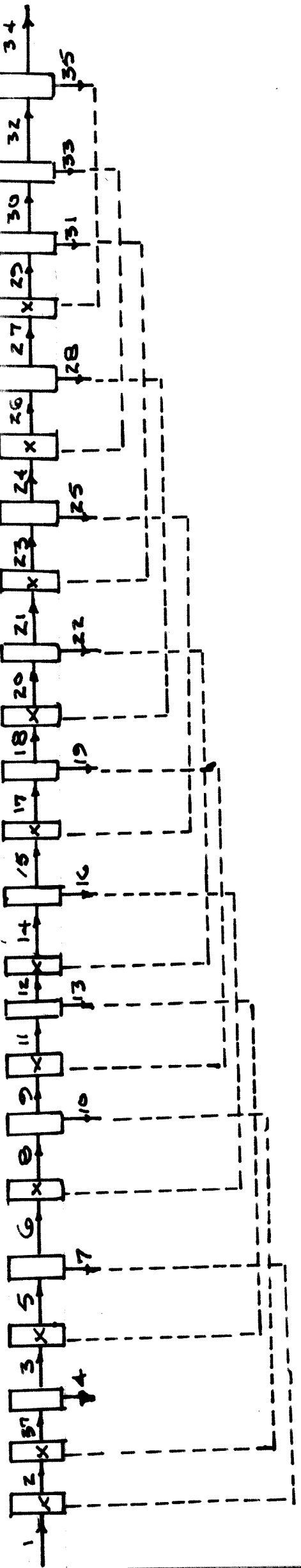

FIGURE 9

Flow Diagram for Example 4

(dotted lines show possible recycled streams)

Process Matrix for Run 18

Equipment				
Number	Name	Associated Streams		
1	UNAME 3	1	10	-2
2	" 2	32	-3	-4
3	" 3	3	13	-5
4	" 2	5	-6	-7
5	" 3	6	16	-8
6	" 2	8	-9	-10
7	" 3	9	19	-11
8	" 2	11	-12	-13
9	" 3	12	22	-14
10	" 2	14	-15	-16
11	" 3	15	25	-17
12	" 2	17	-18	-19
13	" 3	18	28	-20
14	" 2	20	-21	-22
15	" 3	21	31	-23
16	" 2	23	-24	-25
17	" 3	24	33	-26
18	" 2	26	-27	-28
19	" 3	27	35	-29
20	" 2	29	-30	-31
21	" 2	30	-32	-33
22	" 2	32	-34	-35
23	" 3	2	7	-37

TABLE 4

Results for Example 4

Run Number	Streams Recycled	Volume Flow Rate of Product (moles/hr)	Flow Rate of Component 1 in Product (moles/hr)	Mole Fraction of Component 1 in Product	Fraction Component 1 Recovered
6	-	3.2559	0.8502	2.7683	0.0138
7	35	3.0816	0.8510	3.2350	0.0162
8	35,33,31	5.7275	0.8533	4.8875	0.0244
9	35,33,31,28	7.3787	0.8552	6.3100	0.0316
10	35,33,31,28,25	9.6296	0.8571	8.2539	0.0413
11	35,33,31,28,25,22	12.8048	0.8595	11.0053	0.0550
12	35,33,31,28,25,22,19	17.2561	0.8621	14.8762	0.0744
13	35,33,31,28,25,22,19,16	23.3941	0.8648	20.2315	0.1012
14	35,33,31,28,25,22,19,16,13	31.9037	0.8676	27.6812	0.1384
16	33,31,28,25,22,19,16,13,10	31.9958	0.8678	27.7651	0.1388
17	35,33,31,28,25,22,19,16,13,10	43.8358	0.8707	38.1668	0.1908
18	35,33,31,28,25,22,19,16,13,10,7	73.2197	0.8638	63.2476	0.3162

Example 5

This problem illustrates the uses of equipment subroutine UNAME4, a flash distillation simulation. The feed stream has five components. Three different sets of equilibrium data are used. The equilibrium data depends on the conditions of the flash, specifically the temperature and pressure.

Run 3 has data which converged to 26% of the feed being flashed. Run 4 has data which resulted in none of the feed having flashed. Run 5 has data which resulted in all of the feed being flashed. Input data is shown for all three runs. Results are only shown for Run 3. The intermediate printout is shown to illustrate how quickly the Newton-Raphson method converged.

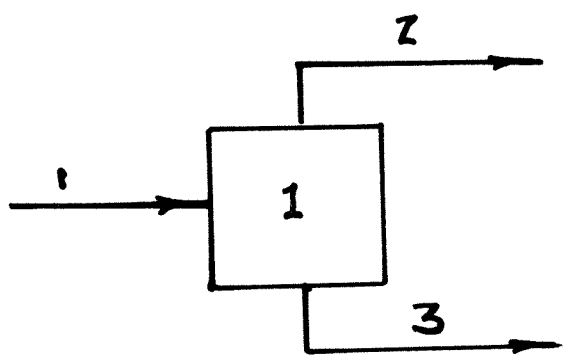

FIGURE 10

Flow Diagram for Example 5

Process Matrix

Equipment		Associated Streams		
Number	Subroutine			
1	UNAME4	1	-2	-3

Data Input for Example 5

```
FLASH DISTILLATION
      3   -1    0   20    1
      1    5    1    3    8    5   13    5    0
      1
      1UNAME4        222          1   -2   -3
      1
1.                  23.                              1.95          1.55
0.65                0.56          0.25
      1
1.                  23.
      3
1.                  1.            1000.          0.3             0.2
0.1                 0.2           0.2            300.            200.
100.                200.          200.
2.0                 2.

3.0                 2.

      3
1.0                 1.0           1.0            5.0
2.0                               1.0            5.0
3.0                               1.0            5.0
      0
      0
      0
NO  FEED  VAPORIZED
      4   -1    1   20    1
      1    5    1    3    8    5   13    5    0
      0
      1
1.                  23.                            0.1           0.1
0.1                 0.1           0.1
      0
      0
      0
      0
      0
      0
ALL FEED VAPORIZED
      5   -1    1   20    1
      1    5    1    3    8    5   13    5    0
      0
      1
1.                  23.                            10.0          10.0
10.0                10.0          10.0
      0
      0
      0
      0
      0
      0
```

TABLE 5

Results of Example 5, Run 3

$$\sum_{i=1}^{NOCOMP} Z_i/K_i = SUMX = 1.594$$

$$\sum_{i=1}^{NOCOMP} Z_i K_i = SUMY = 1.122$$

Newton-Raphson Iteration Calculations

Loop (KLOOP)	Assumed Vapor Fraction (VAPOR)	Calculated Vapor Fraction (CALCV)
1	0.0	0.1978
2	0.1978	0.2566
3	0.2566	0.2612
4	0.2612	0.2613
5	0.2613	0.2613

Final Results

Component	Equilibrium Value (K_i)	Feed Mole Fraction	Overheads Mole Fraction	Bottoms Mole Fraction
1	1.95	0.3	0.4687	0.2403
2	1.55	0.2	0.2711	0.1749
3	0.65	0.1	0.0715	0.1101
4	0.56	0.2	0.1265	0.2260
5	0.25	0.2	0.0622	0.2487

APPENDIX C

Nomenclature

A - equivalent to I. A is an equipment subroutine name in subroutine DREAD.

DELS - stream variables test vector.

DELSC - stream control variables test vector.

EN - equipment parameters matrix.

ENC - equipment control parameters matrix.

I2 - variable for PACER equipment numbers.

I12 - variable for PACER stream numbers.

KBACK - an internal variable which is equal to zero except when the problem parameters are varied internally by CHANGE.

KCLEAN - input indicator which controls the process of setting all matrices to zero before starting the present calculations.

KCV1 - records an unknown stream assumed known in GUESS1.

KCV2 - records an unknown stream assumed known in GUESS1.

KCV3 - records an unknown stream assumed known in GUESS1.

KE1 - same list of equipments as KE3 but in the order of list KE2.

KE2 - first list of equipments formed by GUESS2 in attempting to use an iterative solution.

KE3 - shortest list of equipment with which an iterative solution can be performed.

KEFLAG - equipment flag.

KES - an indicator. KES≠0 only when an equipment was calculated directly during last scan of KPM matrix. This indicates another scan is necessary.

KES2 - indicator used in GUESS2.

KPM - the Process Matrix.

KPMR - dummy Process Matrix into which the input cards are directly read before storing the information.

KPS - preferred streams vector.

KRET - if KRET=2, streams assumed known in GUESS1 were not satisfactory in attempting to use an iterative solution.

KRET2 - if KRET2=2, the calculations failed.

KRET3 - indicator telling if iterative solution converged.

KRUN - run number of problem.

KSEM - stream-equipment matrix.

KSFLAG - stream flag.

KSETS - input indicator controlling the amount of printout.

KUS - unknown stream vector formed in GUESS1.

KVOID - an indicator which will reject the problem if an equipment name has been given incorrectly.

LIMIT - used in TEST to indicate if a stream has not entirely converged for all of its variables.

LIMIT2 - in SETUP of LIMIT2=0, convergence has been obtained.

LIMIT3 - used in TEST to count each stream variable and stream control variable which has not converged.

LOOP - used in SETUP to count the iterative loops.

LOOPS - maximum number of iteration loops allowed in trying to obtain convergence.

N2 - variable for Process Matrix row.

N3 - variable for Process Matrix column.

N2MAX - number of rows in the Process Matrix.

N3MAX - number of columns in the Process Matrix.

NAME - equipment subroutine name of given equipment.

ND3 - used as counting index in GUESS1.

ND4 - used as counting index in GUESS1.

NDELS - an input control variable. A stream variables test vector will be read by DREAD only if NDELS≠0.

NDELSC - an input control variable. A stream control variables test vector will be read by DREAD only if NDELSC≠0.

NE - PACER equipment number.

NE1MAX - number of equipment in list KE1.

NE2 - index for list KE2.

NE2MAX - number of equipment in list KE2.

NE3MAX - number of equipment in list KE3.

NECALL - call number of a given equipment which depends on the equipment subroutine name.

NECLMX - length of equipment control parameters vector.

NELMAX - length of equipment parameters vector.

NEMAX - maximum equipment number in problem.

NEN - variable for EN row number.

NENC - variable for ENC row number.

NEQUIP - number indicating which equipment subroutine to call in calculating an equipment.

NEX - PACER equipment number of the Process Matrix row. NEX=KPMR(K2,1).

NEXEQN - external equipment number.

N1N - number of input streams to a given equipment.

NOEN - number of equipment parameter vectors to be read by DREAD.

NOENC - number of equipment control parameter vectors to be read by DREAD.

NOGO - input indicator determining whether a non-converged set of answers is acceptable if LOOP=LOOPS.

NOKPM - number of Process Matrix vectors to be read.

NOKPS - number of preferred streams to be read by DREAD.

NOSN - number of stream variable vectors to be read by DREAD.

LAST PAGE

NOSNC - number of stream control variable vectors to be read by DREAD.

NOUT - number of output streams from a given equipment.

NPSMAX - maximum number of preferred streams to be supplied in the input.

NS - PACER stream number.

NSCLMX - length of stream control properties vector.

NSL - variable for SN column number.

NSLMAX - length of stream properties vector.

NSMAX - maximum stream number in problem.

NSN - variable for SN row number.

NSNC - variable for SNC row number.

NUS - index of unknown stream vector.

NUS1 - used as counting index in GUESS1.

NUS2 - used as counting index in GUESS1.

NUS3 - used as counting index in GUESS1.

NUSMAX - total number of streams listed in the unknown stream vector.

SN - stream variables matrix.

SNC - stream control variables matrix.

STRM - dummy matrix into which the input data is read before storing the data in its proper matrix. Also used for storage of information when performing an iterative solution.

STRMC - matrix used to store information about the SNC matrix when performing an iterative solution.

STRMCI - matrix for SNC row vectors of streams entering an equipment being calculated.

STRMCO - matrix for SNC row vectors of streams leaving an equipment being calculated.

STRMI - matrix for SN row vectors of streams entering an equipment being calculated.

STRMO - matrix for SN row vectors of streams leaving an equipment being calculated.

TITLE - title of problem.